# THE CONVEX UNIFORM 5-POLYTOPES: A COMPUTER-ASSISTED CLASSIFICATION

P. Graham
https://orcid.org/0009-0007-8493-1505

Abstract. A convex polytope is uniform if its symmetries act transitively on its vertices and its facets are uniform, down to regular polygons. We give a computer-assisted completeness proof for the known convex uniform 5-polytopes: Wythoff constructions of types $A_5$, $B_5$ and $D_5$, prisms over convex uniform 4-polytopes, and products of regular polygons with convex uniform polyhedra. The theorem holds conditional on the cited 3-D and Conway–Guy–Möller 4-D classifications, the correctness of the written reductions in Sections 2–5, and the correctness of the certificate checkers. For faces with at most 16 sides, enumeration covers all possible vertex neighborhoods, identifies 628 known polytopes and rules out one additional candidate; a face with at least 17 sides forces a product. This extends Möller's method to dimension five and produces no new examples.

AI production. The proofs, the Lean formalization and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction. The automated checks described below have been run; no human expert review of the written reductions has taken place.
Scope of verification. This is a computer-assisted classification. The Lean project checks a conditional implication and selected geometric lemmas; the complete theorem is not formalized in Lean. The computational hypotheses and their connection to the certificates remain outside Lean. This work has not been expert-reviewed. Section 6 gives the replay and Section 7 states the formalization's scope.

Contents

1. Introduction 2
1.1. Uniform polytopes 2
1.2. The theorem 3
1.3. History and status 4
1.4. Dependence on the 3- and 4-dimensional classifications 4
1.5. Strategy 5
2. The reduction 6
2.1. Inscribed sphere and face centers 6
2.2. The separation lemma 6
2.3. Vertex figures 7
2.4. Cycles around a polygon and the closure equation 7
2.5. Faces, simple vertices 9
2.6. Stars and local uniqueness 9
2.7. The case split 10
3. The finite library and the known list 10
3.1. The facet library 10
3.2. Facet data 11
3.3. Model constructions 13

*Date*: October 20, 2026.
This work is licensed under Apache-2.0; embedded fonts retain their own licenses, listed in `THIRD_PARTY.md`.

3.4. The known list 14
4. Case A: every 2-face has at most 16 sides 14
4.1. Stage 1: cycles and their radii 14
4.2. Stage 2: vertex figures 17
4.3. Model ridge lists 18
4.4. Stage 3: from vertex figures to polytopes 19
4.5. Independent replays 21
4.6. Proof of Theorem 1.2 25
4.7. Computed coverage 25
5. Case B: a 2-face with at least 17 sides 26
5.1. Setting 26
5.2. Facts about the families 26
5.3. The radius forced by a large 2-face 27
5.4. Step B2: no large-polygon radius is a Case-A radius 28
5.5. Exactly one large 2-face at each vertex 30
5.6. Two or more large 2-faces at each vertex 31
5.7. Proof of Theorem 1.3 40
5.8. Proof of the Main Theorem 41
6. Verification and reproducibility 41
6.1. Arithmetic and software 41
6.2. Computational evidence and shared dependencies 41
6.3. The four-dimensional control 42
6.4. How to reproduce 42
7. What is checked in Lean 42
7.1. External hypothesis evidence map 43
Appendix A. The twelve Lean core hypotheses 51
Appendix B. Names 54
References 55

## 1. Introduction

1.1. **Uniform polytopes.** All polytopes are convex. A *polygon* is uniform if it is regular. For $d \geq 3$, a $d$-polytope $P$ is *uniform* if its symmetry group $\mathrm{Sym}(P)$ acts transitively on its vertices and each of its facets is a uniform $(d-1)$-polytope. Then every face of $P$ is uniform (faces of facets are faces), and all edges have the same length: the edges of a uniform polytope have one length by induction on the dimension, two facets sharing a ridge share its edges, and the facet graph (facets adjacent through ridges) is connected. We always scale to *unit edges*. Uniform 3-polytopes are the uniform polyhedra, uniform 4-polytopes are also called uniform polychora, and uniform 5-polytopes uniform polytera. The Lean predicates `Uniform5.Composition.Uniform` and `UnitUniform` in `lean/Uniform5/CompositionGeometry.lean` use this same geometric definition and unit-edge normalization.

Notation used throughout:

- $\{p\}$ is the regular $p$-gon with unit edges, of circumradius $r_p = 1/(2\sin(\pi/p))$; the distance between the two neighbors of a vertex of $\{p\}$ is $\mathrm{ch}_p := 2\cos(\pi/p)$.
- $P_q = \{q\} \times [0,1]$ is the $q$-gonal prism ($P_4$ is the cube) and $A_q$ the $q$-gonal antiprism ($A_3$ is the octahedron), with unit edges. The height of $A_q$ is $h_A(q)$ with $h_A(q)^2 = 1 - \frac{1}{4\cos^2(\pi/(2q))}$, and its circumradius is $R_{A_q}$ with $R_{A_q}^2 = r_q^2 + h_A(q)^2/4$.

- For polytopes $X \subset \mathbb{R}^a$, $Y \subset \mathbb{R}^b$ the product $X \times Y \subset \mathbb{R}^{a+b}$ is again a polytope; $\{p\} \times \{q\}$ is a *duoprism*, $K \times \{\} := K \times [0,1]$ a *prism*. A product of uniform polytopes with unit edges is uniform with unit edges.
- Sym, Stab, aff, conv, relint have their usual meanings; VF denotes a vertex figure (Section 2).

**Definition 1.1** (Wythoffian polytopes [1, 2])**.** Let $W$ be a finite irreducible Coxeter group of rank $k$ acting on $\mathbb{R}^k$, with simple roots $\alpha_0, \ldots, \alpha_{k-1}$ (nodes of the Coxeter diagram), and let $\emptyset \neq J \subseteq \{0, \ldots, k-1\}$ (the *ringed* nodes). Let $x_J$ be the point at distance $1/2$ from the mirrors of the ringed nodes and on the mirrors of the others, on the positive side of all of them. The *Wythoffian polytope* $W_J = \mathrm{conv}(Wx_J)$ is uniform with unit edges. We name it by the group and the ring pattern, e.g. $B_5$ `00001`; node orders are those of Table 3 in Appendix B.

The two non-Wythoffian non-prismatic convex uniform 4-polytopes are the *snub 24-cell* (the 600-cell minus the vertices of an inscribed 24-cell) and the *grand antiprism* (the 600-cell minus two great decagons in orthogonal planes) [8].

1.2. **The theorem.** To our knowledge, this is the first proof of completeness of the known convex uniform 5-polytopes, conditional on the cited three- and four-dimensional classifications. The proof has explicit computational lemmas and reproduction commands. It is a specialist classification result about the known five-dimensional list, not a classification in arbitrary dimension or a new general method. It is a computer-assisted result conditional on the cited 3-D and 4-D classifications (Facts 1.4 and 1.5); the separate Lean composition checks an implication from named inputs, with the limits stated in Section 7.

**Main Theorem.** *Assume Facts 1.4 and 1.5. Every convex uniform 5-polytope is congruent, after scaling to unit edges, to one of the following:*

(K1) *a Wythoffian polytope $W_J$ of one of the Coxeter groups $A_5$, $B_5$, $D_5$ (19 distinct $A_5$ forms, 31 $B_5$ forms and 8 further $D_5$ forms);*

(K2) *a prism $K \times \{\}$ over a convex uniform 4-polytope $K$; 45 Wythoffian bases, plus the snub 24-cell and grand antiprism, give 47 prisms, of which 46 are not already in (K1), since the tesseract prism is the 5-cube;*

(K3) *a product $\{p\} \times Q$ of a regular polygon, $p \geq 3$, with a convex uniform polyhedron $Q$: a Platonic or Archimedean solid, a prism $P_q$ (giving the* duoprism prisms $\{p\} \times \{q\} \times \{\}$*), or an antiprism $A_q$.*

The descriptions overlap (the 5-cube is a $B_5$ Wythoffian, the prism over the tesseract and $\{4\}\times$cube; prisms over polyhedral prisms, antiprism prisms and duoprisms are of the form (K3)). The theorem asserts coverage, not disjointness. Counting the Coxeter–Johnson constructions [2, 9, 15] with the four-dimensional bases of [18] gives

$$19 + 31 + 8 + 47 - 1 = 104.$$

The 47 fixed-base prisms include the tesseract prism, which is also the $B_5$ 5-cube; the subtraction counts it once. The same object occurs as $\{4\} \times$ cube in the product families. Thus 104 is a finite-sublist convention, not a disjoint count after removing every intersection with the infinite families. The Open Problem Garden page [19], checked October 20, 2026, states two infinite classes and 105 additional examples but gives no component counts or overlap explanation. Its 105 equals the subtotal $19 + 31 + 8 + 47$ before subtracting the cube. This is consistent with an unremoved overlap, but the page does not establish that historical explanation. The number 105 is not used by the proof. The generated congruence classes independently give 19 $A_5$, 31 $B_5$, 8 further $D_5$ forms and 47 fixed-base prisms, with exactly that one overlap. `checks/count_convention.py` extracts this bookkeeping from the audited stage-3 output; it is not another congruence proof. The proof

generates every ring pattern and product and tests congruence (Section 3). Every polytope in (K1)–(K3) is convex and uniform, so the theorem is a classification. It is the union of two theorems, proved in Sections 4 and 5:

**Theorem 1.2** (Case A)**.** *Every convex uniform 5-polytope all of whose 2-faces have at most 16 sides is congruent to one of the 628 distinct polytopes of the list $\mathcal{K}_{16}$ of Section 3 (all members of (K1)–(K3) whose polygons have at most 16 sides).*

**Theorem 1.3** (Case B)**.** *Every convex uniform 5-polytope with a 2-face of at least 17 sides is congruent to $\{n\} \times Q$ with $Q$ a convex uniform polyhedron, or to $\{q\} \times A_n$, for integers $n \geq 17$ and $q \geq 3$.*

Both conclusions of Theorem 1.3 are of type (K3). So Theorems 1.2 and 1.3 give the Main Theorem.

1.3. **History and status.** Gosset [12] enumerated the semiregular polytopes (regular facets) in all dimensions; G. Blind and R. Blind [6] proved that list complete. Elte's 1912 monograph [11] is another early treatment of higher-dimensional semiregular polytopes. Coxeter developed the Wythoff construction and the theory of uniform polytopes [9, 10], and Johnson's thesis [15] set up the general theory. The convex uniform polyhedra are the Platonic and Archimedean solids, prisms and antiprisms; the list of all uniform polyhedra, convex or not, was shown complete by Sopov [23] and Skilling [22]. In dimension four Conway and Guy announced in 1965 a computer-assisted completeness proof, finding one new non-Wythoffian polytope, the grand antiprism [8]; a complete account of a computer-assisted proof is Möller's thesis [17, 18]: 1000 edge neighborhoods, each with a forced circumradius, and 64 vertex neighborhoods, each matched with a known polychoron, with the prism and antiprism families treated by hand.

The present proof builds directly on Möller's classification and its edge-neighborhood, forced-radius and vertex-neighborhood enumeration strategy. The published three-dimensional classification supplies the possible polyhedral faces, and Conway–Guy and Möller supply the possible four-dimensional facets. Coxeter's and Johnson's constructions supply the known families whose completeness is at issue here. These foundational classifications and constructions are credited inputs; the contribution of this paper is the conditional five-dimensional completeness argument and its checked computations.

Schulte's 2017 Handbook chapter states that "a complete list is known only for $d = 4$" [21]. Winter's Appendix E.3 states: "it is unknown whether there exist non-prismatic non-Wythoffian uniform polytopes in dimension five or above." [24, p. 167] The Open Problem Garden and community catalogs also record the question as open [19, 20]. These sources support the status of the problem; they are not completeness premises.

Johnson's 1966 thesis [15] develops constructions in arbitrary dimension, including dimension five; its discussion of known enumeration (pp. 226, 291–296) does not establish completeness of the convex five-dimensional list. Johnson's unpublished 1991 manuscript [16] is cited by Schulte for the four-dimensional list; its full text is unavailable, so no five-dimensional completeness claim is attributed to it here. Bowers's and Klitzing's catalogs [3, 4] document known examples and nomenclature; Bowers's search includes nonconvex polytera and is not presented here as an exhaustive search of convex examples. Neither catalog is a completeness premise, and the construction lists used here are generated from Wythoff and Cartesian-product recipes.

Related verification design. The verification design was developed with Càllisto's honeycomb work [7] as a known methodological precedent, including its division between Lean arguments and external finite certificate checks; none of its theorems or certificates is a premise here. The geometric enumeration method used here is Möller's.

1.4. **Dependence on the 3- and 4-dimensional classifications.** The proof uses two classifications as cited inputs. They enter only through the following two facts.

**Fact 1.4** (3-D classification [21, Section 18.5], [13])**.** *A convex uniform polyhedron is a Platonic solid, one of the 13 Archimedean solids, a prism $P_q$ ($q \geq 3$) or an antiprism $A_q$ ($q \geq 4$). It is determined up to congruence by its vertex configuration (the cyclic sequence of polygon sizes around a vertex). In particular a convex uniform polyhedron with a $q$-gonal face, $q \notin \{3, 4, 5, 6, 8, 10\}$, is $P_q$ or $A_q$.*

**Fact 1.5** (4-D classification [8, 17])**.** *A convex uniform 4-polytope is one of: the Wythoffians of $A_4$, $B_4$, $D_4$, $F_4$, $H_4$; the snub 24-cell; the grand antiprism; a prism $Q \times \{\}$ over a convex uniform polyhedron $Q$; a duoprism $\{p\} \times \{q\}$. In particular a convex uniform 4-polytope with a $q$-gonal 2-face, $q \notin \{3, 4, 5, 6, 8, 10\}$, is a duoprism $\{q\} \times \{p\}$ (including $P_q \times \{\} = \{q\} \times \{4\}$) or the antiprism prism $A_q \times \{\}$.*

The vertex-transitivity in our definition matters: the pseudo-rhombicuboctahedron (elongated square gyrobicupola) has congruent vertex configurations but is not vertex-transitive, and is not uniform [13]. Fact 1.5 is itself computer-assisted. As a check of the method, the certified shared pipeline in dimension four recovers all 152 known objects at family cutoff 14, including the grand antiprism and snub 24-cell (Section 6.3). This bounded positive control is not a link of the 5-D proof or a proof of the cited all-parameter 4-D classification.

1.5. **Strategy.** Let $P$ be a convex uniform 5-polytope with unit edges and circumradius $R$. Section 2 proves the reduction: $P$ is inscribed in a sphere; around every 2-face $G$ the facets form a cycle, and $R$ is a root of an explicit *closure equation* of that cycle; at a vertex $v$ the neighbors of $v$ span the vertex figure VF, a 4-polytope inscribed in a 3-sphere whose facets are the vertex figures of the facets of $P$ at $v$; two distinct vertices are at least one edge apart (the separation lemma), which lets interval enclosures decide vertex identity when they are narrow enough (and report it as undecided otherwise); and the star of one vertex, if it forces the stars of its neighbors, forces $P$.

*Case A* (Section 4): all polygons have at most 16 sides. Then every facet belongs to a finite library of 181 congruence classes of uniform 4-polytopes. Stage 1 enumerates every cycle of library facets around a polygon and certifies all roots of its closure equation (14,965 cycles); radii that can occur together form 563 *radius classes* after a combinatorial filter. Stage 2 enumerates candidate signatures covering every actual vertex figure in each class (629 in total). Stage 3 covers every actual star by retained encodings, certifies the scalar-root assertion, and proves identification and forcing conditional on realization. For 628 encodings, any represented actual star is identified with that of a separately constructed known polytope $Q$ and forces $P \cong Q$. The one extra candidate is excluded from every convex uniform polytope by a development that gets stuck; no exact local realization of that candidate is asserted.

Here **181 counts four-dimensional facet types**, not five-dimensional polytopes. At the same polygon cutoff the known 5-D list has 890 names representing 628 congruence classes (Section 3); without the cutoff it includes infinite families. The facet library is checked by exact construction and incidence checks and by a separate rebuild in `checks/independent-s1-b2/` (Computation 3.5). Neither rebuild proves Möller's classification anew.

*Case B* (Section 5): some polygon $F$ has $n \geq 17$ sides. Around $F$ only three cycle types are possible, each forcing $R$ by a formula in $n$ and finitely many parameters. A certified computation (B2) shows that this $R$ is never a root of any Case-A cycle, so every polygon at a vertex lies in a facet containing a large polygon. A written structural argument, with two more certified computations ((P1) and (W)), then forces $P$ to be a product.

Why the cutoff is 16/17. Seventeen is a sufficient, not an optimal, threshold: it exceeds every exceptional polygon size in the lower-dimensional classifications and satisfies $2\sin(2\pi/17) < 1$ (Lemma 5.4). The B2, P1 and W certificates cover the entire remaining range. The small-face radius cover has $\max \Omega = 63.2195709614\ldots$; it includes closure roots that need not extend to any polytope, so this is not the largest radius of a known polytope with at most 16-sided faces. B2

checks $17 \le n \le 397$ and the corresponding auxiliary parameters, then uses $r_{398} > 63.344 > \max \Omega$ and monotonicity for the tails (Section 5.3).

The computations state the finite or interval claims they check and the written arguments needed to interpret them. Section 6.2 records shared dependencies; Section 7 states the formal scope.
What independence means here. Separate implementations share mathematical constructions, some input lists, model certificates and arithmetic libraries. The exact-exclusion checks also share the stage-1 construction code. P1's sampled tails are non-proof checks; its proof uses written angle, chord and monotonicity arguments. Section 6.2 specifies these dependencies.
Why this does not yet classify dimension six. In dimension six, the facets would be the five-dimensional families proved complete here, but completeness of that input list alone does not supply a finite search. The codimension-three faces are now polyhedra rather than polygons; their neighborhoods require new incidence and metric data, and the five-dimensional vertex figures make assembly and forcing larger. Most importantly, the large-polygon fan argument of Case B does not classify all unbounded families in dimension six. A new parameter reduction and certified bounds would be needed before a finite replay could prove completeness there.

## 2. The reduction

Throughout this section $P$ is a convex uniform $d$-polytope with unit edges, $d \ge 3$ (we use $d = 5$ from Section 2.4 on), with vertex set $V$.

### 2.1. **Inscribed sphere and face centers.**

**Lemma 2.1.** *Let $c$ be the centroid of $V$. Then all vertices lie on the sphere of some radius $R$ about $c$, and $c$ lies in the interior of $P$. For every face $\Phi$ of $P$ the orthogonal projection $o_\Phi$ of $c$ onto* aff $\Phi$ *is the circumcenter of $\Phi$, and lies in the relative interior of $\Phi$. If $\Phi$ has circumradius $R_\Phi$, then $|c - o_\Phi| = \sqrt{R^2 - R_\Phi^2}$.*

*Proof.* Every symmetry permutes $V$ and so fixes $c$; by vertex-transitivity all vertices are equidistant from $c$. The centroid of the vertices of a full-dimensional polytope is an interior point. The vertices of $\Phi$ lie on the intersection of the sphere with aff $\Phi$, a sphere in aff $\Phi$ about $o_\Phi$; they affinely span aff $\Phi$, so $o_\Phi$ is the unique circumcenter of $\Phi$. Since $\Phi$ is uniform, applying the first part to $\Phi$ shows that its circumcenter is its vertex centroid, which is relatively interior. The last formula is Pythagoras. □

### 2.2. **The separation lemma.**

**Lemma 2.2** (Separation lemma)**.** *Let $X$ be a convex polytope inscribed in a sphere, with at least three vertices, let $v$ be a vertex and $w$ a vertex nearest to $v$ among the other vertices. Then $[v, w]$ is an edge of $X$. Consequently, in a convex uniform polytope with unit edges any two distinct vertices are at distance at least* 1*.*

*Proof.* Let $B$ be the closed ball with diameter $[v, w]$, center $m$ and radius $\rho = |v - w|/2$. If a vertex $x \notin \{v, w\}$ lay in $B$, the angle $\angle vxw$ would be at least 90°, so $|v-w|^2 \ge |v-x|^2+|x-w|^2 > |v-x|^2$, contradicting the choice of $w$. Let $S$ be the circumsphere, with center $c$ and radius $R$, and consider the difference of powers $f(x) = (|x - m|^2 - \rho^2) - (|x - c|^2 - R^2)$, an affine function. It vanishes at $v$ and $w$, and at every other vertex $x$ we have $|x - c| = R$ and $|x - m| > \rho$, so $f(x) > 0$. In particular $f$ is not constant, so $H = \{f = 0\}$ is a hyperplane, $X \subseteq \{f \ge 0\}$, and $X \cap H$ is the face $\operatorname{conv}\{v, w\}$. So $[v, w]$ is an edge. If all edges have length 1, the nearest other vertex of any vertex is at distance 1, so all distances between distinct vertices are at least 1. □

**Corollary 2.3** (Decision rule)**.** *Let $x, y$ be two vertices of one convex uniform polytope with unit edges (possibly hypothetical), and let $[\ell, u]$ be a rigorous enclosure of $|x - y|^2$. Then:*

(a) *if $\ell \le 0$ and $u < 1$, then $x = y$;*

(b) *if* $0 < \ell$ *and* $u < 1$, *we have a contradiction (the configuration does not occur);*
(c) *if* $0 < \ell$ *(and* $u \geq 1$*), then* $x \neq y$*;*
(d) *if* $\ell \leq 0$ *and* $u \geq 1$, *nothing is concluded:* $x = y$ *and* $x \neq y$ *both remain possible.*

Case (d) is *undecided*. A search that meets it must either refine the enclosure, keep both possibilities, or stop and report the branch as open; treating the two points as distinct (or as equal) would not be conservative. Every search used by the proof handles (d) as undecided, and the full replays recorded below met it zero times (Sections 4.2, 4.5.2 and 4.5.3): `cert_vf.py` (`identify`) raises `Undecided`, counts it, and refuses to return a complete class; `check_u2` raises `PrecisionError` and exits nonzero; `checks/primary/check_u3` uses the stricter equality bound $u < 1/4$ and turns undecided cases into `OPEN`, never into a refutation.

### 2.3. **Vertex figures.**

**Lemma 2.4.** *Let* $P$ *be a convex uniform* $d$*-polytope with unit edges and circumradius* $R$*, centered at* $0$*, and* $v$ *a vertex. The neighbors of* $v$ *(the other ends of the edges at* $v$*) are exactly the vertices at distance* $1$ *from* $v$*. They lie on the hyperplane* $H = \{x : x \cdot v = R^2 - \frac{1}{2}\}$*, every other vertex except* $v$ *lies strictly on the side* $x \cdot v < R^2 - \frac{1}{2}$*, and* $\mathrm{VF} := H \cap P = \mathrm{conv}(\text{neighbors of } v)$ *is a* $(d-1)$*-polytope inscribed in a sphere of radius* $r = \sqrt{1 - 1/(4R^2)}$ *about* $o_{\mathrm{VF}} = (1 - \frac{1}{2R^2})v$*. The faces of* VF *correspond to the faces of* $P$ *containing* $v$ *(a* $j$*-face* $\Phi \ni v$ *gives the* $(j-1)$*-face* $H \cap \Phi$*). For a facet* $K \ni v$ *of circumradius* $R_K$*,* $\mathrm{VF}(K) := H \cap K$ *is inscribed in a sphere of radius* $r_K = \sqrt{1 - 1/(4R_K^2)}$ *in a hyperplane of* $H$ *at distance* $\sqrt{r^2 - r_K^2}$ *from* $o_{\mathrm{VF}}$*. An edge of* VF *coming from a* $p$*-gon has length* $\mathrm{ch}_p = 2\cos(\pi/p)$*. The point* $o_{\mathrm{VF}}$ *lies in the relative interior of* VF*.*

*Proof.* By Lemma 2.2 the vertices at distance 1 from $v$ are neighbors, and conversely neighbors are at distance 1. For a vertex $x$, $|x - v|^2 = 2R^2 - 2x \cdot v$, so $|x - v| = 1$ iff $x \cdot v = R^2 - \frac{1}{2}$, and $|x - v| > 1$ (every other vertex $\neq v$) iff $x \cdot v < R^2 - \frac{1}{2}$. Hence $H$ separates $v$ from all non-neighbors, and $H \cap P$ is the convex hull of the neighbors; this is the standard vertex figure, whose face lattice is that of the faces through $v$. A neighbor $x$ makes an angle $\theta$ with $v$ at the center with $2R\sin(\theta/2) = 1$, so its distance from the axis $\mathbb{R}v$ is $R\sin\theta = \cos(\theta/2) = \sqrt{1 - 1/(4R^2)}$. The same computation inside $K$ gives $r_K$; the circumcenter of $\mathrm{VF}(K)$ is the projection of $o_{\mathrm{VF}}$, which gives the distance. In a regular $p$-gon the two neighbors of a vertex are $\mathrm{ch}_p$ apart. Finally, write $o_{\mathrm{VF}} = t_0 v$ with $t_0 = 1 - 1/(2R^2)$. Since 0 is the centroid of the $N$ vertices, $q := -v/(N-1)$ is the centroid of the vertices other than $v$, so $q \in P$, and $q \cdot v \leq R^2 - \frac{1}{2}$, since every other vertex $x$ has $x \cdot v \leq R^2 - \frac{1}{2}$, with equality exactly for the neighbors. If some vertex other than $v$ is not a neighbor, the inequality is strict, i.e. $-1/(N-1) < t_0$; also $t_0 < 1$. Then $o_{\mathrm{VF}}$ lies on the open segment from $q$ to $v$, which passes through the interior point 0; every point of this open segment is interior to $P$ (it lies strictly between the interior point 0 and a point of $P$, or is 0). Hence $o_{\mathrm{VF}}$ is an interior point of $P$ lying in $H$, i.e. a relative interior point of $\mathrm{VF} = H \cap P$. If every vertex other than $v$ is a neighbor (as in the 5-simplex), then VF is the convex hull of all other vertices, $t_0 = -1/(N-1)$ and $o_{\mathrm{VF}} = q$ is the centroid of the vertices of VF, which is a relative interior point of VF. □

*Remark* 2.5. The center of the vertex figure need not lie on the segment from $v$ to the center of $P$: when $R < 1/\sqrt{2}$, $t_0 < 0$; for the 5-simplex, $R^2 = 5/12$ and $t_0 = -1/5$. The centroid argument above handles both cases, including 4-simplex facets when applied in Lemma 4.10.

### 2.4. **Cycles around a polygon and the closure equation.**

From now on $d = 5$. Let $G$ be a 2-face of $P$, a $p$-gon with circumcenter $o_G$ and circumradius $\rho = r_p$, and let $N$ be the 3-dimensional orthogonal complement of the direction space of $\mathrm{aff}\, G$. The faces of $P$ containing $G$ are cells (3-faces) and facets; modulo $\mathrm{aff}\, G$ the tangent cone of $P$ at $G$ is a pointed polyhedral 3-cone in $N$, whose extreme rays correspond to the cells $C \supset G$ and whose 2-faces correspond to the facets

$K \supset G$. Going around the cone gives a cyclic sequence $(K_1, C_1, K_2, C_2, \dots, K_k, C_k)$, $k \geq 3$, where facet $K_i$ contains the cells $C_{i-1}$ and $C_i$ (indices mod $k$). Let $u_C \in N$ be the unit vector of the ray of $C$ and $\alpha_i$ the dihedral angle of the 4-polytope $K_i$ at its 2-face $G$, between the cells $C_{i-1}$ and $C_i$; this is the angle between $u_{C_{i-1}}$ and $u_{C_i}$. We call $(p; K_1, C_1, \dots, K_k, C_k)$ the *cycle* of $G$; cycles are taken up to rotation and reversal.

**Lemma 2.6** (Closure equation)**.** *With this notation, let $w = c - o_G \in N$, $h = |w| = \sqrt{R^2 - \rho^2}$, and for each cell $C \supset G$ of circumradius $R_C$ let $d_C = \sqrt{R_C^2 - \rho^2} > 0$. Then:*

(a) $\sum_i \alpha_i < 2\pi$;
(b) $w \cdot u_C = d_C$, *so the angle $\delta_C$ between $w$ and $u_C$ satisfies $\cos\delta_C = d_C/h$;*
(c) *$w$ lies strictly inside the cone spanned by the $u_C$, and with $\theta_i \in (0,\pi)$ the angle at $\hat{w} = w/h$ of the spherical triangle $(\hat{w}, u_{C_{i-1}}, u_{C_i})$,*

$$\cos\theta_i = \frac{\cos\alpha_i - \cos\delta_{C_{i-1}}\cos\delta_{C_i}}{\sin\delta_{C_{i-1}}\sin\delta_{C_i}}, \qquad \sum_{i=1}^{k}\theta_i = 2\pi;$$

(d) *$R > R_K$ for every facet $K \supset G$.*

*Proof.* (a) The intersection of a pointed polyhedral 3-cone with the unit sphere is a convex spherical polygon in an open hemisphere, whose side lengths are the face angles $\alpha_i$; its perimeter is less than $2\pi$. (b) By Lemma 2.1 applied to $C$, $o_C$ is the projection of $c$ on $\operatorname{aff} C$ and $o_G$ the projection of $o_C$ on $\operatorname{aff} G$; so $o_C - o_G \in N$ points along the ray of $C$ (as $o_C \in \operatorname{relint} C$), has length $d_C > 0$, and $w - (o_C - o_G)$ is orthogonal to $\operatorname{aff} C$, hence to $u_C$. (c) The tangent cone of $P$ at $G$ is the sum of the direction space of $\operatorname{aff} G$ and a pointed cone $T_N \subset N$, and the interior of $P$ lies in $o_G$ plus its interior; since $c$ is interior and $w = c - o_G \in N$, $w$ lies in the interior of $T_N$, the cone spanned by the $u_C$. The triangles $(\hat{w}, u_{C_{i-1}}, u_{C_i})$ are then non-degenerate, their angles at $\hat{w}$ fill the full angle around $\hat{w}$, and the formula is the spherical law of cosines for the side $\alpha_i$ opposite $\hat{w}$. (d) $|c - o_K| = \sqrt{R^2 - R_K^2}$ is the distance from the interior point $c$ to the hyperplane of $K$, which is positive. □

In the variable $s = 1/R$ the cosines become $\cos\delta_C(s) = d_C\, s/\sqrt{1-\rho^2 s^2}$, increasing in $s$, and the *closure function* of the cycle is

$$g(s) = \sum_{i=1}^{k}\theta_i(s) - 2\pi, \qquad 0 \leq s \leq 1/R_{\max},$$

where $R_{\max} = \max_i R_{K_i}$ and $s = 0$ stands for $R = \infty$.

*Remark* 2.7. On this domain $\delta_{C_{i-1}} + \delta_{C_i} \geq \alpha_i$, so the argument of the arccosine is at least $-1$: at $s = 1/R_{K_i}$ the point $\hat{w}$ is the direction of $o_{K_i} - o_G$, which lies on the arc from $u_{C_{i-1}}$ to $u_{C_i}$ (equality), and the $\delta$'s decrease in $s$. If instead $|\delta_{C_{i-1}} - \delta_{C_i}| > \alpha_i$ the triangle does not exist; on such $s$ the configuration is impossible. The implementations clip the lower end at $-1$ (sound by this remark) and discard boxes on which a triangle is certified impossible.

Why the closure equation is not an identity. For an admissible cycle, $\theta_i(0) = \alpha_i$ and hence $g(0) = \sum_i \alpha_i - 2\pi < 0$ by Lemma 2.6(a). Thus the closure equation cannot hold identically on its radius domain. For the finite library we have the stronger, checked monotonicity that also rules out an accumulation of roots at an endpoint. Put $a = d_A^2$, $b = d_B^2$, $D = R_K^2 - \rho^2$ and $c = \cos\alpha$ for a word, and let $u = 1/(R^2 - \rho^2)$. Its closure-angle cosine is

$$f(u) = \frac{c - \sqrt{ab}\,u}{\sqrt{(1-au)(1-bu)}}.$$

Since $D > a, b > 0$, the sign of $f'(u)$ on $[0, 1/D]$ is the sign of

$$-\sqrt{ab} + \tfrac{1}{2}c(a+b) + u\big(\tfrac{1}{2}\sqrt{ab}(a+b) - cab\big).$$

This is affine in $u$. The independent stage-1 certificate checks strict negativity at both endpoints for all 558 library words; hence every closure angle, and their sum, increases strictly in $u$ on the nondegenerate domain. Each Case-A cycle has at most one admissible finite root (Computation 4.1). A flat cycle has $g(0) = 0$ but its closure function is still not identically zero: the same strict monotonicity makes the closure sum exceed $2\pi$ for $u > 0$. Flat cycles are excluded in any event by convexity. For Case B, Section 5.3 gives the explicit radius formulas and finite root equations. The finite-radius conclusion therefore uses more than an appeal to analyticity alone.

Two kinds of cycles satisfy the closure equation only spuriously and are excluded exactly:

**Lemma 2.8** (Exact exclusions)**.** (Flat.) *A cycle with $\sum \alpha_i = 2\pi$ exactly is not the cycle of a 2-face of a convex 5-polytope.* (Degenerate.) *A root at $R = R_{\max}$ exactly does not give a convex uniform polytope.*

*Proof.* Lemma 2.6(a) and (d). □

For a flat cycle, $g(0) = 0$, the formal infinite-radius limit. Degenerate roots put the center on a facet hyperplane. Interval arithmetic cannot certify these equalities; they are proved exactly (Computations 4.1 and 3.6). A local angle-sum identity alone does not give a global honeycomb.

2.5. **Faces, simple vertices.**

**Lemma 2.9** (Face lemma)**.** *Let $X$ be a polytope, $\Phi$ a face of $X$ and $p, q$ vertices of $\Phi$. If $[p, q]$ is an edge of $X$, it is an edge of $\Phi$. Conversely every edge of $\Phi$ is an edge of $X$.*

*Proof.* $[p, q] \subseteq \Phi$ by convexity, and $[p, q] = [p, q] \cap \Phi$ is an intersection of two faces of $X$, hence a face of $X$ contained in $\Phi$, hence a face of $\Phi$. The converse holds because a face of a face is a face. □

**Lemma 2.10** (Simple vertices)**.** *A vertex $x$ of degree 4 in a 4-polytope is simple: each facet containing $x$ contains exactly three of the four edges at $x$, and each 3-subset of the edges at $x$ is the set of edges at $x$ of exactly one facet.*

*Proof.* The vertex figure of $x$ is a 3-polytope with four vertices, a tetrahedron; its facets (all four triangles) correspond to the facets through $x$. □

2.6. **Stars and local uniqueness.** The *star* of a vertex $v$ is the set of facets of $P$ containing $v$, each with its full vertex set.

**Lemma 2.11** (Local uniqueness gives global uniqueness)**.** *Let $S$ be the star of $v_0$ in some convex uniform 5-polytope, and suppose: for every convex uniform 5-polytope $P$ with star $S$ at $v_0$ and every neighbor $w$ of $v_0$, the star of $P$ at $w$ is one and the same set $\Sigma_w$, independent of $P$. Then any two convex uniform 5-polytopes $P_1$, $P_2$ with star $S$ at $v_0$ are equal.*

*Proof.* We show by induction on the edge distance from $v_0$ that $P_1$ and $P_2$ have the same vertices and the same stars there. Suppose $P_1$ and $P_2$ have the same star $\Sigma$ at a vertex $w$. By vertex-transitivity of $P_1$ there is an isometry $g_1$ with $g_1(v_0) = w$ and $g_1(S) = \Sigma$. Then $g_1^{-1}(P_1)$ and $g_1^{-1}(P_2)$ are convex uniform 5-polytopes with star $S$ at $v_0$; by hypothesis they have the same star at every neighbor $u'$ of $v_0$. Applying $g_1$, $P_1$ and $P_2$ have the same star at every neighbor $u = g_1(u')$ of $w$ (the neighbors of $w$ are read off the common star $\Sigma$). The edge graph of a polytope is connected (Balinski [5]), and a polytope is the convex hull of its vertices, so $P_1 = P_2$. □

The induction is written out here because a symmetry of $P_1$ mapping $v_0$ to $w$ need not be a symmetry of $P_2$; the hypothesis is applied to $g_1^{-1}(P_i)$, not to $P_i$.

2.7. **The case split.** A 2-face is *large* if it has at least 17 sides. *Case A*: $P$ has no large 2-face. *Case B*: $P$ has a large 2-face. The cases are disjoint and exhaustive. Polygon sizes $7, 9, 11, \ldots, 16$ (which occur only in prisms, antiprisms and duoprisms) belong to Case A, together with the sizes $3, 4, 5, 6, 8, 10$ of the non-prismatic polyhedra and 4-polytopes.

## 3. The finite library and the known list

3.1. **The facet library.** For $3 \le P \le 16$ let $\mathcal{L}_P$ be the following list of names of convex uniform 4-polytopes all of whose 2-faces have at most $P$ sides. For $P = 16$ it has 214 names:

- the Wythoffians `A4`*abcd*, `B4`*abcd*, `D4`*abcd*, `F4`*abcd*, `H4`*abcd* (*abcd* $\in \{0,1\}^4 \setminus \{0000\}$; 75 names; their 2-faces have at most 10 sides);
- `sadi` (snub 24-cell) and `gap` (grand antiprism);
- `pr-`$X$, the prism over $X$, for the 18 Platonic and Archimedean solids $X$;
- `antiprism-prism-`$n = A_n \times \{\}$, $3 \le n \le 16$ (14 names);
- `duoprism-`$p$`-`$q = \{p\} \times \{q\}$, $3 \le p \le q \le 16$ (105 names; the prisms $P_q \times \{\}$ are `duoprism-4-`$q$).

Names, diagrams and cell labels are listed in Appendix B. For smaller $P$, the definition of $\mathcal{L}_P$ retains exactly the names in this list whose every 2-face has at most $P$ sides.
The small control. The $P = 6$ quick control $\mathcal{H}_6$ retains fixed 8- and 10-gon facets and has 136 classes rather than the literal cutoff's 129, whereas $\mathcal{H}_{10} = \mathcal{L}_{10}$ and $\mathcal{H}_{16} = \mathcal{L}_{16}$; we write $P_{\mathrm{ctl}}$ for its family-parameter bound.

**Lemma 3.1** (Library completeness)**.** *Every convex uniform 4-polytope all of whose 2-faces have at most 16 sides is congruent to the polytope of some name in* $\mathcal{L}_{16}$.

*Proof.* Fact 1.5: the Wythoffians and the two snub-type polytopes have only 3-, 4-, 5-, 6-, 8- and 10-gons; the prisms over polyhedra $X$ with at most 16-gonal faces are the `pr-`$X$, the duoprisms `duoprism-4-`$q$ and the antiprism prisms (Fact 1.4); and the duoprisms are listed for $p, q \le 16$. □

Several names denote congruent polytopes (e.g. `A40001` and `A41000` are both the 5-cell, and `B40001`, `pr-cube` and `duoprism-4-4` are the tesseract). The enumeration keeps one *canonical* name per congruence class: 181 classes, with 33 aliases (Table 4). The canonical types are 45 Wythoffians, the two snub-type polytopes, 17 polyhedral prisms, 13 antiprism prisms and 104 duoprisms. A wrong merge would silently drop a facet type, so the merges are proved exactly.

**Certified computation 3.2** (MERGE: exact congruence of the 33 aliases)**.** Each of the 33 aliases is congruent to its canonical name.

*What is checked.* `merge.py` builds both vertex sets exactly in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ (elements stored as four rationals in the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$): Wythoffians from explicit simple roots, first checked exactly against the Coxeter diagram; the point $x_J$ by an exact rational Gram solve, rechecked; the orbit by exact breadth-first search; the products in closed form. A floating-point search proposes a bijection; it is accepted only if the vertex counts agree and every pairwise squared distance is preserved exactly. By polarization the image vectors then have the same Gram matrix, so the bijection extends to an affine isometry of the affine hulls. The program also checks that its alias list equals the alias list of the shared facet file `checks/check_u1/inputs/u1_facets_p16.json` (option `--facets`). Result: 33/33 congruences proved; four corruptions rejected (two false pairs, one vertex moved by $10^{-12}$, two images swapped). *Second implementation.* `checks/independent-merge-filter/` independently derives the Wythoff and product coordinates in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ and certifies all 33 aliases by vertex bijections preserving 465,206 unordered squared distances. A separate affine checker agrees; both reject seven semantic corruptions. The alias list is a shared input; the later library reconstructions still take it as input.

3.2. **Facet data.** For every library type $K$ the computations need: its circumradius $R_K$; for each orbit of 2-faces under $\mathrm{Sym}(K)$, the polygon size $p$, the two cells at the polygon and the dihedral angle $\alpha$ between them; and the *model* of $K$: a base vertex $K[0]$, its neighbors $K[1],\dots$ (the vertex figure $X_K$), its full vertex set, and the list of 2-faces of $X_K$ (the cells at $K[0]$) with their cyclic orders and cell labels. Every polygon orbit has a member through $K[0]$ by vertex-transitivity, so all of this is local data at one vertex.

**Certified computation 3.3** (LIB: facet data)**.** For each of the 181 canonical types, the circumradius and the dihedral angles of all polygon orbits are enclosed in balls, and the model is constructed with all coordinates enclosed in balls.

*What is computed.* `cert_facets4.py`, `cert_star.py`, `cert_full.py` (python-flint/Arb, 200 bits): Wythoffians from rigorous Wythoff coordinates (`cert_wythoff.py`); prisms and antiprism prisms from the vertex-configuration data of the polyhedra (`cert_poly3.py`, below); duoprisms in closed form; the snub 24-cell and the grand antiprism from exact $\mathbb{Q}(\sqrt{5})$ coordinates of the 600-cell; full vertex sets as orbits under exact reflections or (for the snub cube and snub dodecahedron) under their rotation groups, moved into the model frame by isometries verified on balls. `cert_poly3.py` computes a uniform polyhedron's circumradius and edge angles from its vertex configuration alone: the neighbors of a vertex lie on a circle, the central angles of the chords $\mathrm{ch}_{q_i}$ sum to $2\pi$, and the unique solution is enclosed. Exact values: `exact_angles.py` computes $\cos\alpha$ exactly for every non-snub facet angle ($\cos^2\alpha$ lies in $\mathbb{Q}(\sqrt{d})$, $d\in\{1,2,5\}$, computed in the root basis; exact $\mathbb{Q}(\sqrt{5})$ coordinates for `sadi` and `gap`; closed forms for products), and all agree with the balls.

The following certificate supplies a certified correspondence from cells and labels to every polygon-orbit and angle entry used by stage 1, including the word-angle property.

**Lemma 3.4** (A word determines its angle)**.** *Let $G$ be a $p$-gon of a uniform 4-polytope $K$ (circumradius $R_K$), between cells $A$, $B$ of circumradii $R_A$, $R_B$, and let $\rho = r_p$. Then the dihedral angle of $K$ at $G$ is*

$$\alpha = \arccos\sqrt{\frac{R_A^2-\rho^2}{R_K^2-\rho^2}} + \arccos\sqrt{\frac{R_B^2-\rho^2}{R_K^2-\rho^2}}.$$

*In particular the word $(K,p,A,B)$ determines $\alpha$, even when its polygon occurrences lie in different orbits of* $\mathrm{Sym}(K)$.

*Proof.* In the plane through the center $f$ of $G$ perpendicular to $G$, the two cells give rays $u_A, u_B$ bounding a convex wedge of angle $\alpha$, and the center $O$ of $K$ projects strictly inside it (Lemma 2.1 applied to $K$, and Lemma 2.6(b),(c) one dimension down). With $D = |O-f|^2 = R_K^2-\rho^2$ we have $(O-f)\cdot u_A = \sqrt{R_A^2-\rho^2} > 0$ and likewise for $B$, since the projection of $O$ on each cell is its relatively interior circumcenter. The direction of $O-f$ lies between the rays, so the two angles to it add to $\alpha$. □

**Certified computation 3.5** (LINK: certified stage-1 input)**.** For each of the 181 canonical types (assuming the construction properties of Section 3.3), the list of polygon occurrences at the base vertex, with their two cells and labels, is complete and correct (1,277 occurrences, 558 distinct words $(K,p,A,B)$); the exported circumradius encloses $R_K$ and each exported angle encloses the angle of Lemma 3.4. Every radius and angle entry of the stage-1 input tables at $P_{\mathrm{ctl}} = 6$, 10 and 16 contains these rigorous enclosures, and each table lists exactly the words of its corresponding control library $\mathcal{H}_P$.

*What is checked.* Wythoff neighbors are generated in exact $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{5})$ coordinates; `sadi` and `gap` from all 120 vertices of the 600-cell in exact $\mathbb{Q}(\sqrt{5})$ arithmetic; prisms, antiprism prisms and duoprisms from their exact product identities (which replace the small-value zero tests that `checks/check_u1` uses for these types). Floating-point hulls only propose faces; every face then

passes the ridge certificate (Computation 4.8: exact coplanarity or the product identity, strict supporting signs, strict cyclic order, chord labels, two faces per edge), and closure of the face graph proves the list complete. Two separately written link checkers (one reading adjacency from the sides of the certified faces, one from pairwise face intersections and an independent dihedral computation) accept every occurrence and each rejects 15 semantic corruptions. The input-table containment check (`compare_legacy.py`, both checkers) passes at $P_{\mathrm{ctl}} = 6$ (76 radii, 243 words, 627 occurrences), 10 (106/333/815) and 16 (181/558/1,277), each rejecting six further corruptions; the 558 words are also exactly those used by the independent stage-1 checker's cycle file. *Second implementation.* The independent library of `checks/independent-s1-b2/` (Section 4.5.4), written from this paper's statements, reconstructs the 181 canonical types by exact parabolic Coxeter combinatorics, exact supporting planes for `sadi` and `gap` and product identities, and obtains the same 558 words and 1,277 occurrences; all 181 radius and 558 angle enclosures lie inside the recorded input enclosures. The two link checkers above share the ridge certificate and the constructions; the independent library shares the alias table, the construction theorem of Section 3.3 and the written exceptional-model proofs, as well as the Arb and SymPy arithmetic libraries. In particular, `checks/independent-s1-b2/library.py`, `checks/independent-s1-b2/parabolic.py` and `checks/independent-s1-b2/exceptional.py` rebuild the geometry; `checks/independent-s1-b2/library_verify.py` checks its local incidence complex and the comparison checks containment in the consumer intervals. The construction proofs and the external 4-D classification remain common mathematical premises. The alias list is a common inventory input, though all 33 merges have separate exact congruence checks. Arithmetic-library errors and a common misinterpretation of the recursive-uniform definition are also not eliminated by matching counts. The independent rebuild reduces dependence on the original construction program; it does not make every upstream premise independent.

With Computation 3.5, the stage-1 proof uses certified input tables and the two implementations described in Section 4.1.

Exact inputs of the exclusions. LINK certifies interval enclosures. The exact exclusions of stage 1 (Lemma 2.8) need more: the algebraic cosine and radius used in each exclusion must be *exactly* the geometric quantity of the named incident word. A floating-point reconstruction or numerical recognition of a cosine does not prove this exact correspondence. By Lemma 3.4, with $\rho = r_p$, $D = R_K^2 - \rho^2$, $a = R_A^2 - \rho^2$ and $b = R_B^2 - \rho^2$, the cosine of the word's angle is

$$\cos\alpha = \frac{\sqrt{ab} - \sqrt{(D-a)(D-b)}}{D},$$

with the positive square roots fixed by the geometry ($0 < a, b < D$). So exact radii and exact cell labels determine every exact cosine.

**Certified computation 3.6** (EXACT: certified exact exclusion inputs)**.** The squared circumradii of the 181 canonical types and of the 44 named cell types and, for each of the 558 incident words, the algebraic cosine used by the exclusions, are exactly the values of the constructions of Section 3.3; each cosine equals the displayed expression with positive square roots.

*What is checked.* The exact proposals are rebuilt by both checkers from exact constructions (the exact parabolic, quadratic-field, 600-cell and product routines of `checks/independent-s1-b2/`); neither accepts the file by digest or by numerical agreement. Checker A requires the exact residual $Dc^2 - 2c\sqrt{ab} + a + b - D = 0$, decided by an exact minimal-polynomial test, together with $\sqrt{ab} - Dc > 0$, strict positivity of $a$, $b$, $D - a$, $D - b$ by outward arithmetic, and an exact residual for every radius against its construction. Checker B squares once more, verifies $4c^2ab = (Dc^2 + a + b - D)^2$, restores the sign, authenticates the radii by its own exact Gram solves and product identities (for the snub factors, a Sturm count isolating the unique admissible root of the eliminated polynomial), projects the incident word labels from cell intersections, and decides zero by exact number-field embedding. Neither accepts an interval containing zero as an

equality; `nsimplify` and rational-angle guesses are only proposals and need an exact zero residual. The live stage-1 facet table is then authenticated against these values: complete names, aliases and words, containment of every exact radius and angle after the stage-1 parser, and the depth bound $9\alpha > 2\pi$. `exact_angles.py` takes its operands only from this certified source (`exact_inputs.py`). The flat identities are proved using positive sines, the exact product $\prod(c_j + i\,s_j) = 1$, and a certified sum in $(\pi, 3\pi)$; the degenerate exclusions use only that the three-ray Gram matrix is nonsingular and the opposite Cramer numerator is exactly zero, which already rules out a strictly interior center. The consumer entry point `authenticate_exclusions` rechecks every requested deletion on the actual cycle's words and exact operands and rejects duplicate, overlapping, unproved or out-of-range indices. *Result.* Both checkers accept 181 facet radii, 44 cell radii and 558 cosines; a separate direct sweep of the normal-cosine residual passes all 558 words. The exclusions at the three bounds are 790/1,703/7,080 flat cycles and 23/39/57 degenerate roots at $P_{\mathrm{ctl}} = 6/10/16$. *Corruptions.* Twelve test methods pass, rejecting in both checkers perturbations of a facet radius, a cell radius or a cosine by $10^{-40}$, the wrong square-root branch, a lost sign, wrong labels, polygons and multiplicities, missing and duplicate words and facets, floating operands, false flat and degenerate claims (four near-right angles, a $4\pi$ winding, a singular Gram matrix), malformed exclusion lists and root records, and $\sqrt{2}+10^{-40}$ replaced by $\sqrt{2}$. *Shared input and second implementation.* Both `check_inputs.py` and `check_inputs_b.py` import `exact_source.py`. That bridge calls `checks/independent-s1-b2/library.py` and `checks/independent-s1-b2/library_verify.py`, with the parabolic, exceptional and product constructions behind them. This is one shared construction implementation, not two independent library reconstructions. Checker B supplies its own radius solve in `radii_b.py` and incidence projection; it does not thereby make the common constructions independent. A common error in that construction layer could survive both input checks; their agreement alone does not rule it out. The exclusions themselves are proved twice: by `exact_angles.py` with the independent certificate check `check_exclusions_b.py`, and by the separately written stage 1 of `checks/independent-s1-b2/`, which finds the same 7,080 flat cycles and 57 degenerate roots with its own exact Gram matrices and exact phase products (it uses the same construction code).

3.3. **Model constructions.** The computations use explicit constructions ("models") of the library facets and of the known 5-polytopes. It is not enough that these are *named* correctly: the proof uses that each constructed point set is the vertex set of a uniform polytope with unit edges, with the stated facet types and complete vertex orbits, since the separation lemma is applied to model vertices. These properties are supplied as follows (`checks/model_audit/`).

- *Wythoff constructions, antiprisms and products* (`checks/model_audit/wythoff_proof.md`): a written proof, from standard facts on finite Coxeter groups, of the supporting-face theorem, unit edges and recursive uniformity of $W_J$ for every binary ring pattern of the groups used, of the product and antiprism constructions, of the facet rule and of the named aliases; and an explicit contract under which an indexed interval orbit enumeration returns exactly the orbit, index by index.
- *Snub 24-cell and grand antiprism*: full convex-hull uniformity of the 96- and 100-vertex subsets of the 600-cell and containment of the actual indexed coordinates, checked by two independent implementations (rational-pair and number-field), each rejecting eight corruptions.
- *Snub cube and snub dodecahedron*: complete algebraic constructions with exact generator and index identities and coordinate containment, checked independently with interval/Krawczyk and rational-dyadic arithmetic, each rejecting nine corruptions.
- *Runtime guard*: `checks/primary/model_audit/model_guard.py` enforces in every run the midpoint, width and grid conditions of that contract for every point-set and lookup query, using exact rational endpoints. It is installed before any model is built in the independent stage-2 and stage-3 drivers, the equivalence checkers and the template auditor, and kept through their searches; a guard failure is reported as `OPEN`, never as an exclusion. All 181 facet models (79,445

vertices; 273,802 point-set and 2,066 lookup queries) pass, and every final stage-2 and stage-3 run below records the guard active with zero failures.

Conventional names matter only for reporting. The two independent packages of Section 4.5.4 build their own models in exact number fields (Coxeter orbits by exact reflections without rounding, products and antiprisms from exact recurrences, the snub factors from isolated algebraic roots, the two 600-cell deletions from the 120 exact points, recovered from certificate data and checked against them in `checks/independent-s23/`). Both rely on the same written construction theorem and exceptional-model proofs `checks/model_audit/snub_PROOF.md` and `checks/model_audit/q5_PROOF.md`; `checks/independent-s23/` also uses the two exceptional model certificates.

3.4. **The known list.** For $P \le 16$, $\mathcal{K}_P$ is the list of names

- `A5`*abcde*, `B5`*abcde*, `D5`*abcde*, all 31 ring patterns each (93 names);
- `prism-`$K$ for the canonical types $K$ of $\mathcal{L}_P$;
- $\{p\} \times X$ for $3 \le p \le P$ and $X$ a Platonic or Archimedean solid, $P_q$ or $A_q$, all of whose faces have at most $P$ sides.

For $P = 16$ this gives $93 + 181 + 14 \cdot 44 = 890$ names. They denote 628 pairwise non-congruent polytopes: the stage-3 identification (Section 4.5.3) assigns every one of the 890 names to one of 628 vertex-figure signatures with verdict `KNOWN`. Names with the same signature are congruent, because actual stars with that signature have the same rooted encoding and radius, and their star forces the polytope (Proposition 4.14); existence comes from the named constructions, not from retaining a search row; names in different radius classes cannot be congruent, because the closed consumer intervals are pairwise disjoint (checked with exact rational endpoints by `checks/check_filter.py`); within one class, different full cycle-word signatures cannot be congruent, because the signature of a vertex figure is a congruence invariant. This count is not taken from the literature. The audited name classes also give the exact decomposition $628 = 104 + 420 + 105 - 1$: 420 products from 14 polygon parameters and 30 non-prismatic polyhedral bases, 105 unordered polygon-pair products, and 104 fixed-list classes. The first set is disjoint from the others; the latter two meet only in the 5-cube. The 30 bases are four non-cubic Platonic solids, 13 Archimedean solids and 13 antiprisms of parameters 4 through 16. The 47 fixed-list prisms include the grand-antiprism prism. These are checked class-set bijections, not merely arithmetic subtotals. By construction $\mathcal{K}_{16}$ contains every polytope of (K1)–(K3) whose 2-faces have at most 16 sides: the Wythoffians have only 3-, 4-, 5-, 6-, 8- and 10-gons, a prism over a prismatic 4-polytope is of the form (K3), and every (K2) prism over a non-prismatic 4-polytope is a `prism-`$K$.

## 4. Case A: every 2-face has at most 16 sides

In this section $P$ is a convex uniform 5-polytope with unit edges, circumradius $R$, center $c$, all of whose 2-faces have at most 16 sides. By Lemma 3.1 every facet of $P$ is congruent to a canonical type of $\mathcal{L}_{16}$, and every cell is a uniform polyhedron with at most 16-gonal faces (Fact 1.4). We write $P_{\max} = 16$; the same pipeline was also run with $P_{\mathrm{ctl}} = 6$ and 10 as controls, with the counts of Table 1.

4.1. **Stage 1: cycles and their radii.** A *library cycle* is a cycle $(p; K_1, C_1, \dots, K_k, C_k)$ with $p \le 16$, $k \ge 3$, each $K_i$ a canonical type having a $p$-gon orbit between cells of types $C_{i-1}$, $C_i$ (dihedral angle $\alpha_i$), and $\sum \alpha_i$ not certified to exceed $2\pi$. By Lemma 2.6, the cycle of every 2-face of $P$ is a library cycle. By Lemma 3.4 the word $(K, p, A, B)$ of a polygon orbit determines its dihedral angle, so the word $(p; K_1, C_1, \dots, K_k, C_k)$ determines the angles and the closure function; stage 2 and stage 3 treat cycles as such words.

**Certified computation 4.1** (S1: stage 1)**.** (i) The list $\mathcal{C}_{16}$ of library cycles, up to rotation and reversal, has 14,965 elements, and it is complete.

(ii) For each cycle the zeros of its closure function $g$ on $[0, 1/R_{\max}]$ (with the domain extended to the upper end of the ball enclosing $1/R_{\max}$) lie in a listed finite set of *root intervals*, formed by merging retained leaves of width below $10^{-20}$ in $s$; this threshold does not bound the width of a merged interval. Of the cycles, 14,428 have at least one root interval, 537 have none, and none is undecided.

(iii) Exactly 7,080 cycles have $\sum \alpha_i = 2\pi$ exactly (flat), and exactly 57 root intervals are the root $R = R_{\max}$ exactly (degenerate); each is proved by exact algebra.

*What is computed.* `cert_stage1_5d.py` (Arb, 160–200 bits) enumerates cycles as closed walks in the multigraph whose vertices are cell types and whose edges are the $p$-gon orbits of the facet types, pruning when the certified angle sum exceeds $2\pi$. The smallest facet angle in the library is $\pi/3$ (in $\{3\} \times \{q\}$ at its $q$-gons, the interior angle of the triangle), so a non-flat cycle has at most 5 facets and a library cycle at most 6 (six angles $\pi/3$ are flat); the search goes to 8. On $[0, 1/R_{\max}]$ the closure function is evaluated on boxes by ball arithmetic, using the monotonicity of $\cos \delta_C(s)$ for the range of each $\delta_C$ and the clipping of Remark 2.7; a box is discarded when the enclosure of $g$ excludes 0 or a triangle is certified impossible, and the remaining boxes are bisected below width $10^{-20}$ and merged into root intervals. Adjacent retained leaves can produce a merged interval wider than $10^{-20}$; no upper bound of that size is needed for coverage or the later whole-class root check. Root intervals are written as outward-rounded 45-digit decimal bounds with a margin of $10^{-32}$; the merge of adjacent boxes is exact. `exact_angles.py` proves flatness, on the exact cosines of Computation 3.6 only: angles that are rational multiples of $\pi$ are verified as such exactly (a rational-angle guess is accepted only after an exact cosine identity), the others are combined into the claim $\exp(i \sum \alpha_j) = 1$, which is verified by an exact minimal-polynomial zero test (sympy), and the ball of $\sum \alpha_j$ lies in $(\pi, 3\pi)$; so $\sum \alpha_j = 2\pi$. The output records 30/31/32 proved cosine tuples at $P_{\text{ctl}} = 6/10/16$, including $2 \arccos(-\sqrt{6}/4) + \arccos(-1/4) = 2\pi$, and $\arccos(-\sqrt{1/3 - 2\sqrt{5}/15}) + \arccos(-\sqrt{1/3 + 2\sqrt{5}/15}) + \arccos(-1/\sqrt{5}) = 2\pi$ (pentagonal antiprism, icosidodecahedron and dodecahedron angles). All degenerate roots occur in 3-cycles. For a 3-cycle the link is a rigid spherical triangle and the closure has one explicit solution: with $\Gamma$ the Gram matrix of the three cell rays ($\Gamma_{ij} = \cos \alpha$) and $d_i = \sqrt{R_{C_i}^2 - \rho^2}$, the unnormalized center offset is $w = \sum \lambda_i u_i$ with $\lambda = \Gamma^{-1} d$ (for the unit direction $\hat{w}$ multiply by the positive scale $1/|w|$); the center lies on the hyperplane of facet $K$ iff the coefficient of the ray opposite $K$ vanishes, a Cramer determinant, which is proved exactly zero. Since $\Gamma$ is nonsingular, the center offset is unique, and an exactly zero coefficient already excludes a strictly interior center, whatever the signs of the other two; the exclusion uses nothing more. A positive rescaling of the center does not change the vanishing determinant.

*Proof route.* The stage-1 run uses input tables certified by Computation 3.5, with the exclusions on the exact inputs of Computation 3.6. *Second implementation.* The stage 1 of `checks/independent-s1-b2/` (Section 4.5.4) enumerates all closed walks of length three to six on its own library (every angle is at least $\pi/3$, so longer walks exceed $2\pi$ and length six is flat), and isolates roots using a strict monotonicity proved for each word: in the variable $u = 1/(R^2 - \rho^2)$ the derivative of the cosine of each closure angle has the sign of an affine function of $u$, certified strictly negative at both ends of the admissible interval, so each angle increases strictly in $u$ (decreases in $R$) and every non-flat cycle has at most one root. It reproduces the 14,965 cycles with identical words and statuses, the 7,080 flat cycles, 57 degenerate roots and 537 cycles without roots, and 7,291 retained roots, each contained in the recorded interval of the same cycle; its root intervals are narrower. *Shared with the proof route:* the alias table, the construction theorem and, for the exclusions, the exact construction code of Computation 3.6; both use Arb (python-flint) and SymPy.

**Definition 4.2.** $\Omega$ is the set of root intervals of all cycles in $\mathcal{C}_{16}$ other than the flat cycles and the degenerate roots: 7,291 intervals, all bounded, with $\min \Omega = 0.6454972\ldots$ (the 5-simplex) and $\max \Omega = 63.2195709614\ldots$.

The export (`export_omega.py`, function `project_intervals`, shared by both loaders) removes a flat cycle as a whole, as Lemma 2.8 allows, but Lemma 2.8 excludes only the degenerate *root*, not the whole cycle. The export therefore removes a cycle carrying a degenerate root only after checking that it has exactly one finite, positive certified root interval, and raises otherwise. All 23/39/57 degenerate cycles at $P_{\text{ctl}} = 6/10/16$ pass this check; the exports contain 1,629/3,516/7,291 intervals, and an added second root on an actual excluded cycle is rejected by both loaders at each $P_{\text{ctl}}$. The export entry point validates the records structurally before exporting, regenerates the complete cycle inventory against the actual facet table, and calls the exact authentication of Computation 3.6 on the actual excluded cycles, requiring the returned index sets to equal the requested ones; `project_intervals` implements the structural step. Nine entry-point controls (malformed authentication results and failing guards) are rejected at all three bounds.

**Corollary 4.3.** *Let $X$ be any convex uniform 5-polytope with unit edges and circumradius $R_X$, and $G$ a 2-face of $X$ all of whose facets (around $G$) belong to $\mathcal{L}_{16}$, with $G$ at most 16-gonal. Then $R_X$ lies in a root interval of the cycle of $G$, and that interval belongs to $\Omega$.*

*Proof.* The cycle of $G$ is a library cycle (Lemma 2.6(a)), hence in $\mathcal{C}_{16}$; it is not flat (Lemma 2.8); $1/R_X$ is a zero of $g$ in $[0, 1/R_{\max})$ by Lemma 2.6(c),(d), so it lies in a root interval, which is not a degenerate one (Lemma 2.8). □

The facets around $G$ need not be in $\mathcal{L}_{16}$ when $X$ has larger polygons elsewhere; Corollary 4.3 is stated for any $X$ because Case B uses it.

Radius classes and the closure filter. A *radius class* is a connected component of the overlap graph of the intervals of $\Omega$ (closed intervals that meet are joined); its hull is the class interval. A class carries the set of cycles with an interval in it. The *closure filter* then removes, repeatedly until nothing changes, every cycle of a class containing a facet type $K$ such that some polygon orbit $(p, C_a, C_b)$ of $K$ is not a side ($K$ between $C_a$ and $C_b$ at a $p$-gon) of any remaining cycle of the class. A class *survives* if cycles remain.

**Lemma 4.4.** *All cycles of the 2-faces of $P$ at a vertex $v$ lie in one radius class, the class containing $R$, and none of them is removed by the closure filter.*

*Proof.* By Corollary 4.3 each of these cycles has a root interval in $\Omega$ containing $R$, so all these intervals lie in the class of $R$. Let $K$ be a facet of $P$ at $v$ and $(p, C_a, C_b)$ a polygon orbit of $K$. Since $K$ is vertex-transitive, the orbit has a member $G \ni v$, a 2-face of $P$ whose cycle contains $K$ between $C_a$ and $C_b$; this cycle is in the class of $R$. So the set of cycles of $P$'s 2-faces at $v$ is never touched by the filter (by induction over the iterations). □

**Certified computation 4.5** (FILTER)**.** The intervals of $\Omega$ form 4,224 radius classes, of which 563 survive the closure filter.

The program `class_filter.py` applies this combinatorial filter to the stage-1 output. The program rejects unresolved, unknown, missing or malformed statuses, malformed or non-finite surviving intervals, and invalid, duplicate or overlapping exclusion lists; it reconstructs the complete canonical cycle inventory from the facet table and compares it with the input (so a dropped record of either status is detected); it compares endpoints and forms class hulls in exact decimal arithmetic; and before forming classes it runs both LINK checkers on the actual table and the exact authentication of Computation 3.6. Rerun on the pinned stage-1 roots with regenerated exact exclusions, it gives 760/1,790/4,224 radius classes and 136/278/563 surviving classes at $P_{\text{ctl}} = 6/10/16$, with all class files byte-identical to their pins; 23 controls pass. *Second implementation.* `checks/independent-merge-filter/` independently computes exact rational interval

components and fixed-point deletions, with explicit missing-word witnesses. Its separate certificate checker imports no search code; both checkers reject the same three semantic corruptions. Complete surviving class cycle sets match the $\mathcal{H}_6$, $\mathcal{H}_{10}$ and $\mathcal{H}_{16}$ controls (136/278/563); the literal $\mathcal{L}_6$ result is 129 (Section 3.1). The certified stage-1 intervals, statuses, facet words and aliases are supplied inputs, not re-proved by FILTER. The stage-1 checker `checks/check_u1` also finds the same 4,224 classes and 563 survivors from its own roots, but its inputs carry the uncertified decisions listed in Section 6.2. The independent stage-2 and stage-3 programs of `checks/independent-s23/` take these classes as input and do not recompute them.

4.2. **Stage 2: vertex figures.** Let $I$ be a surviving class with interval $[R_{\mathrm{lo}}, R_{\mathrm{hi}}]$. By Lemmas 2.4 and 4.4, if $R \in I$ then the vertex figure VF of $P$ at $v$ satisfies:

(a) VF is a convex 4-polytope inscribed in the 3-sphere of radius $r = \sqrt{1 - 1/(4R^2)}$; each facet is $\mathrm{VF}(K)$ for a facet type $K$ that occurs in a cycle of $I$, lying in a hyperplane at distance $\sqrt{r^2 - r_K^2}$ from the center, with the center on its inner side;
(b) every ridge (2-face) of VF is the vertex figure of a cell and lies in exactly two facets of VF, with the same cell label;
(c) around every edge of VF (a polygon of $P$ at $v$) the facets and ridge labels, read cyclically, form a cycle of $I$;
(d) distinct vertices of VF are at distance at least 1 (Lemma 2.2).

A *signature* of a vertex figure is its multiset of facet types together with the multiset of canonical cyclic words around its edges.

**Certified computation 4.6** (S2: stage 2)**.** For each of the 563 surviving classes, an exhaustive search, sound for every $R$ in the class interval, outputs a finite list of complete candidate encodings; their signatures are 629 in total (one class, the class at $R = 1.0505014948\ldots$ formed by a cycle of three 5-cells and a 16-cell around a triangle, has none). For every $R$ in the class and every vertex figure satisfying (a)–(d), its signature is one of the signatures listed for the class. This is a coverage statement: retaining an encoding does not prove that all its assembly equations have a simultaneous exact solution. Existence of the known models is supplied separately by Section 3.3 and their named constructions.

*Why such a search is exhaustive.* Fix the least facet type $t_0$ present: the first facet is $X_{t_0}$ in a fixed position, which is no loss of generality after an isometry. Repeatedly take an open ridge (a ridge in only one placed facet). In the true vertex figure exactly one other facet contains it (by (b)). Trying every facet type $\geq t_0$, every ridge of that type with the same label and size, every cyclic and reversed correspondence of the ridge polygon, and both positions of the hinge (the two hyperplanes through the ridge at the right distance from the center; only the one on the side opposite the existing facet can occur, by step 2 of the proof of Lemma 4.10) therefore includes the true facet. A branch may be cut only for a certified violation of (a)–(d): a vertex certified strictly outside a facet hyperplane; two vertices at a certified distance in $(0, 1)$; three facets at a ridge or a label clash; a closed cycle around a VF-edge that is not a cycle of the class. These rejection rules are exhaustive. Rounded or approximately symmetric equivalences are insufficient for this purpose. The proof route below removes candidates only by *exact* equivalences.

**Lemma 4.7** (Exact equivalence of placements)**.** *Let $g$ be an exact isometry of the model vertex figure $X_K$ fixing its center, preserving its labeled ridges and VF-edge polygon sizes, and mapping an ordered source ridge correspondence $o$ to another one $o'$ (that is, $g(o_i) = o'_i$). Then for every admissible $R$ in the class ($R > R_K$) the facets placed over a given target ridge by $o$ and by $o'$, with the hinge on the side opposite to the existing facet, coincide as labeled point sets.*

*Proof.* $g$ fixes the template center, so it preserves the distance of the source ridge from it and carries its outward normal to that of $o'$. With the hinge on the opposite side, the new facet's normal is

computed from $d_K(R)$, that distance and the target ridge alone, so it is the same for both options. If $A_o$, $A_{o'}$ are the two gluing maps, $A_{o'} \circ g$ and $A_o$ have the same ordered ridge images and the same outward normal; they agree on three non-collinear ridge vertices and on the facet center, which is not in their affine plane, and so they are equal on the facet hyperplane. Hence the placed facets coincide as functions of $R$, whether or not $g$ reverses orientation, and $g$ preserves the adjacent labels used by later filters. □

*Proof route* (`check_u2` with `stage2`; see). The independent stage-2 search (Section 4.5.2) builds every VF($K$) directly from the guarded exact construction models of Section 3.3 (`exact_types.py`): metric constants by directed interval operations, cell lists, cyclic orders and labels from the model-backed cell certificates, no inverse cosine. The only reductions are exact stabilizer permutations of the construction model, restricted to the neighbors of the base vertex and checked to preserve labeled ridges and polygon sizes; a subgroup suffices, since fewer symmetries only retain more options. The *equivalence certificate* `equivalences-p16.json` lists, for each of the 181 types, every ordered ridge correspondence (cyclic and reversed): 5,108 in total, 1,437 retained and 3,671 discarded, each discarded one with an explicit exact permutation carrying it to a retained one, so that Lemma 4.7 applies. Two checkers (`audit_certificates.py` and the separately written `check_equivalences.py`, which imports neither the search nor the certificate producer) check ordered correspondence, labels, edge types, the hinge flag, full coverage of the options, exact group membership and that every representative is retained; each accepts all 5,108 correspondences and rejects all 905 deliberate corruptions (five per type). Vertex identity uses the stricter bound $u < 1/4$ and an undecided distance raises `PrecisionError`, so the class is reported undecided and the run exits nonzero (a control widens a radius interval deliberately and checks this). *Result.* On the certified stage-1 class intervals the guarded searches complete every class with zero undecided, error or non-strict outputs: $P_{\mathrm{ctl}} = 6$: 136 classes, 168 signatures, 2,081 nodes; $P_{\max} = 10$: 278, 332, 3,556; $P_{\max} = 16$: 563, 629, 5,869 nodes (no node cap). The same 629 signatures are found on the independent stage-1 checker's broader class intervals. Both certificate checkers accept all 5,108 correspondences and reject all 905 corruptions. They share template, interval and model-guard code, so they check one certificate without providing two independent proofs of the model library. The strict-completion validator in `checks/primary/stage2/replay_all.py` requires zero non-strict completions and checks every per-figure strict flag and the complete class accounting. C9 applies that validator to the regenerated P16 search. The earlier recorded strict controls at $P_{\mathrm{ctl}} = 6, 10, 16$ cover 977 classes in total; the supported public entry point is C9, not the legacy multi-cutoff command-line interface of `replay_all.py`.

*Second implementation* (`checks/independent-s23/`, Section 4.5.4). The implementation uses exact number-field models, Arb at 320 bits, the opposite hinge of Lemma 4.10, its own exact equivalence certificate (all 181 models and 5,108 ordered options, checked by a separate checker), cuts only for certified separation, supporting planes, ridge multiplicity, labels, polygon sizes or a partial edge link that extends no allowed cycle, and no search cap (finiteness from a packing bound). On the same class intervals it completes 136/278/563 classes for $\mathcal{H}_6$, $\mathcal{H}_{10}$ and $\mathcal{H}_{16}$ with 168/332/629 signatures, equal to the proof route's, with no undecided class. *Shared with the proof route:* the stage-1 classes, the facet-name table and alias table, the construction theorem, the exceptional model certificates and the separation lemma.

## 4.3. **Model ridge lists.**

**Certified computation 4.8** (RIDGES: certified ridge lists)**.** For each of the 181 canonical types $K$, the list of 2-faces of the model vertex figure $X_K$ (which point sets, their cyclic order, their cell labels) is correct.

*What is checked* (`cert_ridges.py`), ridge by ridge:

(1) *Face.* A plane through three consecutive points of the ridge; every other point of $X_K$ certified strictly on one side; the ridge's remaining points on the plane *exactly*: for Wythoffian types the neighbors of $K[0]$ are recomputed in exact arithmetic ($\mathbb{Q}$ for $A_4$, $D_4$; $\mathbb{Q}(\sqrt{2})$ for $B_4$, $F_4$; $\mathbb{Q}(\sqrt{5})$ for $H_4$), matched to the model points by the separation lemma inside $K$, and the differences have exact affine rank 2; `sadi` and `gap` likewise in $\mathbb{Q}(\sqrt{5})$; the polyhedral base ridge of a prism or antiprism prism lies in a plane by construction (its points and $K[0]$ have last coordinate exactly $1/2$, and lie on two spheres); all other ridges are triangles.
(2) *Cyclic order.* For consecutive points $a, b$ every other ridge point is certified strictly on one side of the line $ab$ in the ridge plane; so consecutive pairs are sides, hence edges of $X_K$.
(3) *Completeness.* Every side lies in exactly two listed ridges and every point in a listed ridge; since the face graph of a 3-polytope is connected, the list contains every 2-face.
(4) *Label.* A side of the ridge of a cell is the vertex figure of a $q$-gon, of length $\mathrm{ch}_q$; $q$ is pinned by the certified bracket $4\cos^2(\pi/(q-1)) < |a-b|^2 < 4\cos^2(\pi/(q+1))$ (the function $4\cos^2(\pi/q)$ increases with $q$). The cyclic sequence of sizes is the cell's vertex configuration, which determines the cell (Fact 1.4); it must match exactly one label.

Result: 181/181 types certified; 823 ridges (747 triangles, 55 exactly coplanar by exact arithmetic, 21 by construction), 1,277 VF-edges, each in exactly two ridges. Five wrong lists (a wrong label, a dropped ridge, a wrong cyclic order, a square split into two triangles, a non-coplanar four-point set) are rejected, and a non-coplanar set has exact affine rank 3. This certifies the local data of the models; that the model points are the vertex figure of a uniform polytope with unit edges is supplied by Section 3.3, and the correspondence of these certified cells and labels with every stage-1 input entry by Computation 3.5. *Independent replay:* the stage-3 checker certifies its own templates (cells, neighbors and symmetries) from exact models (Section 4.5.3).

4.4. **Stage 3: from vertex figures to polytopes.** Stage 3 is checked by `checks/primary/check_u3/` (Section 4.5.3; Proposition 4.14) and `checks/independent-s23/` (Section 4.5.4). Both certify a unique radius on the whole class interval, cover every actual star by candidate encodings, identify known polytopes by elimination, and refute the remaining encoding by the development of Computation 4.12.

4.4.1. *The exact radius.*

**Certified computation 4.9** (ROOT: unique radius)**.** For every one of the 629 vertex-figure signatures, one of its edge cycles has a closure function $g$ with exactly one zero on the *whole* class interval $[1/R_{\mathrm{hi}}, 1/R_{\mathrm{lo}}]$ of the surviving stage-1 class: $g$ has certified nonzero values of opposite signs at the two ends, and $g'$, enclosed over the whole interval, excludes 0.

*Proof route* (`closure.py`, step 2 of Section 4.5.3): sign change at $R_{\mathrm{lo}}$ and $R_{\mathrm{hi}}$ and a forward-mode derivative excluding 0 on the whole exported stage-1 class interval, evaluated with certified template angles. The inverse cosine used there (`checks/primary/check_u3/ivla.py`, `iacos`) is a proved enclosure: for an input interval $[a, b] \subseteq [-1, 1]$, an ordinary `mp.acos` evaluation only *proposes* bounds $l \le u$, and each is returned only after the outward-rounded interval cosine certifies $0 \le l \le \pi$, $\cos l \ge b$ and $0 \le u \le \pi$, $\cos u \le a$; since cosine decreases on $[0, \pi]$ this gives $l \le \arccos b \le \arccos a \le u$. A proposal that cannot be certified is replaced by the safe domain bound 0 or $\pi$, which can only make a caller report uncertainty, never a false sign. *Coverage*: the audit `audit_radius_route.py` checks, by exact decimal comparison, that every surviving stage-1 class is exported, that each exported interval contains its certified input interval and equals the interval the run used, and that every signature is accounted for with zero radius, verdict and development uncertainty. In the final guarded runs all 168/332/629 radii at $P_{\mathrm{ctl}} = 6/10/16$ are unique on their whole classes; a separate check confirms these outputs. *Second implementation:* `checks/independent-s23/` (`s3_radius.py`, Section 4.5.4) proves a unique root on the same whole class intervals for all 629 signatures, with Arb inverse cosine and its own closure constants from the exact models.

4.4.2. *Congruence from discrete data. Discrete data.* A vertex figure's data are, for each of its facets, the facet's point set, its library type $K$, and a correspondence $c$ (model point $t \mapsto$ VF point $c(t)$). Correspondences are taken modulo a set $E_K$ of *exact* isometries of $X_K$ fixing $K[0]$: the permutations induced by generators of $\mathrm{Stab}_K(K[0])$ (exact reflections and diagram automorphisms, matched by the separation lemma inside $K$), and, for prisms over polyhedra, the relabellings $t \mapsto s \pm t$ of the polyhedron's vertex-figure polygon that preserve its vertex configuration (a rotation or reflection of the circle, exactly; this adds the mirror symmetry of the snub vertex figures, which Stab does not contain). Elements that do not map labeled ridges to ridges with the same label are dropped. A missing element of $E_K$ can only make the check below fail, never pass.

**Lemma 4.10** (Congruence from discrete data)**.** *Let $X$, $X'$ be convex uniform 5-polytopes with the same circumradius $R$ and vertex figures $V$, $V'$ indexed by the same set, with the same facets: the same index sets, the same types, and correspondences that agree modulo $E_K$. Then a linear isometry $A$ with $AV_i = V'_i$ for all $i$ exists.*

*Proof.* (1) Take a facet $f$ with $V[S_f] = g(X_K)$ and $V'[S_f] = g'(X_K)$ and $c' = c \circ e$ with $e \in E_K$ realized by the isometry $\varepsilon$. Then $V[c(t)] \mapsto V'[c(t)]$ is $g'\varepsilon^{-1}g^{-1}$, an isometry. Both facet sets lie in hyperplanes at the same distance $\sqrt{r^2 - r_K^2}$ from the center $O$, so it extends to a linear isometry $A$ of $\mathbb{R}^4$. (2) Let $f'$ share a ridge $\rho$ (at least three non-collinear points) with $f$. Project to the 2-plane $W$ orthogonal to the directions of $\rho$: $\rho$ becomes a point $p$, and the hyperplanes through $\rho$ become lines through $p$, tangent at their feet $T = dn$ to circles about $O$. The center $O$ is interior to the vertex figure (Lemma 2.4), so it lies in the open wedge at $p$ bounded by the two facet lines (angle below $\pi$). Each facet's circumcenter lies on its ray; it is relatively interior to the facet (the facet is the vertex figure of a uniform 4-polytope), so $T \neq p$. Hence the two feet lie strictly on opposite sides of the line $pO$, and the hyperplane of $f'$ is the tangent line from $p$ at distance $d_{K_{f'}}$ on the side opposite to $f$'s foot; it is the same for $AV$ and $V'$. (3) Inside that hyperplane $AV[S_{f'}]$ and $V'[S_{f'}]$ are related by an index-preserving isometry, as in (1), which fixes $\rho$ pointwise and the common circumcenter, which is not in the plane of $\rho$; so it is the identity. (4) The facet graph of $V$ through ridges is connected, so $AV = V'$. □

**Lemma 4.11** (Coset comparison)**.** *Let $S$ be the full star at $v_0$ of a hypothetical convex uniform polytope of circumradius $R$, centered at the origin, and let $T$ be a group of exact linear isometries fixing $S$. At a neighbor $w$, let $g_1, \dots, g_k$ be a complete, nonempty list of candidate linear isometries for its star, with a true congruence $g_i$ among them. Suppose that for every $j$ there is $\tau_j \in T$ and a certified Frobenius-norm bound $\|g_1\tau_j - g_j\|_F \le \varepsilon_j$, and $2R\max_j \varepsilon_j < 1/2$. Suppose each $g_j$ has anchor vertices $a$ in $S$ spanning $\mathbb{R}^5$, and every image $g_j(a)$ belongs to a known facet through $w$. Then every candidate gives the same full star at $w$.*

*Proof.* The identity

$$g_i\tau_i^{-1}\tau_j - g_j = (g_i - g_1\tau_i)\tau_i^{-1}\tau_j + (g_1\tau_j - g_j)$$

bounds the operator norm by $\varepsilon_i + \varepsilon_j$. On an anchor of norm $R$, the two images are less than $1/2$ apart. Both are vertices of the hypothetical polytope: one lies in the true star $g_i(S)$, the other in the known facet. By Lemma 2.2 they coincide. The spanning anchors determine the isometry, so $g_j = g_i\tau_i^{-1}\tau_j$ and $g_j(S) = g_i(S)$. □

4.4.3. *The one other vertex figure.* In the class whose scalar root is $R = 1$, with interval

$$[0.99999999999999999206,\ 1.00000000000000000867],$$

the class has two candidate vertex-figure signatures with the same facet multiset: two 5-cells (`A40001`), six $\{3\} \times \{3\}$ and eight tetrahedral prisms (`pr-tet`), 8 vertices. One is the vertex figure of the stericated 5-simplex `A510001`; the other is the only candidate signature in Case A not identified

with a known polytope. No exact star with the latter encoding is asserted to exist; the refutation below applies to any hypothetical actual star it represents throughout the class interval.

**Certified computation 4.12** (DEV: the extra vertex figure is refuted)**.** No convex uniform 5-polytope has an actual star represented by this extra candidate encoding in its class interval.

*What is checked* (independent checker, `develop.py`, Section 4.5.3). A *development*: starting from the star of $v_0$ (all facets with their full vertex sets), at the next vertex $w$ without a star every congruence $g$ with $g(v_0) = w$ mapping a facet of the star onto a known facet through $w$ is tried (all of them: a star facet of the right type composed with every certified template symmetry taking the base to the right vertex); $g$ is kept unless a known facet through $w$ is certified not to lie in $g$(star) or adding $g$(star) gives a certified violation (separation, convexity, a cell in three facets, two facets with five affinely independent common vertices). Kept stars are compared by exact encodings, not by interval overlap; distinct stars give branches. Result: the signature has one retained rooted encoding. Any actual star represented by it, if one existed, would force its neighbors; it matches no listed polytope (`A510001`, with the same facet multiset, is excluded for all 48 frames; the other names of the class have 12 or 6 neighbors), the development fixes the star at the first neighbor, and at the second neighbor none of the 48 candidate congruences contains every facet already known there (a certified mismatch of a facet center or of a template map). So the vertex figure is `REFUTED`. *Uncertainty.* The development raises a distinct exception for an undecided identity, which contributes `OPEN` (also when another candidate survives) and is counted with its reason, so it can never yield `REFUTED`; candidate-point lookup tests every potentially matching point. The final guarded runs refute this vertex figure at $P_{\text{ctl}} = 6$, 10 and 16 with zero search, development and model-guard uncertainty. At $P = 16$ the primary development records two branches. The independent trace records four nodes, not four branches; these counters describe different algorithms. The chosen extra-candidate cycle has cosines $1/5, -1/5, 1/5, -1/5$ at $R = 1$, hence angle sum $2\pi$. Indeed, it is the triangular cycle `A40001/tet`, `pr-tet/P3`, `duoprism-3-3/P3`, `pr-tet/tet`. Here $\rho^2 = 1/3$, the successive sectional-angle cosines are $1/4, 0, 1/2, 0$, and the successive shared-cell squared radii are $3/8, 3/8, 7/12, 7/12$. Substitution in the closure formula gives the four displayed cosines, whose opposite pairs sum to $\pi$ under arccos. Whole-class uniqueness therefore fixes the root at 1. Development still uses interval enclosures for the whole class; it does not substitute a rounded point estimate. *Second implementation:* `checks/independent-s23/` refutes the same signature at $P_{\text{ctl}} = 6$, 10 and 16 by a finite development tree in which every branch dies (Section 4.5.4); its restricted $R = 1$ control (5-cell, $\{3\}\times\{3\}$ and tetrahedral prism only) identifies the other signature with `A510001` and refutes this one.

4.5. **Independent replays.** The implementations use the shared geometric specifications and inputs listed in Section 6.2. The stage-2 and stage-3 checkers below use mpmath.iv; the second implementation uses Arb. Agreement therefore checks distinct code and arithmetic paths, subject to those shared mathematical premises.

4.5.1. *Stage 1.* The stage-1 proof is replayed by `checks/independent-s1-b2/`, using exact constructions and certified closure bounds as described in Section 4.5.4.

4.5.2. *Stage 2:* `check_u2/`. The proof route of Computation 4.6. An orthogonal-frame gluing search with fail-first ridge choice in `mpmath.iv` at 200 bits. Each VF($K$) is built from the guarded exact construction model (`exact_types.py`, Section 3.3); reductions are only the exact equivalences of Lemma 4.7, certified by `equivalences-p16.json` (Section 4.2). Branches are cut only for certified separation ($0 < |x - y|^2 < 1$), certified convexity violations or combinatorial contradictions; two points are identified only when $|x - y|^2 < 1/4$ is certified, and an enclosure that permits both equality and distinctness stops the class as undecided; a facet type is dropped when $R_K \geq R$ is certified. Every completed figure is rechecked (every ridge paired, every edge cycle allowed, every

point not on a facet certified strictly inside: 0 exceptions). Result: 168, 332 and 629 signatures at $P_{\mathrm{ctl}} = 6$, 10 and 16, with zero undecided classes (Section 4.2). Corruptions: deleting an allowed cycle removes exactly the vertex figure using it; substituting a facet type in a cycle adds no vertex figure; 905 equivalence-certificate corruptions (Section 4.2).

4.5.3. *Stage 3:* `check_u3/`. The stage-3 checker (Proposition 4.14 below) compares stars by exact encodings; the following lemma says what an encoding records and why equal encodings give equal stars.

**Lemma 4.13** (Rooted determinism)**.** *Fix a common circumcenter $c$ and an admissible radius $R$ (a radius in the class interval with $R > R_K$ for every facet type $K$ used). A* rooted encoding *of a star at $v_0$ records the full template type of an initial facet and its root frame (the images of the template's base vertex and its neighbors), and, along a connected traversal of the facet adjacency graph of the vertex figure (facets adjacent through cells containing $v_0$), the type of each further facet and the ordered correspondence of its model neighbor indices, taken modulo the exact stabilizer of the base vertex in the* full *template. If two stars at the same $R$ have equal rooted encodings in root frames whose identification also identifies $c$, they are equal as sets of full facets.*

*Proof.* The root frame fixes the template map on the base vertex and its neighbors, which affinely span the facet hyperplane, so it fixes the full root facet. Suppose a facet $F$ is fixed and $C \ni v_0$ is a cell of $F$ at which the traversal reaches the next facet $F'$ of type $K'$. The affine hull of $C$ is fixed. The center of $F'$ is at distance $\sqrt{R^2 - R_{K'}^2} > 0$ from the fixed common center $c$, so the hyperplane of $F'$ is one of the two hyperplanes through $\operatorname{aff} C$ at that distance, and, as in step (2) of the proof of Lemma 4.10 one dimension up (the center of the polytope is interior, the facet centers are relatively interior by Lemma 2.1), it is the one whose foot lies on the side opposite to that of $F$. The ordered correspondence fixes the images of $v_0$ and of the vertex-figure points of $C$; together with the center of $F'$, which is not in $\operatorname{aff} C$, they affinely span the hyperplane of $F'$, so the isometric template map, and hence the full facet $F'$, is unique. Changing a correspondence by an element of the full-template base stabilizer does not change the placed full facet. The facet graph of the vertex figure is connected, so induction reaches every facet. If the stars are given as functions of $R$ on an interval, the conclusion is equality of the placement functions at each common admissible $R$; the unique-root step then selects one value. □

Neither the coarse signature nor the larger group $E_K$ of isometries of the vertex figure $X_K$ would do here: for the snub-cube prism a mirror fixes the vertex figure but not the full facet (after reflecting one snub-cube-prism facet of $\{3\}\times$ snub cube, the two facets contain distinct vertices with squared distance less than 0.296). `checks/primary/check_u3` (`canon.py`, `search.py`) uses full-template stabilizers and records the ordered model-neighbor indices, as the lemma requires. Its canonical code minimizes the complete incidence and ordered-template record over initial facets of the least present type and their full-template base stabilizers. Equality of codes therefore exhibits such a correspondence: undo the two minimizing relabelings, fix the common initial template and apply Lemma 4.13 along the connected facet graph. Conversely a rooted congruence supplies one of the enumerated relabelings. Thus the canonical code cannot identify distinct full stars at a common admissible radius; minimizing a coarse facet multiset would not suffice.

(1) *Realization coverage.* For each signature, every actual star at a radius in the class interval is represented by a retained candidate encoding: facets are placed copies of templates through $v_0$ at the right distance; at an open cell every type with a cell of the same label is glued on the other side (convexity forces the side), under every rotation or reflection of the cell's vertex-figure polygon (reflections give mirror images of chiral facets); options equal under the stabilizer of the base are merged exactly; cuts only for certified reasons (separation, convexity, non-orthogonality, polygon or cell-patch mismatch, cell in three facets, two cells sharing three

vertex-figure points, type counts, word multisets). A facet is rejected for a non-positive squared height only when the *upper* end of its enclosure is non-positive; an enclosure straddling zero is undecided. Every undecided exit, at the start or deeper in the search, is counted and makes the class `OPEN`; a class with an unresolved realization search cannot be used for identification by elimination. Point resolution tests every candidate enclosure by interval distance. Complete candidate states are grouped by an exact rooted encoding; actual stars with equal encodings in identified root frames are equal at every common admissible radius (Lemma 4.13). Result: one retained encoding per signature (629; 633 complete search states). Compatible interval constraints and the final closure/off-facet checks do not prove simultaneous exact satisfaction of all assembly equations. In particular, retaining a row is not an existence theorem.

(2) *Radius.* For one polygon, $f(R_{\mathrm{lo}})$ and $f(R_{\mathrm{hi}})$ have certified opposite signs and $f'$ (forward-mode derivative on the whole class interval) excludes 0: a unique root in the class interval. The template angles and closure values use the proved inverse-cosine enclosure, and the class intervals are the exported stage-1 classes, with the coverage audit of Computation 4.9. This certifies the scalar root, not existence of an assembled star at that root.

(3) *Identification by elimination.* A listed name $Q$ enters a class only if its radius lies in the class interval and every facet type of $Q$, derived from its name by construction rules, is a facet type of the input; otherwise $Q$ is only reported. For each retained encoding $e$ of the class, possible matches to $Q$ are tested: `NO` if for every frame (isometry fixing the center with $v_0 \mapsto q_0$, fixed by $v_0$ and four vertex-figure points sent to every neighbor 4-tuple of $q_0$ with the same distances) some star point is certified apart from every vertex of $Q$. A frame is skipped only when an exact interval Gram entry is disjoint from the required one (the double-precision Gram comparison only chooses which entry to test); a star point is certified apart only by coordinate-interval disjointness from every local vertex of $Q$ or a positive interval lower bound on its squared distance; local vertices of $Q$ are discarded only when an interval lower bound exceeds the cutoff; a frame that cannot be inverted or decided is `UNDECIDED`, never `NO`. `NO-COUNT` if $e$ has a number of vertex-figure points different from the exact neighbor count of $q_0$ (an upper bound from enclosures containing 1, a matching lower bound exhibited by the exact model of $Q$'s construction). An actual star represented by $e$ is identified with $Q$ when every other encoding of the class is excluded for $Q$: the separately constructed $Q$ has an actual star represented in this class by completeness of stage 2 and step 1, so its encoding must be $e$. Step 2 fixes the common radius and Lemma 4.13 gives equality in the identified root frames. No interval overlap is used as an equality; the overlap test `MATCH` is reported as a consistency check only.

(4) *Local uniqueness.* At each neighbor $w$, every congruence $g = [L_{F^*}S \mid \cdot][L_F \mid \cdot]^T$ ($F$ a star facet of the type of a facet $F^*$ through $v_0$ and $w$, $S$ a template symmetry taking the base to $w$'s template vertex, all of them) not certified to miss a facet through $w$ is kept; all kept $g$ must give one exact encoding, hence at most one actual full star at $w$ at the fixed radius (Lemma 4.13); an empty kept set is not accepted as forcing. For an actual polytope the star at $w$ supplies existence. Whether the facets of $g$(star) can match the known facets through $w$ is decided by a perfect matching in the complete bipartite graph of interval-compatible pairs (`locuniq.py`); a candidate is excluded only if no perfect matching exists.

(5) *Development* for any actual star represented by the unidentified encoding, as in Computation 4.12.

Template facts (neighbors, cells, symmetries of the 181 facet types) are certified against exact models of the constructions: symmetries by construction, vertices at exact distance 1, exactly flat cell vertex sets, `sadi` and `gap` in $\mathbb{Q}(\sqrt{5})$; all symmetry candidates are enumerated and rejected only for certified reasons, and the accepted sets equal the exact groups (orders $|\mathrm{Stab}| \cdot |V|$: 120 or 240 for $A_4$ types, 384 or 1152 for $B_4$, 1152 or 2304 for $F_4$, 14400 for $H_4$, $4pq$ or $8p^2$ for duoprisms, $8n$ for antiprism prisms, 48, 96, 120 or 240 for polyhedral prisms, 400 for `gap`, 576 for `sadi`).

Double precision is used only to propose, except in one place covered by a stated error lemma with asserted preconditions (a coordinate lemma: a double within $1.1 \cdot 10^{-6}$ of an exact coordinate of size at most 64, used to certify a rejected symmetry candidate's image apart from distant template vertices). Frames and local vertices are excluded by interval evidence only. All models are built and used under the runtime guard of Section 3.3. Result at $P_{\max} = 16$: 629 retained encodings, 628 `KNOWN` (628 distinct constructed polytopes, all 890 names matched, none outside the library), 1 `REFUTED` (the extra candidate), 0 `OPEN`; scalar-root uniqueness and conditional forcing for all 629; 230 `NO-COUNT` and 32 `NO` eliminations. Corruptions: shifting every class interval by $10^{-12}$ leaves no retained encoding and no `KNOWN`; mirroring one snub-cube-prism facet changes `KNOWN` to `REFUTED`. *Recorded controls with the model guard active* (the public primary full-mode driver replays $P = 16$): $P_{\text{ctl}} = 6$: 136 classes, 168 encodings, 167 `KNOWN`, one `REFUTED`; $P_{\max} = 10$: 278, 332, 331, one; $P_{\max} = 16$: 563, 629, 628, one; all roots and all forcing checks certified, and zero `OPEN`, search, development and model-guard uncertainty. The checker's inputs are the completeness of stage 2 (Computation 4.6), the construction properties of Section 3.3, the separation lemma, and the standard facts used by its certificates.

**Proposition 4.14** (stage 3, proof route)**.** *Let $P$ be a convex uniform 5-polytope with all 2-faces at most 16-gonal, and let $\sigma$ be the signature of its vertex figure at $v_0$, in the surviving class $I$ containing $R$. If $\sigma$ has verdict `KNOWN` with name $Q$, then $P \cong Q$; if $\sigma$ has verdict `REFUTED`, $P$ does not exist.*

*Proof.* Let $X$ be the actual star of $P$ at $v_0$. By step 1 it is represented by the single retained encoding of $\sigma$; by step 2, $R$ is the unique root on the whole class interval. If the verdict is `KNOWN`, then $Q$'s own star is a realization of a signature of the class (stage 2 and step 1 applied to $Q$, whose facets lie in the library) and every other encoding of the class is excluded for $Q$ (step 3; $Q$ entered the class because its radius lies in $I$). Its star therefore has the same encoding as $X$ and, by step 2, the same radius; Lemma 4.13 makes the stars equal in identified root frames. Thus after an isometry $P$ and $Q$ have the same star $X$ at $v_0$. By step 4 any convex uniform 5-polytope with star $X$ at $v_0$ has one and the same star at every neighbor of $v_0$; Lemma 2.11 gives $P = Q$ after the isometry. If the verdict is `REFUTED`, the development of step 5 excludes every possibility (Computation 4.12). □

4.5.4. *Independently written second implementations.* The packages `checks/independent-s1-b2/` and `checks/independent-s23/` implement separate algorithms using Arb (python-flint) and SymPy. The latter shares stage-1 classes, model certificates, known-name lists and the facet-name table with the other stage-2/3 implementation; neither package uses `mpmath.iv`.

*Stage 1 and B2:* `checks/independent-s1-b2/`. Library: exact parabolic Coxeter combinatorics with rational coefficient-pair keys, exact supporting planes for the two 600-cell deletions, product identities; 181 types, 558 words, 1,277 occurrences, all enclosures inside the recorded ones. Stage 1: closed walks of length three to six, exact Gram matrices for three rays, the two-endpoint monotonicity certificate of Section 4.1, exact phase products for flatness; the same 14,965 cycles and statuses, 7,080 flat, 57 degenerate, 537 without roots, 7,291 roots each inside the recorded interval. B2: Section 5, Computation 5.8. The derivative of the closure-angle cosine is checked symbolically and its strict sign certified at both ends of the admissible interval for all 558 words; a synthetic word with a sign change is rejected.

*Stages 2 and 3:* `checks/independent-s23/`. Stage 2 as described in Section 4.2. Stage 3: closure constants derived from the full exact models; radius uniqueness on the whole exported class interval by Arb inverse cosine and forward-mode derivatives in $s = 1/R$; full five-dimensional stars rebuilt from the base and four model neighbors, with every facet expanded by all exact right cosets of the full-template base stabilizer in the isometry group of its vertex figure (restoring chiral orientation alternatives); candidate states grouped only by correspondence isomorphisms whose facet maps are certified full-template stabilizers; identification by exhaustive elimination (certified Gram disjointness or strict apartness from every vertex of $Q$, exact neighbor counts); local

uniqueness by all congruences mapping a star facet to a known facet through $w$, complete bipartite matching and the coset bound of Lemma 4.11 with Arb norms; development by a finite tree in which every branch must die; a grid index on exact dyadic endpoints that falls back to all points rather than excluding. Undecided identities, signs or resource limits make a class `OPEN`. *Result*: at $P_{\mathrm{ctl}} = 6$, 10 and 16 all 136/278/563 classes complete with 168/332/629 signatures; stage 3 finds 167/331/628 signatures `KNOWN` (235/455/890 names matched) and one wholly refuted signature at each bound (the extra vertex figure), with no `OPEN` class or realization. Its raw candidate-state counts are 176/348/657 independent realization groups against the primary 168/332/629 retained encodings (633 complete primary search states at $P = 16$): the restored mirror placements (8/16/28) each have a strict full-star separation contradiction; all classification conclusions are the same. *Shared with* `check_u2/checks/primary/check_u3:` the stage-1 class intervals (regenerated as black boxes and matched to the 30 input pins, with class membership and interval containment additionally checked by C8), the known-name lists and facet-name table, the construction theorem (`checks/model_audit/wythoff_proof.md`), the two exceptional model certificate files, the alias table and the separation lemma.

4.6. **Proof of Theorem 1.2.** Let $P$ be a convex uniform 5-polytope with all 2-faces at most 16-gonal, scaled to unit edges, with circumradius $R$ and a vertex $v_0$. By Lemma 4.4 all cycles at $v_0$ lie in the surviving class $I$ containing $R$ (Computations 4.1, on the input certified by Computation 3.5, and 4.5). Its vertex figure satisfies (a)–(d) of Section 4.2, so its signature is one of the signatures listed for $I$ (Computation 4.6, proof route `check_u2` with the exact equivalence certificate). Each listed signature has verdict `KNOWN` with a name $Q \in \mathcal{K}_{16}$ or is the refuted extra vertex figure, so Proposition 4.14 gives $P \cong Q$ or a contradiction. Each of these computations has a second rigorous implementation that reaches the same result, sharing the inputs listed in the shared-input discussion in Section 6.2 (see also Section 4.5.4). □

TABLE 1. Counts of the finite computation for the control libraries $\mathcal{H}_6$, $\mathcal{H}_{10}$ and $\mathcal{H}_{16}$ from the recorded control and full replays. The public wrapper replays P6 independent controls and both P16 proof routes; P10 is a recorded control, not a command in that wrapper. Retained signatures are not exact-existence certificates for every row.

| | $\mathcal{H}_6$ | $\mathcal{H}_{10} = \mathcal{L}_{10}$ | $\mathcal{H}_{16} = \mathcal{L}_{16}$ |
|---|---|---|---|
| canonical facet types | 76 | 106 | 181 |
| cycles | 2,559 | 5,471 | 14,965 |
| flat (exact) / degenerate roots (exact) | 790 / 23 | 1,703 / 39 | 7,080 / 57 |
| radius classes / surviving the filter | 760 / 136 | 1,790 / 278 | 4,224 / 563 |
| surviving classes with a known radius | 135 | 277 | 562 |
| vertex-figure signatures | 168 | 332 | 629 |
| distinct known polytopes | 167 | 331 | 628 |

4.7. **Computed coverage.** The recorded guarded searches at $P_{\mathrm{ctl}} = 6, 10, 16$ give 136/278/563 surviving classes and 168/332/629 strict vertex figures, with zero undecided or non-strict results. The independent stage-1/B2 replay and stage-2/3 replays of Section 4.5.4 agree. The full $P = 16$ computation covers all 563 surviving classes; its 629 candidate figures lead to 628 known polytopes and the refuted extra candidate. These are coverage statements, not assertions that every retained local encoding has an exact realization.

At $P_{\mathrm{ctl}} = 6$ ten known radii lie in no surviving class and four known facet multisets are not found; all of them have octagons or decagons, in facets outside $\mathcal{H}_6$ (the variable families remain capped), so these lists are non-empty by design; at $P_{\max} \geq 10$ both are empty.

## 5. Case B: a 2-face with at least 17 sides

5.1. **Setting.** In this section $P$ is a convex uniform 5-polytope with unit edges, circumradius $R$, center $c$, having a large 2-face (at least 17 sides). By vertex-transitivity every vertex lies in a large 2-face. Fix a vertex $v$; VF is its vertex figure, a convex 4-polytope inscribed in a 3-sphere of radius $r = \sqrt{1 - 1/(4R^2)} < 1$ (Lemma 2.4). Faces of VF correspond to faces of $P$ at $v$: VF-vertex = edge of $P$ at $v$ (we identify it with the neighbor), VF-edge = 2-face, VF-ridge = cell, VF-facet = facet. A VF-edge is *long* if its 2-face is large; its length is $\mathrm{ch}_n \geq \mathrm{ch}_{17} = 2\cos(\pi/17) = 1.96595\ldots$. A facet is *large* if it contains a large 2-face. For a large 2-face $F \ni v$ with VF-edge $e$, the *fan* of $F$ (or of $e$) is the set of VF-facets $\mathrm{VF}(K)$ with $K \supset F$, and its vertices are the *fan vertices*.

5.2. **Facts about the families.**

**Fact 5.1** (F2)**.** *Let $q \notin \{3,4,5,6,8,10\}$. A convex uniform polyhedron with a $q$-gonal face is $P_q$ or $A_q$; a convex uniform 4-polytope with a $q$-gonal 2-face is a duoprism $\{q\} \times \{p\}$ ($p \geq 3$, including $P_q \times \{\} = \{q\} \times \{4\}$) or $A_q \times \{\}$ (Facts 1.4 and 1.5). In particular this holds for every large $q$, and for $q \in \{7, 9, 11, \ldots, 16\}$ ("family polygons"), which (P1) uses.*

**Fact 5.2** (F3)**.** *In $\{n\} \times \{q\}$ and in $A_n \times \{\}$ every vertex lies on an $n$-gon. At a vertex $v$: $\mathrm{VF}(\{n\} \times \{q\})$ is the tetrahedron $\mathrm{conv}(a, b, g, g')$, where $a, b$ are the neighbors on the $n$-gon and $g, g'$ those on the $q$-gon: $|ab| = \mathrm{ch}_n$, $|gg'| = \mathrm{ch}_q$, $ab \perp gg'$, and the other four edges have length $\sqrt{2}$. $\mathrm{VF}(A_n \times \{\})$ is the pyramid with apex $z$ (the prism edge) over the trapezoid $(a, b, y, x) = \mathrm{VF}(A_n)$: $ab$ is the VF-edge of the $n$-gon, $|by| = |yx| = |xa| = 1$ (the three triangles at $v$), the diagonals $ay$, $bx$ have length $\sqrt{1 + \mathrm{ch}_n}$, and the four apex edges have length $\sqrt{2}$.*

These follow directly from the stated product and antiprism constructions.

**Lemma 5.3** (F5: cycle types)**.** *Let $F$ be a 2-face of $P$ with $n \notin \{3,4,5,6,8,10\}$ sides. Then the cycle of $F$ is one of*

(i) $\{n\} \times \{q_1\}, \ldots, \{n\} \times \{q_k\}$ *(all cells $P_n$), with $k \in \{3,4,5\}$ and $\sum_i (1 - 2/q_i) < 2$;*
(ii) $A_n \times \{\}$, $A_n \times \{\}$, $\{n\} \times \{q\}$;
(iii) $A_n \times \{\}$, $A_n \times \{\}$, $\{n\} \times \{q_1\}$, $\{n\} \times \{q_2\}$ *with $\{q_1, q_2\} \in \{\{3,3\}, \{3,4\}, \{3,5\}\}$.*

*Proof.* By Fact 5.1 the cells at $F$ are $P_n$ or $A_n$ and the facets $\{n\} \times \{q\}$ or $A_n \times \{\}$. The face angles of the tangent cone at $F$ (Lemma 2.6) are $\pi(1 - 2/q)$ for $\{n\} \times \{q\}$ (between two $P_n$) and $\pi/2$ for $A_n \times \{\}$ (between $A_n$ and $P_n$), with sum below $2\pi$. Both neighbors of an $A_n$ ray are $A_n \times \{\}$, each containing one $A_n$ cell at $F$, so $j$ rays of type $A_n$ cost $2j$ distinct right angles, and $j \leq 1$. If $j = 0$ we get (i), with $k \geq 3$ and $k \leq 5$ since each term is at least $1/3$. If $j = 1$ there are $m \geq 0$ duoprisms between the two $P_n$ rays, and the cycle has $m + 2$ rays. The tangent cone at $F$ is a full-dimensional pointed cone in the 3-dimensional space $N$, so a section of it by a positive affine functional is a convex polygon and the cone has at least three extreme rays; hence $m = 0$ is impossible. $m = 1$ is (ii); $m = 2$ needs $1/q_1 + 1/q_2 > 1/2$, i.e. (iii); $m \geq 3$ gives a sum of at least $\pi + 3\pi/3$, too large. □

**Lemma 5.4** (Long edges are disjoint)**.** *Two long VF-edges share no vertex.*

*Proof.* A long edge $[a, b]$ subtends at the center of the VF sphere an angle $\theta$ with $\sin(\theta/2) = \cos(\pi/n)/r \geq \cos(\pi/17)$, so $\theta \geq \pi - 2\pi/17$ (158.82°). If $[a, b]$ and $[a, b']$ are both long, $b$ and $b'$ lie within $2\pi/17$ (21.18°) of the antipode of $a$, so $|b - b'| \leq 2r\sin(2\pi/17) < 0.7225 < 1$, contradicting Lemma 2.2. □

Lemma 5.4 is used in Section 5.6.2: the Arb certificate of (W) leaves out the ends of the long edges themselves, and this lemma covers them.

5.3. **The radius forced by a large 2-face.** Let $F$ be an $n$-gon of $P$ at $v$, $n \notin \{3,4,5,6,8,10\}$ (so that Lemma 5.3 applies), with center $o_F$ in the plane $L$, and write $c = o_F + W$ with $W \perp L$; so $R^2 = r_n^2 + |W|^2$. A neighbor $v+u$ of $v$ with $u \perp L$ is on the circumsphere iff $u \cdot W = 1/2$ (expand $|v+u-c|^2 = R^2$ using $u \cdot (v - o_F) = 0$).
Type (i). The cells at $F$ are $F \times [v, v+u_i]$ with unit $u_i \perp L$; consecutive $u_i, u_{i+1}$ span the $q_i$-gon of $\{n\} \times \{q_i\}$ at $v$, so $|u_i - u_{i+1}| = \mathrm{ch}_{q_i}$. All $u_i$ satisfy $u_i \cdot W = 1/2$, so they lie on a circle of radius $S = \sin\alpha$ on the cone of half-angle $\alpha$ about $W$, $\cos\alpha = 1/(2|W|)$. By Lemma 2.6(c), $W$ is strictly inside the cone spanned by the $u_i$, i.e. the circle's center is strictly inside the inscribed polygon of the $u_i$: all central angles $2\arcsin(\cos(\pi/q_i)/S)$ are below $\pi$ and they sum to $2\pi$. Let $f_c(S) = \sum_i 2\arcsin(\cos(\pi/q_i)/S)$ on $[c_{\max}, 1]$, $c_{\max} = \max_i \cos(\pi/q_i)$; $f_c$ is strictly decreasing and $f_c(1) = \sum \pi(1-2/q_i) < 2\pi$. Hence:

**Lemma 5.5** (Type (i) radius and closability)**.** *A configuration $c = (q_1, \dots, q_k)$ occurs in a type-(i) cycle only if it is* closable*: $f_c(c_{\max}) > 2\pi$. Then $f_c(S) = 2\pi$ has a unique solution $S \in (c_{\max}, 1)$, the* configuration radius *$R_c = 1/(2\sqrt{1-S^2})$ depends only on the multiset $c$, and*

$$R^2 = r_n^2 + R_c^2.$$

*Moreover $R_c \geq r_{q_{\max}}$. For $k = 3$, closability is equivalent to the triangle with sides $\mathrm{ch}_{q_1}, \mathrm{ch}_{q_2}, \mathrm{ch}_{q_3}$ being acute; for $k = 4$ to the circle's center being strictly inside the inscribed quadrilateral. If $f_c(c_{\max}) = 2\pi$ exactly (*degenerate*), the center lies on the hyperplane of a facet, which is impossible.*

*Proof.* All but the formula $R_c \geq r_{q_{\max}}$ and the triangle criterion is above. $S \geq \cos(\pi/q_{\max})$ gives $R_c^2 = 1/(4(1-S^2)) \geq 1/(4\sin^2(\pi/q_{\max}))$. For three rays: the foot of the perpendicular from 0 to the plane of $u_1u_2u_3$ is the circumcenter $o_T$ of the chord triangle, and $o_T \cdot u_i = |o_T|^2$ for each $i$, so $W = o_T/(2|o_T|^2)$ (both satisfy the three equations $x \cdot u_i = 1/2$, which have a unique solution); so $W$ is interior to the cone iff $o_T$ is interior to the triangle iff the triangle is acute, and a right triangle puts $W$ in the plane of a facet. The same argument with the circle gives the $k = 4$ form. □

*Remark* 5.6 (four or five rays). For $k \geq 4$ a configuration is never degenerate, and it is closable if and only if $\sum_i (1-2/q_i) < 2$. Indeed, at $S = c_{\max}$ one maximal term of $f_c$ equals $\pi$, and each of the other $k-1 \geq 3$ terms is strictly larger than $2\arcsin(\cos(\pi/q_i)) = \pi(1-2/q_i) \geq \pi/3$ because $c_{\max} < 1$; so $f_c(c_{\max}) > 2\pi$ even when maximal entries are tied. Since $f_c$ is strictly decreasing, a root in $(c_{\max}, 1)$ exists exactly when $f_c(1) = \pi\sum(1-2/q_i) < 2\pi$ (at equality the only root is $S = 1$, which gives no finite $R_c$). This makes the closability classification used in B2 and (P1) checkable by hand for $k \geq 4$. The exact angle condition already excludes $k \geq 6$, so the finite enumerations need $k = 3, 4, 5$ only; the B2 checker rejects cycles with fewer than three or more than five entries, and the (P1) closure routines reject fewer than three, explicitly.

Exactly three configurations with entries at most 16 are degenerate: $(3,3,4)$, $(3,4,6)$, $(3,5,10)$, since $\mathrm{ch}_3^2 + \mathrm{ch}_3^2 = \mathrm{ch}_4^2$, $\mathrm{ch}_3^2 + \mathrm{ch}_4^2 = \mathrm{ch}_6^2$ and $\mathrm{ch}_3^2 + \mathrm{ch}_5^2 = \mathrm{ch}_{10}^2$ (the last is $1+(3+\sqrt{5})/2 = (5+\sqrt{5})/2$). The configuration $(3,5,q)$ is not closable for $q \geq 11$, since $1+\mathrm{ch}_5^2 = 3.618\ldots < \mathrm{ch}_q^2$; similarly $(3,3,q)$ for $q \geq 5$ and $(3,4,q)$ for $q \geq 7$. For $c''$ a sequence such that $(n, c'')$ is closable, write

$$\delta_{c''}(n) := R^2_{(n,c'')} - r_n^2.$$

**Lemma 5.7** (closed forms)**.** *Let $n \geq 17$, $\theta = \pi/n$, $c = \cos\theta$ and $x = c^2$. Then*

$$\delta_{(4,4)}(n) = \tfrac{1}{4}, \qquad \delta_{(3,6)}(n) = \frac{x}{4x-1} = \frac{1}{4-\sec^2\theta}, \qquad \delta_{(3,3,3)}(n) = \frac{2c+1}{8(1+c)} = \frac{h_A(n)^2}{4}.$$

*$\delta_{(3,6)}$ decreases and $\delta_{(3,3,3)}$ increases with $n$, so for $n \geq 17$, including the limit $n = \infty$, $\delta_{(3,6)}(n) \in [1/3,\ \delta_{(3,6)}(17)]$ with $\delta_{(3,6)}(17) = \cos^2(\pi/17)/(4\cos^2(\pi/17) - 1) = 0.3372617\ldots < 0.337262$, and $\delta_{(3,3,3)}(n) \in [h_A(17)^2/4,\ 3/16]$ with $h_A(17)^2/4 = 0.1869633\ldots$; the value $3/16$ is attained only in the limit $n = \infty$.*

*Proof.* For a chord triangle with squared sides $A, B, C$ the circumradius satisfies $S^2 = ABC/((A+B+C)^2 - 2(A^2+B^2+C^2))$, and $R_c^2 = 1/(4(1-S^2))$, $r_n^2 = 1/(4(1-x))$. For $(n,3,6)$ the sides are $2c, 1, \sqrt{3}$, which gives $S^2 = 3x/(-1+8x-4x^2)$ and $\delta_{(3,6)} = x/(4x-1)$; the triangle is acute since $1/2 < x < 1$, so this is the closable solution. For $(n,4,4)$ the sides are $2c, \sqrt{2}, \sqrt{2}$, giving $S^2 = 1/(2-x)$ and $\delta = 1/4$ (the prism $P_n$); this triangle is also acute. For $(n,3,3,3)$ put $S^2 = 1/(3-2c)$ and $\sin\alpha = 1/(2S)$; then $\pi/6 < \alpha < \pi/4$, the three unit chords subtend $2\alpha$ each, the remaining arc $2\beta = 2\pi - 6\alpha$ lies in $(\pi/2, \pi)$, and $\sin\beta = \sin 3\alpha = \sin\alpha(3 - 4\sin^2\alpha) = 2c\sin\alpha$, so its chord is $2S\sin\beta = 2c$: all four arcs are below $\pi$ and sum to $2\pi$. Substitution gives $\delta_{(3,3,3)} = (2c+1)/(8(1+c)) = \big(1 - 1/(2(1+c))\big)/4 = h_A(n)^2/4$ (the antiprism $A_n$), using $\cos^2(\theta/2) = (1+c)/2$. Finally $d\delta_{(3,6)}/dc = -2c/(4c^2-1)^2 < 0$ and $d\delta_{(3,3,3)}/dc = 1/(8(1+c)^2) > 0$, and $c$ increases to 1 with $n$. □

The limiting value 3/16 is included in the closed range used by (P1). The partner filter uses the valid bound 0.3373 for $\delta_{(3,6)}(17)$; the exact rational function, rather than a rounded constant, is evaluated throughout the parameter interval.

Type (ii). The $A_n$ cell has its second $n$-gon at height $h_A = h_A(n)$ along a unit vector $\hat{h} \perp L$, rotated by $\pi/n$. The edge condition gives $h_A^2 = 1 - 1/(4\cos^2(\pi/(2n)))$, and equidistance from the two $n$-gons gives $W \cdot \hat{h} = h_A/2$, so $R_{A_n}^2 = r_n^2 + h_A^2/4$. The other two rays $z_1, z_2$ (lateral edges of the $P_n$ cells) make right angles with $\hat{h}$ and the angle $\pi(1-2/q)$ with each other, so they lie in the plane orthogonal to $\hat{h}$ in $L^\perp$; the component $W'$ of $W$ there satisfies $W' \cdot z_j = 1/2$ and is the center offset of the $q$-gon, $|W'| = r_q$. Hence

$$R^2 = R_{A_n}^2 + r_q^2.$$

Type (iii). The rays are $\hat{h}, z_1, w, z_2$. As in (ii), $W = (h_A/2)\hat{h} + W'$ with $W' \in \operatorname{span}(z_1, z_2)$, $W' \cdot z_j = 1/2$, so with $\phi$ the angle between $z_1$ and $z_2$ and $\gamma := \cos\phi$, $|W'|^2 = 1/(2(1+\gamma))$ and $R^2 = R_{A_n}^2 + 1/(2(1+\gamma))$. Writing $w = \alpha z_1 + \beta z_2 + \eta\hat{h}$ with $w \cdot z_j = c_j := \cos(\pi(1-2/q_j))$, the remaining condition $W \cdot w = 1/2$ reads $h_A\eta = 1 - (\alpha+\beta)$, where $\alpha + \beta = (c_1+c_2)/(1+\gamma)$ and $\eta^2 = 1 - |\alpha z_1 + \beta z_2|^2$. Convexity of the cone $(\hat{h}, z_1, w, z_2)$ puts $w$ and $\hat{h}$ on opposite sides of $\operatorname{span}(z_1, z_2)$, so $\eta < 0$ (the *geometric* branch). Squaring, with $N = 1 - \gamma^2 - c_1^2 - c_2^2 + 2\gamma c_1 c_2$ (so that $\eta^2 = N/(1-\gamma^2)$) and $M = 1 + \gamma - c_1 - c_2$, gives the polynomial equation

$$h_A^2\, N\,(1+\gamma) - M^2\,(1-\gamma) = 0,$$

to be solved on the domain $-1 < \gamma < 1$ together with the norm condition $N \geq 0$ and the unsquared sign $\eta < 0$. When $c_1 = c_2$ (the pair $(3,3)$) the polynomial has the spurious factor $1-\gamma$, which is canceled, leaving a quadratic; otherwise it is a cubic. The radius identity above holds at every geometric root. At $n = 17$, for instance, the roots in $(-1,1)$ are one for $(3,3)$ and two each for $(3,4)$ and $(3,5)$, and exactly one of them is geometric in each case. B2 checks *all roots in* $(-1,1)$ (Computation 5.8), so no claim about their number is needed. The (W) checker parametrizes the type-(iii) fan by $\cos(\phi/2)$ and recovers $n$ from it, without root finding.

### 5.4. **Step B2: no large-polygon radius is a Case-A radius.**

**Certified computation 5.8** (B2)**.** Let $n \geq 17$. None of the following radii lies in an interval of $\Omega$ (Definition 4.2):

- $\sqrt{r_n^2 + R_c^2}$ for every closable configuration $c$ (any entries $\geq 3$, $k \in \{3,4,5\}$, $\sum(1-2/q_i) < 2$);
- $\sqrt{R_{A_n}^2 + r_q^2}$ for every $q \geq 3$;
- $\sqrt{R_{A_n}^2 + 1/(2(1+\gamma))}$ for every root $\gamma \in (-1,1)$ of the type-(iii) polynomial equations, geometric or not.

*Reduction to a finite check.* $r_q$ increases with $q$ and $r_{398} > 63.344 > \max\Omega$. Every radius above exceeds $r_n$, so only $17 \le n \le 397$ matter (381 values). For type (i), $R_c \ge r_{q_{\max}}$ (Lemma 5.5), so configurations with an entry $\ge 398$ give $R > \max\Omega$; for type (ii), $R \ge r_q$, so $q \le 397$; for type (iii), $q_1, q_2 \le 5$. Both `b2check.py` (`QMAX=397`, function `run`) and `checks/independent-s1-b2/b2.py` (function `run`) certify the $r_{398}$ inequality before the enumeration. The boundary certificate `checks/independent-s1-b2/evidence/b2-comparison.json`, `cutoff`, records `last_finite_n=397`, `first_tail_n=398` and $r_{397} < \max\Omega < r_{398}$.

*Proof route* (`b2check.py`). This uses `mpmath.iv` at 200 bits, with its own derivation of the three radius formulas (as above). Type (i): all 1,304 closable multisets with entries 3..397, $k = 3, 4, 5$ and exact rational angle sum below 2 (1,316 cyclic arrangements; $R_c$ depends only on the multiset), $R_c$ by a verified bracket, for each of the 381 values of $n$ (496,824 radii); closability by interval arcsine and the exact degenerate cases in $\mathbb{Q}(\sqrt{5})$. Type (ii): $q = 3, \ldots, 397$ (150,495 radii). Type (iii): every root in $(-1, 1)$, isolated from certified nonzero endpoint values by subdivision; a box is discarded only for a nonzero interval polynomial value or a derivative of constant sign with equal strict endpoint signs, opposite strict signs with a derivative excluding zero certify one root, every split point is certified non-root, and an unresolved box is a failure. There is no positive-root or geometric-branch prefilter: 1,905 roots (1,143 geometric, 762 non-geometric), all compared with $\Omega$. The overlap query is complete (sorted lower endpoints with a prefix maximum of upper endpoints). Result: 649,224 radii, **0 coincidences, 0 undecided boxes**. The closest approach is a relative gap of $1.66 \cdot 10^{-7}$ (type (i), $n = 24$, $c = (5, 5, 8)$, $R = 4.8302\ldots$), against interval widths of about $10^{-45}$ (radii) and $10^{-20}$ ($\Omega$). The closability classification: not closable exactly $(3, 3, q)$, $q \ge 5$; $(3, 4, q)$, $q \ge 7$; $(3, 5, q)$, $q \ge 11$; degenerate exactly $(3, 3, 4)$, $(3, 4, 6)$, $(3, 5, 10)$; all others closable (Remark 5.6). *Root coverage audit* (`check_root_audit.py`, importing neither enumeration program): it reconstructs each quadratic or cubic in 320-bit Arb arithmetic, counts its roots in $(-1, 1)$ by Sturm sequences with all signs certified and no repeated roots, and checks that the certified brackets are disjoint, have opposite endpoint signs, the right derivative sign and branch label, and are as many as the Sturm count; so no root in $(-1, 1)$ is omitted. It rejects four corruptions (a dropped negative non-geometric root, a duplicate, a changed branch, an endpoint bracket). The B2 check also returns zero coincidences against the independent stage-1 intervals.

*Inverse sine.* The lower-bound proposal loop of `rasin` uses an explicit guard: for a positive proposal require $0 \le t_{\mathrm{lo}} \le L$, where $L$ is the lower endpoint of the interval for $\pi/2$. Hence $t_{\mathrm{lo}}$ is in the principal domain before sine monotonicity is used; a nonpositive proposal is replaced by the safe lower bound zero, and an unproved domain raises an error. The wrong-branch proposal $t_{\mathrm{lo}} = 4$ at $x = 1/2$ is rejected.

*Second implementation.* Written from this paper only and run against the independent $\Omega$ of that package (its own stage 1): all 381 values $17 \le n \le 397$, all 1,304 closable multisets (1,171 not closable and 3 degenerate, classified exactly), $q = 3, \ldots, 397$, and every root in $(-1, 1)$ of the three type-(iii) polynomials found by exhaustive subdivision and counted by a separate Sturm sequence (1,143 geometric, 762 non-geometric); the cutoff $r_{397} < \max\Omega < r_{398}$ and the parameter tails are certified. Result: 649,224 radii, 0 coincidences, 0 undecided; the smallest absolute separation is $7.9977\ldots \cdot 10^{-7}$, again at $n = 24$, $c = (5, 5, 8)$. Positive controls 1,750/1,750, 196/196, 42/42 geometric and 2/28 non-geometric; 16 semantic corruptions behave as expected. *Shared with `check_b2`:* the reductions of this section only (the two use different $\Omega$ files, built by different stage-1 implementations).

*Positive control.* Every type-(i), (ii) and geometric type-(iii) radius with $n \le 16$ and entries at most 16 is a stage-1 root, so it must lie in $\Omega$: 1,750/1,750, 196/196 and 42/42 of them are found in $\Omega$. In addition 2 of the 28 non-geometric type-(iii) roots in that range also lie in $\Omega$ (44/70 type-(iii) roots in total); these are not required. These are non-proof checks of the radius formulas. *Corruptions:* three intervals inserted into $\Omega$ around computed radii give exactly three coincidences; the slip $r_n = 1/(2\sin(2\pi/n))$ gives 1,000 coincidences (the radii of $\{n/2\} \times Q$). *Input identity:* $\Omega$

is exported from the unfiltered stage-1 output by `export_omega.py` with the degenerate-cycle guard described after Definition 4.2; the checker refuses an $\Omega$ whose SHA-256 differs from `c7773fab…89bc33` and also checks its content; all 7,080 infinite upper endpoints belong to flat cycles, so no surviving interval is unbounded (an unbounded interval would make B2 vacuous for large $n$).

**Corollary 5.9.** *Every 2-face $G$ of $P$ lies in a large facet. Consequently every VF-edge at $v$ lies in a VF-facet $\mathrm{VF}(K)$ with $K$ large, and $K$ contains a large 2-face through $v$.*

*Proof.* If $G$ is large there is nothing to prove. Otherwise, suppose no facet around $G$ is large; then all of them have only polygons of at most 16 sides, so they are in $\mathcal{L}_{16}$ (Lemma 3.1) and $R \in \Omega$ by Corollary 4.3. But $R$ is also the radius forced by the cycle of a large 2-face of $P$, which is of type (i), (ii) or (iii) (Lemma 5.3) with the parameters of Computation 5.8 (type (i) configurations are closable by Lemma 5.5; in type (iii), $\gamma = \cos\phi$ is a root of the polynomial in $(-1, 1)$, and B2 covers every such root), so $R \notin \Omega$. For the second part, a large facet $K$ is $\{m\} \times \{q\}$ or $A_m \times \{\}$ with $m \geq 17$ (Fact 5.1), and every vertex of $K$, in particular $v$, lies on an $m$-gon of $K$ (Fact 5.2). □

Let $s \geq 1$ be the number of large 2-faces at a vertex (the same at every vertex).

### 5.5. Exactly one large 2-face at each vertex.

**Theorem 5.10.** *If $s = 1$, then $P \cong \{n\} \times Q$ for a convex uniform polyhedron $Q$, or $P \cong A_n \times \{q\}$.*

Fix $v$, its large 2-face $F$ (an $n$-gon with plane direction $L$) and $e = [a, b]$ its VF-edge. By Corollary 5.9 and $s = 1$, every VF-edge lies in a VF-facet $\mathrm{VF}(K)$ with $K \supset F$; so *every VF-edge, and hence every VF-vertex, lies in a facet of the fan of $e$*. The fan is given by the cycle type of $F$ (Lemma 5.3).

*Type (i).* The fan facets are $T_i = \mathrm{conv}(a, b, x_i, x_{i+1})$, where $[v, x_i]$ is the lateral edge of the $i$-th cell $F \times [v, x_i]$; so $x_i - v \perp L$, and the neighbors of $v$ are exactly $a, b, x_1, \ldots, x_k$. The $x_i$ are distinct (two $P_n$ cells at $F$ with the same lateral edge span the same 3-space).

*Product lemma.* For a vertex $t$ let $F_t$ be its unique large 2-face. By vertex-transitivity, the local picture at $t$ is the one at $v$ (a symmetry mapping $v$ to $t$ maps $F$ to $F_t$, by uniqueness). For a neighbor $t'$ of $t$, either $t' \in F_t$, and then $F_{t'} = F_t$; or $t' - t$ is a lateral edge, orthogonal to the plane of $F_t$, and the duoprism $\{n\} \times \{q_i\}$ containing the edge $tt'$ contains the translate $F_t + (t' - t)$, a large 2-face at $t'$, so $F_{t'} = F_t + (t' - t)$. These translates come from the actual prism cells at the edge, not from the shape of the vertex figure. The edge graph of $P$ is connected [5], so every $F_t$ is a translate $F + \tau_t$ with $\tau_t \perp L$ (each step adds 0 or a vector orthogonal to $L$). Hence the vertex set is $F_0 + X$, with $F_0$ the polygon $F$ centered at the origin of $L$ and $X$ a finite set in an affine copy of $L^\perp$, and $P = \mathrm{conv}\, F_0 + \mathrm{conv}\, X \cong \{n\} \times Q$ with $Q = \mathrm{conv}\, X$, a 3-polytope since $\dim P = 5$. Every point of $X$ is a vertex of $Q$ (a point $(f, x)$ of a product is a vertex iff both factors are vertices). $Q$ has unit edges and regular faces, because its edges and 2-faces $G$ give edges and 2-faces $\{f\} \times G$ of $P$; a symmetry $g$ of $P$ maps $F_t$ to $F_{g(t)}$, so its linear part preserves $L$ and $L^\perp$ and $g$ induces an isometry of $Q$; $\mathrm{Sym}(P)$ is transitive on the $F_t$, hence on $X$. So $Q$ is a convex uniform polyhedron. □

*Type (ii).* The fan consists of the pyramids $\Pi_1 = (z_1; a, b, y, x)$, $\Pi_2 = (z_2; a, b, y, x)$ (the two $A_n \times \{\}$ sharing the $A_n$ cell $C$, $\mathrm{VF}(C) = (a, b, y, x)$) and $T = \mathrm{conv}(a, b, z_1, z_2)$. The neighbors of $v$ are $a, b, x, y$ (in $C$) and $z_1, z_2$, with $z_j - v$ orthogonal to the 3-space $M$ of $C$. *Cell partition:* for a vertex $t$ let $C_t$ be the unique $A_n$ cell containing $F_t$ (one $A_n$ ray per cycle). For $t' \in C_t$, $t'$ lies on exactly one of the two $n$-gons of $C_t$ ($n \geq 4$), which is then $F_{t'}$, so $C_{t'} = C_t$. The other two neighbors $t'$ of $t$ satisfy $t' - t \perp M_t$, and the $A_n \times \{\}$ containing $tt'$ gives $C_{t'} = C_t + (t' - t)$. So the vertex set is $C_0 + X$ with $X$ in a 2-plane orthogonal to $M$, $P = A_n \times \mathrm{conv}\, X$, and $\mathrm{conv}\, X$ is a vertex-transitive equilateral convex polygon with unit edges, i.e. a regular polygon $\{q\}$. □

*Type (iii) does not occur.* The fan consists of $\Pi_1$, $\Pi_2$ as in (ii) and $T_1 = \mathrm{conv}(a, b, z_1, w)$, $T_2 = \mathrm{conv}(a, b, w, z_2)$, where $w - v$ is the lateral edge of the $P_n$ cell between the two duoprisms. The seven points $a, b, x, y, z_1, z_2, w$ are distinct: $a - v, b - v \in L$; $x - v$ and $y - v$ are neither in $L$ nor orthogonal to $L$ ($x, y$ lie on the other $n$-gon of the antiprism, rotated by $\pi/n$ and at height $h_A > 0$), and $x \neq y$; $w - v$, $z_1 - v$, $z_2 - v$ are orthogonal to $L$ and are the lateral edges of three different $P_n$ cells. Since every VF-edge lies in a fan facet, the VF-edges are exactly the edges of these four facets:

$$ab,\ ax,\ xy,\ yb,\ z_j a,\ z_j b,\ z_j x,\ z_j y\ (j = 1, 2),\ wa,\ wb,\ wz_1,\ wz_2.$$

In particular $z_1 z_2$ is not a VF-edge, and $x$ has exactly the VF-edges $xa, xy, xz_1, xz_2$, $y$ exactly $yb, yx, yz_1, yz_2$, and $w$ exactly $wa, wb, wz_1, wz_2$; each of $x, y, w$ is simple (Lemma 2.10). The triangle $(x, y, z_1)$ is a 2-face of $\Pi_1$, hence a VF-ridge, lying in $\Pi_1$ and exactly one other VF-facet $K'$. At $x$, $\Pi_1$ has the edges $\{xa, xy, xz_1\}$, so $K'$ has $\{xy, xz_1, xz_2\}$; thus $xa$ is not an edge of $K'$, and by Lemma 2.9 $a \notin K'$. Likewise, at $y$, $b \notin K'$. If $w$ were in $K'$, its $K'$-edges would be among $wz_1, wz_2$ (the others go to $a, b$), degree at most 2 in a 3-polytope, impossible. So $K' = \mathrm{conv}(x, y, z_1, z_2)$, a tetrahedron, and its edge $z_1 z_2$ is an edge of VF (Lemma 2.9, converse), which it is not. □

5.6. **Two or more large 2-faces at each vertex.** The fan of a large $n$-gon $F$ at $v$ is *paired* if one of its facets contains a second large 2-face through $v$: a type-(i) cycle with exactly one $q_i = m \geq 17$ (two such entries would give an angle sum of at least $2(1 - 2/17) + 1/3 > 2$), or type (ii) with $q = m \geq 17$. Otherwise it is *unpaired*: type (i) with all $q_i \leq 16$, type (ii) with $q \leq 16$, or type (iii) ($q_1, q_2 \leq 5$). In a paired fan the second large 2-face's VF-edge (the *partner edge*) is $[x_1, x_2]$ in type (i) and $[z_1, z_2]$ in type (ii).

The paired type-(i) configurations are, by the angle sum and Lemma 5.5: $(m, 4, 4)$, $(m, 3, 3, 3)$, $(m, 3, 6)$ for all $m \geq 17$; $(m, 3, 7)$ for $m \leq 41$; $(m, 3, 8)$ for $m \leq 23$; $(17, 3, 9)$; $(m, 4, 5)$ for $m \leq 19$; and $(m, 3, 3)$, $(m, 3, 4)$, $(m, 3, 5)$ are not closable.

5.6.1. *(P1): the family-polygon filter.* Let $c$ be an unpaired type-(i) configuration (entries at most 16) around $F$ containing a family polygon size $q \in \{7, 9, 11, \ldots, 16\}$. The $q$-gon $Q_0$ of the fan is a 2-face of $P$ lying in the fan facet $\{n\} \times \{q\}$, so its cycle (Lemma 5.3 with $q$ in place of $n$) contains $\{q\} \times \{n\}$. Type (iii) would need both duoprisms at $Q_0$ to have parameter at most 5, so the cycle of $Q_0$ is of type (i) with a configuration $(n, c'')$, or of type (ii) with the duoprism $\{q\} \times \{n\}$. Comparing radii with $R^2 = r_n^2 + R_c^2$:

- type (ii) at $Q_0$: $R^2 = R_{A_q}^2 + r_n^2$, so $R_c^2 = R_{A_q}^2$, i.e. $R_c^2 - r_q^2 = h_A(q)^2/4$;
- type (i) at $Q_0$: $R^2 = r_q^2 + R_{(n,c'')}^2 = r_q^2 + r_n^2 + \delta_{c''}(n)$, so $R_c^2 - r_q^2 = \delta_{c''}(n)$, where $(n, c'')$ is closable with $(1 - 2/n) + \sum_{p \in c''}(1 - 2/p) < 2$. For $n \geq 17$ the angle sum leaves $c'' \in \{(3,3), (3,4), \ldots, (3,9), (4,4), (4,5), (3,3,3)\}$ and restricts $(3,7)$ to $n \leq 41$, $(3,8)$ to $n \leq 23$, $(3,9)$ to $n = 17$ and $(4,5)$ to $n \leq 19$; closability (Lemma 5.5) removes $(3,3)$, $(3,4)$, $(3,5)$ (for $(3,5)$: $1 + \mathrm{ch}_5^2 < \mathrm{ch}_n^2$ for $n \geq 11$, so the chord triangle is obtuse), and the remaining ones are closable in these ranges.

**Certified computation 5.11** (P1)**.** Of the 168 cyclic configurations with entries in 3..16 and angle sum below $2\pi$, 137 are closable, 28 not closable and 3 degenerate. Of the 137, 83 contain a family polygon. For 67 of these 83 there is a family polygon $q$ in $c$ such that $X := R_c^2 - r_q^2$ is certified different from $h_A(q)^2/4$, from $1/4$ ($c'' = (4,4)$), from the range $[0.186963, 3/16]$ of $\delta_{(3,3,3)}$, from the range $[1/3, 0.337262]$ of $\delta_{(3,6)}$, and from each of the finitely many values $\delta_{(3,7)}(n)$, $n \leq 41$, $\delta_{(3,8)}(n)$, $n \leq 23$, $\delta_{(3,9)}(17)$, $\delta_{(4,5)}(n)$, $n \leq 19$. The 16 others are $(4, 4, q)$ and $(3, 3, 3, q)$, $q \in \{7, 9, 11, \ldots, 16\}$.

Hence the 67 configurations do not occur as fans in a convex uniform 5-polytope. The 16 survivors are genuine ($\{n\} \times P_q$ and $\{n\} \times A_q$: $\delta_{(4,4)} = 1/4 = R_{P_q}^2 - r_q^2$, and $(3, 3, 3, q)$ is the vertex configuration of $A_q$). *What is computed.* The first implementation, `caseb_famfilter.py` (Arb, 200 bits; $R_c$ by certified bisection of $f_c$; closability by the sign of $f_c(c_{\max}) - 2\pi$; the three degenerate configurations

by the exact right-triangle identities; the $c''$ list hard-coded with its $n$ ranges, which the program checks against the angle sum). Independent checker `checks/check_caseb/p1check.py` (`mpmath.iv` at 200 bits, sympy for the exact identities) recomputes the $c''$ list from the angle condition with closability tested, uses closed forms for $(4,4)$, $(3,6)$, $(3,3,3)$ over all $n \geq 17$ including $n = \infty$, and gets the same 168/137/28/3/83/67 and the same 16 survivors, with no sporadic $(c'', n)$ coincidence. The Arb implementation is a second rigorous implementation only together with the written rational angle and chord arguments above for its hard-coded $c''$ list and $n$ ranges; its sampled tails are non-proof checks. Corruptions: changing $\delta_{(4,4)}$ by $10^{-30}$, or using $h^2/2$ for $h^2/4$ in $R^2_{A_q}$, makes the corresponding survivors disappear (75 excluded).

5.6.2. *(W): the wedge lemma.* A point $p$ is *admissible* for a fan if it lies on the VF sphere ($|p-v| = 1$, $|p-c| = R$), on the closed inner side of every fan facet's hyperplane, and at distance at least 1 from every fan vertex. Every VF-vertex that is not a fan vertex is admissible (VF is convex and the fan facets are VF-facets; Lemma 2.2). An *allowed* fan vertex is a fan vertex other than $a$, $b$ and, for a paired fan, the partner ends. The partner ends are distinct from the other fan vertices: in type (ii) the apexes satisfy $z_j - v \perp M$ while $x - v, y - v \in M$; in type (i) the $x_i$ are distinct lateral edges.

**Certified computation 5.12** (W)**.** For each fan in the list below and every real $n \geq 17$, including the limit $n = \infty$ (and, in (c), every real $m \geq 17$ including $m = \infty$; in (d), every *integer* $m$ in the stated range), there are no two points $p, p'$ with $|p - p'| \geq \mathrm{ch}_{17}$, each admissible or an allowed fan vertex, other than two fan vertices. The list:

(a) type (i): the 70 closable configurations with entries at most 16 that are not excluded by (P1);
(b) type (ii), $q = 3, \ldots, 16$; type (iii), $(3,3)$, $(3,4)$, $(3,5)$;
(c) paired: $(m,4,4)$, $(m,3,3,3)$, $(m,3,6)$ and type (ii) with $q = m$, all $m \geq 17$ and $m = \infty$;
(d) paired, bounded, with $m$ an integer: $(m,3,7)$, $17 \leq m \leq 37$; $(m,3,8)$, $17 \leq m \leq 22$; $(m,4,5)$, $17 \leq m \leq 18$, for all real $n \geq 17$; and $(m,3,7)$, $38 \leq m \leq 41$, $(23,3,8)$, $(17,3,9)$, $(19,4,5)$ at the integer point $n = m$ only.

Moreover, for each fan in the list, every pair of fan vertices other than $e$, the partner edge and the diagonals $x_i x_j$ of a type-(i) fan has distance below $\mathrm{ch}_{17}$ (the largest is the 16-gon chord $1.961571\ldots$; in type (iii) $|z_1 - z_2| \leq 1.943206$).

*Scope of the two implementations.* The Arb certificate tests pairs in which each point is admissible or an *allowed* fan vertex, so it omits pairs with one end in $\{a, b\}$ or at a partner end, and its fan-vertex distance bound covers pairs not involving $a$ or $b$; the omitted cases are supplied by Lemma 5.4 and by the explicit lengths of Fact 5.2 (Lemma 5.14). The independent checker tests every fan vertex, including $a$, $b$ and the partner ends (its test V), and every pair of fan vertices other than $e$ and the partner edge (its test F), and certifies them. Its certification flag (`CERTIFIED`) requires test F only for the pairs of the "Moreover" clause; the type-(i) diagonals are tested in the same run and reported separately, and in the full replays none of them is long ("long type-(i) diagonals: none"). The proof does not need that supplementary result, since Lemma 5.14 excludes diagonals geometrically.

*Parameter domains.* The continuous parameter $n$ (and $m$ in (c)) is covered by compact charts that include the limits; the bounded families of (d) are generated at the 36 integer slices $(m,3,7)$, $17 \leq m \leq 41$; $(m,3,8)$, $17 \leq m \leq 23$; $(17,3,9)$; $(m,4,5)$, $17 \leq m \leq 19$, and are *not* certified for non-integer $m$. Integer $m$ suffices for polytopes. Both implementations enforce these 36 integer slices and the seven integer points $n = m$ (fractional or out-of-range $m$ is refused before any arithmetic, and the checker's output records the parameter domain of every row).

Finite geometric realizations and the chart. We give the correspondence for the exact finite input `Composition.Wedge.RealizesWFan`; it is not an assumption that a geometric fan already has the program's coordinates. Translate the root to 0. Write its normal and offset as $u, k$. The definition gives $|u| = 1$ and $0 < k < 1$, and every fan vertex satisfies $|p| = 1$, $p \cdot u = k$. Set $c = u/(2k)$ and

$R = 1/(2k)$. On the unit sphere, the latter scalar-product equation is equivalent to $|p - c| = R$. A supporting functional $g$ in `SupportingHalfspace` is nonzero, has kernel exactly the span of the root and the whole specified fan facet, is nonpositive on every fan vertex, and satisfies $g(u) < 0$. These are the properties used below.

Put $s = \sin(\pi/n)$ and $C = \cos(\pi/n)$. For finite $n \geq 17$, $s, C > 0$. The unit vectors $a, b$, with $|a - b| = 2C$, are independent, and an orthogonal coordinate change gives

$$a = (s, C, 0, 0, 0), \qquad b = (s, -C, 0, 0, 0), \qquad L = \operatorname{span}(e_1, e_2).$$

Their sphere equations give $c_1 = 1/(2s)$ and $c_2 = 0$. All fan-facet supports vanish on $L$. We write $c = e_1/(2s) + \gamma$, $\gamma \in L^\perp$.

*Type (i).* The defining distances are $|x_i| = 1$, $|a - x_i| = |b - x_i| = \sqrt{2}$, and $|x_i - x_{i+1}| = \mathrm{ch}_{q_i}$. The first three give $x_i \cdot a = x_i \cdot b = 0$, so the lateral points lie in $L^\perp$. Their affine plane has dimension two and lies in $\gamma \cdot x = 1/2$; in particular it misses 0. Thus, for some unit axis $z$ and $d > 0$, this plane is $z \cdot x = d$, the circle center is $dz$, the circle radius is $S = \sqrt{1 - d^2}$, and $\gamma = z/(2d)$. The exact kernel of the $i$th support is $L + \operatorname{span}(x_i, x_{i+1})$. Its restriction to the lateral plane is the supporting line of that polygon edge. Moreover $g(c) = g(u)/(2k) < 0$ implies $g(dz) < 0$. Hence the circle center lies strictly inside every edge half-plane. The defining `vertices`, `injective` and `cyclic_edges` conditions say that all polygon vertices are traversed once by polygon edges; their order is the convex boundary order or its reversal. Consequently every successive circle arc is a minor arc, and

$$\sum_i 2 \arcsin\bigl(\mathrm{ch}_{q_i}/(2S)\bigr) = 2\pi.$$

The strictly decreasing closure equation fixes $S$ uniquely (Lemma 5.5). A rotation and, if needed, reflection of the lateral plane now gives exactly `FanTypeI`'s coordinates. The explicit lateral-cone argument below proves the nonnegative axial coordinate used by `FanTypeI.free_box`; the other two coordinates lie in $[-1, 1]$ by the unit norm.

*The common trapezoid for types (ii) and (iii).* Here the six prescribed distances are $|a - b| = 2C$, $|x - y| = |a - x| = |b - y| = 1$, and $|a - y|^2 = |b - x|^2 = 1 + 2C$, together with the four unit norms. Taking scalar products determines

$$d = \frac{s}{2(1 + C)}, \qquad x = de_1 + \tfrac{1}{2}e_2 + he_3, \qquad y = de_1 - \tfrac{1}{2}e_2 + he_3, \qquad h = \sqrt{1 - \frac{1}{2(1 + C)}} > 0.$$

Indeed the first two coordinates follow from the products with $a, b$; $|x - y| = 1$ makes the two transverse residuals equal, and their norm is $h$. This orients $e_3$ toward the far edge, explicitly $e_3 = (x + y - 2de_1)/(2h)$, and gives $x - y = (a - b)/(2C)$. Thus the trapezoid is affinely planar, whereas the span of its vertices together with the root is the three-dimensional space $M = L + \mathbb{R}e_3$. There is no assertion that the trapezoid's affine plane has dimension three. Every apex $z_j$ is unit and has distance $\sqrt{2}$ from each of $a, b, x, y$. Its scalar products with all four therefore vanish; hence $z_j \perp M$, including $z_j \perp e_3$.

In type (ii), $|z_1 - z_2| = \mathrm{ch}_q$, with finite $q \geq 3$, fixes two independent unit vectors in the two-dimensional space $M^\perp$. Up to an orthogonal coordinate change they are $z_1 = (0, 0, 0, \sin(\pi/q), \cos(\pi/q))$ and $z_2 = (0, 0, 0, \sin(\pi/q), -\cos(\pi/q))$. Their equations and that of $x$ fix $\gamma = (h/2)e_3 + r_q e_4$, precisely `FanTypeII`. The whole facet through $a, b, z_1, z_2$ has kernel $L + M^\perp$; since $x$ lies on its inner side and has positive $e_3$ coordinate, that side is $p_3 \geq 0$. This proves the first-coordinate restriction $[0, 1]$ in its lateral free box. Neither an extra apex orientation nor a reflected trapezoid produces an omitted shape: these change only the orthogonal frame or exchange the two apex labels.

*The branch in type (iii).* The extra point $w$ is unit and satisfies $|w - a| = |w - b| = \sqrt{2}$, so $w \perp L$. The apex pair is distinct by definition and is not antipodal, since $u \cdot z_1 = u \cdot z_2 = k > 0$. Thus $e_3, z_1, z_2$ form a basis of $L^\perp$; write $w = \alpha e_3 + \beta z_1 + \eta z_2$. For the pyramid containing $z_1$, the exact

supporting kernel is $L + \operatorname{span}(e_3, z_1)$. The other apex is outside this kernel and on the nonpositive side, so its value is strictly negative. Applying this functional to $w$ gives $\eta \geq 0$; the other pyramid gives $\beta \geq 0$. For the support through $a, b, z_1, w$, denote the functional by $g$. Because $g$ vanishes on $L$, $g(e_3) = g(x)/h \leq 0$; this deduction uses the actual vertex $x$, not an assumption that the direction $e_3$ is a vertex. Also $g(z_2) \leq 0$. If $\alpha > 0$ and $\eta > 0$, then $0 = g(w) = \alpha g(e_3) + \eta g(z_2)$ forces both values to vanish. Together with $g|_L = 0$ and $g(z_1) = 0$, this makes $g = 0$, a contradiction. Thus $\alpha > 0$ implies $\eta = 0$, and the support through $a, b, w, z_2$ similarly implies $\beta = 0$. Unit norm would then give $w = e_3$. But $q_1, q_2$ are the two duoprism polygon parameters, $|w - z_j| = \mathrm{ch}_{q_j}$, and their unordered pair is $(3,3)$, $(3,4)$ or $(3,5)$. At least one required distance is $\mathrm{ch}_3 = 1$, whereas $|e_3 - z_j| = \sqrt{2}$; this is impossible. If $\alpha = 0$, the sphere equation gives $\beta + \eta = 1$. A convex combination of two distinct unit vectors has unit norm only at an endpoint, so $w$ would equal an apex, contradicting its positive prescribed chord distance. We have proved $\alpha < 0$.

For completeness, the parameter domain and endpoint exclusions in `FanTypeIII` can be read directly from this argument. Choose coordinates in $M^\perp$ such that

$$z_1 = (0, t, \sqrt{1-t^2}), \quad z_2 = (0, t, -\sqrt{1-t^2}) \quad \text{in } L^\perp, \qquad 0 < t < 1,$$

and put $b_j = -\cos(2\pi/q_j)$. The two chord equations and the proved negative sign give

$$w_4 = \frac{b_1 + b_2}{2t}, \qquad w_5 = \frac{b_1 - b_2}{2\sqrt{1-t^2}}, \qquad w_3 = -\sqrt{1 - w_4^2 - w_5^2} < 0.$$

The center is $c = e_1/(2s) + (h/2)e_3 + e_4/(2t)$, so $w_4/t - 1 = h|w_3|$. Thus the height, and then $C = 1/(2(1-h^2)) - 1$, recover the original $n$. For each of the three pairs the constructor certifies $(b_1 + b_2)2^{19} > 1$, excluding $t \leq 2^{-20}$ by $w_4 > 1$. When $q_1 \neq q_2$, it certifies $|b_1 - b_2|/(2\sqrt{2\,2^{-40}}) > 1$, excluding $t \geq 1 - 2^{-40}$ by $|w_5| > 1$. For $(q_1, q_2) = (3,3)$ the same upper interval gives $w_4/t - 1 = 1/(2t^2) - 1 < 0$, contrary to $h|w_3| > 0$. The search covers all of $[2^{-20}, 1 - 2^{-40}]$, retaining every box compatible with the height interval $[h(17), \sqrt{3}/2]$ and the unit-norm equations. It selects no individual root of a squared polynomial. Every geometric root therefore occurs in the searched chart; reversing $q_1, q_2$ is an isometric relabeling.

*Parameters, free points and supporting inequalities.* For the one-parameter type-(i) and type-(ii) charts the map $n \mapsto s = \sin(\pi/n)$ covers exactly $(0, \sin(\pi/17)]$; its continuous limit $s = 0$ is included in the checked box. Given lateral coordinates $\xi$, every figure-sphere point has

$$p_1 = s(1 - 2\gamma \cdot \xi), \qquad p_2 = \pm\sqrt{1 - p_1^2 - |\xi|^2}.$$

Both signs are searched, and the unit-norm bounds and the justified nonnegative-coordinate restrictions give the stated free boxes. Type (iii) uses the full lateral cube $[-1,1]^3$.

For the four two-parameter charts, the code labels `44`, `333`, `36`, `ii` mean, respectively, $(m,4,4)$, $(m,3,3,3)$, $(m,3,6)$ and type (ii) with $q = m$. Their explicit points in `FanFamily.structure` satisfy the same unit norms, chords and center equations; the uniqueness arguments above identify them with the finite geometric fans. Put $\sigma = \sin(\pi/m)$, $\tau = \max(s, \sigma)$ and

$$\vartheta = \begin{cases} s/\sigma & s \leq \sigma, \\ 2 - \sigma/s & s \geq \sigma, \end{cases} \qquad \lambda = \min(1, \vartheta), \quad \omega = \min(1, 2 - \vartheta).$$

Then $s = \lambda\tau$, $\sigma = \omega\tau$, and every finite pair $n, m \geq 17$ has $0 < \tau \leq \sin(\pi/17)$, $0 < \vartheta < 2$ and $\lambda, \omega > 0$. Writing $\gamma = e/(2\sigma) + \gamma_0$ and $\xi = \omega\Pi e + f$, the sphere equation becomes $p_1 = \lambda(\tau(1 - 2\gamma_0 \cdot f) - \Pi)$. The orthogonal decomposition and all four point lists are given below, where an elementary rational bound proves that the searched box $[-1.5, 1.5] \times [-1,1]^2$ contains every admissible point. Dividing facet functions by $\omega$ or $\sigma$ is legitimate for finite parameters; the continuous extensions are established below.

For every chart, its facet functions vanish on exactly the prescribed whole-facet spans. Their orientations are fixed by the inner sides: for types (iii) and the four families, `check_orient_iii` and

`check_orient_family` certify the off-facet vertex signs on interval boxes covering the full parameter domain; an initial sampled orientation is only a proposal. The exact equality of the support set in `Realizes` means there is no additional choice of supporting halfspace. Thus every checker facet inequality is implied by geometric admissibility. Its remaining feasibility inequalities are precisely the nonnegative square-root radicand and $p\cdot v \le 1/2$ for each unit fan vertex $v$, the latter equivalent to $|p-v| \ge 1$. No direction-test inequality is imposed as a feasibility constraint. Tests V and P prove, respectively, the admissible–vertex and admissible–admissible distance bounds; test F proves the required fan-vertex bounds. The reflection $p_2 \mapsto -p_2$ preserves the fan set and its inequalities, exchanging $a, b$ and, where present, $x, y$. Together with exchanging the pair of points, it reduces the four sign pairs to the two searched pairs $(+,+)$ and $(+,-)$. The auxiliary D test is not an acceptance requirement.

Finally, the explicit full inventory is $70+14+3+36+4 = 127$: the P1-retained unpaired type-(i) configurations, $q = 3, \dots, 16$ in type (ii), the three type-(iii) pairs, the 36 bounded integer slices, and the four two-parameter families just named. The seven exceptional slices are accepted only with their additional $n = m$ runs and the recomputed partner-radius singleton. The other bounded slices keep the whole real $n \ge 17$ domain. The compact boundary charts enlarge the finite domains for certification; they do not assert additional ambient infinite-radius instances of `RealizesWFan`.

Admissible points, allowed vertices, and the exact tests. These are different domains. In `lean/Uniform5/CompositionWedgeInput.lean`, `Geometry.Admissible` requires the figure-sphere equations, membership in every fan halfspace, and distance at least 1 from *every* fan vertex. Such a point cannot itself be a fan vertex. By contrast, `Geometry.Allowed` means exactly a fan vertex outside the distinguished endpoint set; it includes no separation condition. Indeed an allowed vertex has distance zero from itself. The statement `W.no_long_pair` says that if each of $p, q$ is admissible *or* allowed, and $|p-q| \ge \mathrm{ch}_{17}$, then both belong to the fan's vertex set. It does not assert that every such pair is short.

Put $\kappa = \cos(2\pi/17)$. For unit points, $|p-q| \ge \mathrm{ch}_{17}$ is equivalent to $p\cdot q \le -\kappa$. The checker implements the statement by cases. P proves $p\cdot q > -\kappa$ for two admissible points; V proves this for one admissible point and each fixed fan vertex, hence in particular for an allowed vertex. When both points are allowed, the conclusion of `no_long_pair` already holds, and neither is submitted to the admissible-point search. Its feasibility tests $p\cdot v \le 1/2$ for all fan vertices $v$ are therefore justified exactly on the domain where they are used. P may also certify the equivalent strict bound $|p+q|^2 > 2-2\kappa$. The separate field `W.fan_distances` requires $|p-q| < \mathrm{ch}_{17}$ for fan vertices whose ordered pair is not `specialPair`. The special pairs are the large edge, its partner when present, and the type-(i) diagonals. F tests all distinct nonspecial pairs; coincident points have distance zero. Diagonal tests are reported but are not required for F acceptance. These are precisely the V/P/F obligations constructed by `checks/check_caseb/inventory.py`, `obligations`, and checked by `summary.validate_records`. The auxiliary D test is neither a feasibility condition nor a required acceptance field.

The lateral cone and coordinate conventions. For type (i) let $N$ be the number of lateral vertices; `TypeIShape` requires $3 \le N \le 5$, affine dimension two, injectivity, exactly these hull vertices, and a hull edge between every successive pair. The boundary graph of a convex polygon is a cycle. An injective cyclic listing of all its vertices using only hull edges therefore follows that boundary once, in one of its two orientations. It cannot wind repeatedly or use a star-polygon ordering. To make the homogenization argument explicit, write the lateral plane as $z\cdot\xi = d > 0$, with its circle center $dz$, and write its complete edge inequalities in the transverse plane as $a_i\cdot y \le b_i$, $b_i > 0$. Take $a_i$ to be unit outward edge normals and $\ell_i > 0$ the corresponding edge lengths. Closure of the polygon gives $\sum_i \ell_i a_i = 0$. The homogeneous fan inequalities are $a_i\cdot y \le (b_i/d)t$, where $t = z\cdot\xi$. Summing gives $0 \le t\sum_i \ell_i b_i/d$, hence $t \ge 0$. At $t = 0$ all inequalities are equalities, and the normals span the transverse plane, so $y = 0$. For $t > 0$, the inequalities say exactly $dy/t$ belongs to the polygon. Thus the halfspaces give its conical hull and exclude negative axial coordinates. In `FanTypeI`, this

axis is the *third lateral coordinate*, i.e. global coordinate $p_5$ in the displayed five-coordinate chart. In `FanTypeII`, the antiprism axis $e_3$ is the *first lateral coordinate*, global $p_3$. These are the respective $[0,1]$ restrictions in the two free boxes.

For type (ii), write the lateral coordinates as $(\xi_3,\xi_4,\xi_5)$ and put $s_q = \sin(\pi/q) > 0$, $C_q = \cos(\pi/q) > 0$. The three inner-side inequalities in `FanTypeII.evaluate` are

$$C_q\xi_4 - s_q\xi_5 \geq 0, \qquad C_q\xi_4 + s_q\xi_5 \geq 0, \qquad \xi_3 \geq 0.$$

The first vanishes on the pyramid with apex $z_1 = (0, s_q, C_q)$ and has value $2s_qC_q > 0$ at $z_2$; the second has the reversed roles. The third vanishes on the tetrahedron with both apices and has value $h > 0$ at $x, y$. Their kernels and inner orientations thus equal the three whole-facet supports required by `Realizes`. For type (iii), the four raw normal labels are $e_3 \times z_1$, $e_3 \times z_2$, $z_1 \times w$, $w \times z_2$; `FanTypeIII.orient` supplies their signs, certified throughout the retained parameter interval by `check_orient_iii`. For the four families the labels are the ordered vertex pairs in each branch of `FanFamily.structure`; their signs are certified by `check_orient_family`. These checks use interval boxes over the full domain, not just the sample proposing the signs. For the equal type-(iii) pair $(3,3)$, the omitted upper endpoint interval $t \geq 1-2^{-40}$ satisfies $t > 1/\sqrt{2}$ and hence $1/(2t^2)-1 < 0$, whereas the sphere equation requires this quantity to be $h|w_3| > 0$. This is the explicit upper-end exclusion for that pair.

The four family charts and their bounds. Here is the meaning of the coordinates used in `checks/check_caseb/fans.py`, `FanFamily.structure`, `vertices`, `facet_value`, and `evaluate`. In $L^\perp$, let $e$ be the unit direction of the midpoint of the partner edge: $(x_1+x_2)/(2\sigma)$ in the three type-(i) families, and $(z_1+z_2)/(2\sigma)$ in type (ii), where $\sigma = \sin(\pi/m) > 0$. Let $E = L^\perp \cap e^\perp$ and choose the orthonormal ordered basis of $E$ used below. Thus $E$ is a two-dimensional Euclidean space. The center component is $\gamma = e/(2\sigma) + \gamma_0$ with $\gamma_0 \in E$. A free lateral vector is written $\xi = \omega\Pi e + f$, where $f = f_1e_1' + f_2e_2' \in E$; $\Pi$ is a real coordinate, not an additional geometric direction. Write $C_m = \sqrt{1-\sigma^2}$, $h_j = \sqrt{1-1/(2(1+\cos(\pi/j)))}$, $d_m = \sqrt{1-1/(4C_m^2)}$ and $\chi = 1/(2(1+C_m))$. In the notation $(\xi\cdot e; f_1, f_2)$, the lateral vertices are For label 44 the vertices and center component are

$$(\sigma; C_m, 0),\ (\sigma; -C_m, 0),\ (0; 0, 1), \qquad \gamma_0 = (0, 1/2).$$

For label 333 they are

$$(\sigma; C_m, 0),\ (\sigma; -C_m, 0),\ (\sigma\chi; -1/2, h_m),\ (\sigma\chi; 1/2, h_m),$$
$$\gamma_0 = (0, h_m/2).$$

For label 36 they are

$$(\sigma; C_m, 0),\ (\sigma; -C_m, 0),\ (0; -1/(2C_m), d_m), \qquad \gamma_0 = (0, 1/(2d_m)).$$

For label ii the apices and center component are

$$z_1 = (\sigma; 0, C_m), \quad z_2 = (\sigma; 0, -C_m), \qquad \gamma_0 = (h_n/2, 0).$$

For label ii the remaining vertices $x, y$ have lateral component $(0; h_n, 0)$ and $L$ components $(d, \pm 1/2)$ from the trapezoid formula. Thus $e_1'$ is the antiprism axis for that label. Direct substitution gives $|\gamma_0| = 1/2$ for 44, $|\gamma_0| = h_m/2 \leq \sqrt{3}/4$ for 333, $|\gamma_0| = h_n/2 \leq \sqrt{3}/4$ for ii, and $|\gamma_0| = 1/(2d_m)$ for 36.

For an explicit rational common bound, use $\tau \leq \sin(\pi/17) < 23/125$. For example the last strict inequality follows from $\pi < 22/7$ and $\sin x \leq x - x^3/6 + x^5/120$ at $x = 22/119$. It gives $C_m^2 > 15096/15625$ and therefore, in family 36,

$$|\gamma_0|^2 < \frac{15096}{44759} < \frac{625}{1849}, \qquad |\gamma_0| < \frac{25}{43}.$$

The other three displayed bounds are smaller. For $\vartheta \le 1$, $\omega = 1$, so $|\Pi| = |\xi \cdot e| \le 1$. For $\vartheta \ge 1$, $\lambda = 1$ and the sphere equation gives

$$|\Pi| \le |p_1| + \tau(1 + 2|\gamma_0||f|) < 1 + \frac{23}{125}\left(1 + \frac{50}{43}\right) = \frac{7514}{5375} < \frac{141}{100}.$$

Here $|p_1|, |f| \le 1$ follows from the unit norm and the orthogonal coordinate decomposition. This proves the full free-box coverage; `certify_family_chart` checks the corresponding bounds on the live formulas before searching.

The rescaled expressions have no singular divisions at compact chart boundaries. To see this algebraically, for a facet pair write its lateral vertices as $u_A = (\sigma\chi_A, a_1, a_2)$ and $u_B = (\sigma\chi_B, b_1, b_2)$. Put $N_e = a_1b_2 - a_2b_1$ and $N_t = (\chi_B a_2 - \chi_A b_2, \chi_A b_1 - \chi_B a_1)$. Then, since $\sigma = \omega\tau$,

$$(u_A \times u_B) \cdot \xi = \omega\big(N_e\Pi + \tau N_t \cdot f\big).$$

For the partner facet $N_e = 0$, and dividing instead by $\sigma$ gives $N_t \cdot f$. These are exactly the two branches of `facet_value`. After these divisions, legitimate for every finite parameter pair, the displayed formulas extend continuously to $\tau = 0$ or $\vartheta = 0, 2$. The quantities $C_m, h_m, d_m, \chi, \gamma_0$ also extend continuously: their displayed denominators and square-root arguments stay bounded away from zero on the compact domain. Point coordinates, $p_1 = \lambda(\tau(1 - 2\gamma_0 \cdot f) - \Pi)$, $\xi = \omega\Pi e + f$, and the nonnegative square-root branches for $p_2$ contain no division by a vanishing parameter. Hence all unit-point dot products and squared distances used by V/P/F extend continuously on their feasible domains. Compact boundary points are only an enlargement used for bounds, not extra finite ambient fan instances.

Unpaired families and bounded slices. A fan about a large $n$-gon is called paired when one of its facets contains a second large polygon through the vertex. In type (i) this means an entry $m \ge 17$ in the configuration; in type (ii) it means $q = m \ge 17$. Otherwise the fan is unpaired. The term "family polygon" in P1 has a different, explicit meaning: its size belongs to $\{7, 9, 11, 12, 13, 14, 15, 16\}$. These are all at most 16, even though the lower-dimensional classification forces product facets at them. Among the 137 closable small configurations, 83 contain such a size, 67 are excluded, and the other 16 are $(4, 4, q)$ and $(3, 3, 3, q)$ for those eight $q$ values. They remain *unpaired* about the large $n$-gon. Together with the 54 closable configurations without a family polygon, they give the 70 unpaired rows in the W inventory.

The 36 bounded paired rows are precisely $(m, 3, 7)$ for $17 \le m \le 41$ (25 rows), $(m, 3, 8)$ for $17 \le m \le 23$ (7), $(17, 3, 9)$ (1), and $(m, 4, 5)$ for $17 \le m \le 19$ (3). Each is constructed by `FanBounded(m,rest)`, a `FanTypeI` chart with the listed integer $m$ fixed and the whole continuous $s = \sin(\pi/n)$ domain retained. Only $(38, 3, 7), \ldots, (41, 3, 7), (23, 3, 8), (17, 3, 9), (19, 4, 5)$ require the separate `restrict_n(m)` chart. They are accepted at $n = m$ only after `summary.validate_records` recomputes the partner filter's matching singleton. No continuous extension in the bounded parameter $m$, and no unrestricted conclusion for those seven rows, is asserted.

*The seven exceptional fans.* Over the full range of $n$, (W) fails for $(m, 3, 7)$, $m = 38, \ldots, 41$, $(23, 3, 8)$, $(17, 3, 9)$, $(19, 4, 5)$, with rigorous interval witnesses at large finite $n$ (e.g. $n = 547$ for $(41, 3, 7)$ and $(19, 4, 5)$), and also for the (P1)-excluded configurations $(3, 11, 13)$, $(4, 7, 9)$, $(5, 6, 7)$. For the seven paired fans, the partner $m$-gon's own cycle restricts $n$:

**Lemma 5.13** (Partner-radius filter)**.** *If the fan of $F$ (an $n$-gon) is paired of type (i) with configuration $(m, c_1)$, $c_1$ one of $(3, 7)$, $(3, 8)$, $(3, 9)$, $(4, 5)$, then the fan of the partner $m$-gon is of type (i) with configuration $(n, c_2)$ and $\delta_{c_1}(m) = \delta_{c_2}(n)$, or of type (ii) with $\delta_{c_1}(m) = h_A(m)^2/4$. For the seven exceptional fans this forces $c_2 = c_1$ and $n = m$.*

*Proof.* From $F$: $R^2 = r_n^2 + R^2_{(m,c_1)} = r_n^2 + r_m^2 + \delta_{c_1}(m)$. The partner's cycle contains $\{m\} \times \{n\}$, so it is of type (i) with a configuration $(n, c_2)$ or of type (ii) with $q = n$ (type (iii) needs parameters at most 5); this gives $R^2 = r_m^2 + r_n^2 + \delta_{c_2}(n)$ or $R^2 = R^2_{A_m} + r_n^2 = r_m^2 + h_A(m)^2/4 + r_n^2$. For the seven

fans $\delta_{c_1}(m) > 99$ (from 99.67 for $(17,3,9)$ to 1040.7 for $(41,3,7)$), while $h_A^2/4 < 1/4$, $\delta_{(4,4)} = 1/4$, $\delta_{(3,3,3)} < 3/16$, $\delta_{(3,6)} \le 0.3373$, so $c_2$ is one of the bounded families, and a certified comparison of the finitely many values (`checks/check_caseb/partner.py` and `wpoint.py`) leaves only $c_2 = c_1$, $n = m$. □

*What is computed.* The first implementation (`caseb_wedge`, Arb): with $\kappa = 1/(2R)$ and $\hat{c} = c/|c|$ (coordinates with $v = 0$), a VF-vertex is $p = \kappa\hat{c} + \sqrt{1-\kappa^2}\sum_j s_j f_j$ with $s \in S^3$ and $(f_j)$ an orthonormal basis of $\hat{c}^\perp$; then $|p-p'| \ge L$ iff $|s+s'|^2 \le 4 - L^2/r^2$, the separation constraints are linear in $s$, and each fan facet is a linear inequality. A dual-tree branch-and-bound over dyadic $s$-boxes in $[-1,1]^4$ discards box pairs on which $\min|s+s'|^2 > 4 - L^2/r^2$ is certified, or a constraint certainly fails, and tests allowed fan vertices against single boxes. Parameters ($\theta = \pi/n \in [0, \pi/17]$; for unbounded paired families polar parameters $(\beta, \rho)$ continuous at $n = m = \infty$; the facet inequalities of these families are rescaled by the vanishing factor so that they do not degenerate at $m = \infty$) are split until certified. Type (iii) roots $\phi$ are enclosed for every $\theta$-box by interval bisection and Newton steps. Result (228 fans): 184 `CERTIFIED`, 31 `NOT CLOSABLE`, 3 `DEGENERATE`, 10 `FAILED` (the seven and the three above, each with a floating witness above $\mathrm{ch}_{17}$); the seven certify at $n = m$ (`wpoint.py`). Independent checker `checks/check_caseb/` (own outward-rounded binary64 interval arithmetic, constants from `mpmath.iv` at 200 bits; a different chart: $\xi \in \mathbb{R}^3$ for the $L^\perp$ part, $p_1$ from the VF equation, $p_2 = \pm\sqrt{\cdot}$ on two branches; tests V (no long edge from an admissible point to any fan vertex, including $a$, $b$ and the partner ends), P (no long admissible pair, via $\sum_i (p_i + p_i')^2 > 2 - 2\cos(2\pi/17)$), F (fan-vertex pairs); a piecewise linear chart for the unbounded families; type (iii) parametrized by $\cos(\phi/2) \in [2^{-20}, 1 - 2^{-40}]$ with $n$ recovered from it, the two omitted ends excluded by certified inequalities, without root finding): 120 fans certified over the full range, the same 7 failing (with rigorous interval witnesses at integer $n$), all 7 certified at $n = m$; the 67 (P1)-excluded configurations: 64 certified, $(3,11,13)$, $(4,7,9)$, $(5,6,7)$ fail with rigorous witnesses. The verdicts agree fan by fan between the two implementations. Corruptions: in the checker, dropping the separation constraints makes $(4,4,7)$ fail; in the Arb implementation, dropping them makes $(4,6,10)$ fail, lowering the threshold below a fan's float maximum makes it fail, and dropping the fan-facet constraints makes type (ii) $q = 5$ fail. The interval searches of both implementations are rigorous; the independent checker is the one that tests every fan vertex directly.

*Completeness of computed results.* The Arb implementation requires every listed squared fan-vertex distance to be strictly below the threshold in Arb before certifying (equality or uncertainty splits the box or fails), and converts Arb endpoints to binary64 in the outward direction at every precision; the independent summary requires an explicit inventory (full, supplementary or quick), rejects duplicates, omissions, extra fans, wrong domains and missing or pending V/P/F tests, and accepts an exceptional fan's point $n = m$ only with the partner filter's singleton recomputed. Corruption tests exercise these requirements. The full replay checks 228 Arb records, partitioned as $127 + 67 + 31 + 3$: required fans, P1-excluded supplementary fans, nonclosable rows and degenerate rows. The 31 nonclosable rows are 28 bounded type-i configurations and three paired nonclosable families. The outcomes are 184 `CERTIFIED`, 31 `NOT CLOSABLE`, 3 `DEGENERATE`, 10 expected `FAILED`, with seven point repairs; independent 127 full-domain fans (120 certified, 7 expected failures), seven certified points. The independent supplementary 64-of-67 result and 1,642-test diagnostic summary are recorded supplementary evidence; full mode does not rerun that supplementary inventory. The primary 228-row replay does include those supplementary configurations. So both implementations prove the whole of Computation 5.12, including the "Moreover" clause; the pairs with an end at $a$, $b$ or a partner end are still left by the Arb implementation to Lemma 5.4 and Fact 5.2.

*The chart box of the unbounded families.* In the checker's chart for the unbounded paired families, with $s = \sin(\pi/n)$, $\sigma = \sin(\pi/m)$, $\tau = \max(s, \sigma)$, $\lambda = \min(1, \vartheta)$, $\omega = \min(1, 2-\vartheta)$, $s = \lambda\tau$, $\sigma = \omega\tau$ ($0 \le \vartheta \le 2$), the free coordinate $\Pi$ is searched in $[-1.5, 1.5]$. This box contains every admissible

point: the rescaled VF equation is $p_1 = \lambda(\tau(1 - 2\gamma_0 \cdot f) - \Pi)$, and a unit point has $|p_1|, |f| \le 1$; for $\vartheta \le 1$, $|\Pi| = |\xi_e| \le 1$; for $\vartheta \ge 1$, $\lambda = 1$ and $|\Pi| \le 1 + \tau(1 + 2|\gamma_0|) < 7514/5375 < 1.41$, using $|\gamma_0| < 25/43$ for all four families and $\tau \le \sin(\pi/17) < 23/125$. No fan-facet or separation constraint is used, and the bound holds at the endpoint charts $\vartheta = 0, 2$ and $\tau = 0$ without division by zero. The certificate rejects four corrupted certificates, and an explicit admissible point with $\Pi = 33/32$ shows that narrowing the box to $[-1, 1]$ would lose coverage. The live family construction checks the bound before using the box.

**Lemma 5.14.** *Let $F$ be a large 2-face at $v$ whose fan is in the list of Computation 5.12 (with the actual $n$, $m$ of $P$). Then the only long VF-edges with an end among the fan vertices or an admissible point are $e$ and, for a paired fan, the partner edge. In particular every long VF-edge is $e$ or the partner edge.*

*Proof.* Let $[p, p']$ be a long VF-edge other than $e$ and the partner edge. Its ends are VF-vertices, each a fan vertex or admissible. Neither end is $a$, $b$ or a partner end, by Lemma 5.4 (those are already ends of a long edge). So both are admissible or allowed fan vertices, and Computation 5.12 leaves only the case that both are fan vertices. Pairs of fan vertices other than diagonals are shorter than $\mathrm{ch}_{17}$ (Computation 5.12, last part; pairs with $a$ or $b$ have the explicit lengths $\sqrt{2}$, 1 or $\sqrt{1 + \mathrm{ch}_n} < \sqrt{3}$ of Fact 5.2). A diagonal $x_i x_j$ of a type-(i) fan is not a VF-edge: the $x_i$ satisfy $(x_i - v) \perp L$, $|x_i - v| = 1$ and lie on the VF hyperplane, so they lie on a circle; the fan edges $x_i x_{i+1}$ are VF-edges, so the circle order is the fan order (otherwise two fan edges would cross in their relative interiors, and a face containing the crossing point would contain all four ends); for $k \ge 4$ a diagonal crosses another diagonal in their relative interiors, so a face containing it would contain four vertices; for $k = 3$ there are no diagonals. Since every VF-vertex is a fan vertex or admissible, the last claim follows. □

*Remark* 5.15. The Arb certificate of (W) does not test pairs in which one point is $a$, $b$ or a partner end, and Lemma 5.4 supplies exactly those cases. The independent checker tests them directly (test V over all fan vertices) and certifies them, so (W) as used here holds with or without the lemma.

5.6.3. *The reduction to two paired 2-faces.*

**Proposition 5.16.** *If $s \ge 2$, then $s = 2$ and the two large 2-faces $F_1$ ($n$-gon) and $F_2$ ($m$-gon) at $v$ lie in a common facet $D = \{n\} \times \{m\}$.*

*Proof.* Let $F$ be a large 2-face at $v$. If its fan is unpaired: a type-(i) configuration excluded by (P1) cannot occur (Computation 5.11); every other unpaired fan is in the list of Computation 5.12, and Lemma 5.14 shows that $e$ is the only long VF-edge, so $s = 1$, a contradiction. So every large 2-face at $v$ has a paired fan. By Lemma 5.13 the paired fan of $F$ is in the list of Computation 5.12 at the actual $n$, $m$, and Lemma 5.14 shows that the long VF-edges are exactly $e$ and the partner edge. So $s = 2$, and the two large 2-faces lie in the fan facet $\{n\} \times \{m\}$ that makes the fan paired. □

The paired type-(i) configurations not in the list of Computation 5.12, $(m, 3, 3)$, $(m, 3, 4)$, $(m, 3, 5)$, are not closable. The closability criterion of Lemma 5.5 excludes $(3, 5)$: a spherical triangle of rays exists for $(n, 3, 5)$, $n < 30$, but the center lies outside it. The paired families are covered by Computation 5.12(d) and Lemma 5.13.

5.6.4. *The paired theorem.*

**Theorem 5.17.** *If every vertex lies in exactly two large 2-faces $F_1$ ($n$-gon) and $F_2$ ($m$-gon), and they lie in a common facet $D = \{n\} \times \{m\}$, then $P$ is $\{n\} \times \{m\} \times \{\}$, $A_m \times \{n\}$ or $A_n \times \{m\}$.*

*Proof.* Let $e_1 = [a, b]$ and $e_2 = [c, d]$ be the VF-edges of $F_1$, $F_2$; then $\mathrm{VF}(D) = \mathrm{conv}(a, b, c, d)$. By Corollary 5.9 every VF-edge lies in $\mathrm{fan}(e_1)$ or $\mathrm{fan}(e_2)$, so every VF-vertex is a vertex of one of the two fans. The fan of $F_1$ is of type (i) with configuration $(m, q_2, \ldots, q_k)$, $x_1 = c$, $x_2 = d$,

or of type (ii) with apexes $\{z_1, z_2\} = \{c, d\}$ (as $T = \mathrm{VF}(D)$); likewise for $F_2$ with $n$. The angle sum allows only $k = 3$, or $k = 4$ with configuration $(m, 3, 3, 3)$ (the configurations $(m, 4, p, p')$ and $(m, 3, p, p') \neq (m, 3, 3, 3)$ exceed $2\pi$).

*Simple-vertex step.* A vertex $z$ of $\mathrm{fan}(e_1)$ not in $\mathrm{fan}(e_2)$ has all its VF-edges in $\mathrm{fan}(e_1)$. If it has degree 4, each 3-subset of its edges spans a VF-facet (Lemma 2.10), and a VF-facet containing $c$ and $d$ contains the edge $e_2$ (Lemma 2.9), hence belongs to $\mathrm{fan}(e_2)$.

*(i)/(i), $k = 3$.* $z = x_3$ has the VF-edges $za, zb, zc, zd$ in $\mathrm{fan}(e_1)$. If $z \notin \mathrm{fan}(e_2)$, the facet spanned by $\{za, zc, zd\}$ contains $e_2$, so $z \in \mathrm{fan}(e_2)$ after all. In a type-(i) fan of $e_2$ every vertex other than $c, d$ is at distance $\sqrt{2}$ from $c$ and $d$ (Fact 5.2), so $\mathrm{ch}_{q_2} = |z - d| = \sqrt{2} = |z - c| = \mathrm{ch}_{q_3}$: the configuration is $(m, 4, 4)$. The fan of $F_2$ is not of type (ii) (case (i)/(ii) with $k = 3$, below) and not of type (i) with $k = 4$ (case (i)/(i), $k = 4$, below, with the roles of $F_1$ and $F_2$ exchanged), so by the same argument it is $(n, 4, 4)$, with the same third vertex $z$. *Translate argument:* for a vertex $t$ let $D_t$ be the facet containing its two large 2-faces. For $t' \in D_t$, both large 2-faces of $D_t$ at $t'$ are large 2-faces of $P$, so $D_{t'} = D_t$. The $\{n\} \times \{4\}$ and $\{m\} \times \{4\}$ facets at the edge $vz$ contain $F_1 + \tau$ and $F_2 + \tau$, $\tau = z - v \perp \operatorname{span} D$, so $D_z = D + \tau$. Hence the vertex set is $D_0 + X$ with $X$ on a line, every point of $X$ is a vertex, so $|X| = 2$ and $P = \{n\} \times \{m\} \times \{\}$.

*(i)/(i), $k = 4$: both fans of type (i), that of $F_1$ with configuration $(m, 3, 3, 3)$.* Here $|x_3 - d| = |x_4 - c| = \mathrm{ch}_3 = 1 \neq \sqrt{2}$, so neither $x_3$ nor $x_4$ lies in a type-(i) fan of $e_2$. Then both are simple with all edges in $\mathrm{fan}(e_1)$: $x_3$ has $x_3a, x_3b, x_3d, x_3x_4$ and $x_4$ has $x_4a, x_4b, x_4x_3, x_4c$. The facet $G$ at $x_3$ spanned by $\{x_3a, x_3d, x_3x_4\}$ contains $x_4$ and $a$, hence the edge $x_4a$ (Lemma 2.9); it does not contain $b$ (else $x_3b$ would be a fourth edge of $G$ at the simple vertex $x_3$). So at $x_4$ its edges are $\{x_4a, x_4x_3, x_4c\}$, and $G$ contains $c$ and $d$, so $G \in \mathrm{fan}(e_2)$ and $x_3 \in \mathrm{fan}(e_2)$, a contradiction. Impossible.

*(i)/(ii).* $\mathrm{fan}(e_2)$ is the pair of $A_m \times \{\}$ with apexes $a, b$ and trapezoid $(c, d, y', x')$, with $|dy'| = |y'x'| = |x'c| = 1$ and diagonals $\sqrt{1 + \mathrm{ch}_m} \in [1.7222, \sqrt{3})$, strictly between $\mathrm{ch}_5$ and $\mathrm{ch}_6$, hence not a chord $\mathrm{ch}_q$ of any polygon. If $k = 3$, $x_3$ must lie in $\mathrm{fan}(e_2)$ (as above) at distances $\mathrm{ch}_{q_2}$, $\mathrm{ch}_{q_3}$ from $d$, $c$; but the vertices of $\mathrm{fan}(e_2)$ other than $a, b, c, d$ are $x', y'$, each at distance 1 from one of $c, d$ and at the diagonal distance from the other, and the diagonal is not a chord; impossible. If $k = 4$, $(m, 3, 3, 3)$: by the argument of the previous case one of $x_3, x_4$ lies in $\mathrm{fan}(e_2)$; by symmetry say $x_3$; $|x_3 - d| = 1$ forces $x_3 = y'$. If $x_4 \notin \mathrm{fan}(e_2)$, $x_4$ is simple and its facet spanned by $\{x_4a, x_4x_3, x_4c\}$ meets the pyramid $\Pi(a; c, d, y', x')$ in a face containing $a, c, y'$; since $c, y'$ are opposite vertices of the base, no proper face of the pyramid contains these three, so the facet is the pyramid and $x_4 \in \mathrm{fan}(e_2)$. Hence $\{x_3, x_4\} = \{y', x'\}$ and the neighbors of $v$ are those of $\mathrm{fan}(e_2)$. *Cell partition:* at every vertex exactly one large 2-face has a type-(ii) fan (symmetries preserve fan types; this also holds when $n = m$); let $C_t$ be its $A_m$ cell. For $t' \in C_t$ the $m$-gon of $C_t$ through $t'$ has $C_t$ in its cycle, so it is not of type (i): it is the type-(ii) 2-face at $t'$, and $C_{t'} = C_t$. The other neighbors $a, b$ of $v$ are the apexes of $\mathrm{fan}(e_2)$, so $a - v$ and $b - v$ are orthogonal to the 3-space of $C_v$ and $C_a = C_v + (a - v)$. So $P = A_m \times \operatorname{conv} X$ with $\operatorname{conv} X$ a regular polygon, here $F_1$: $P = A_m \times \{n\}$. The case (ii)/(i) gives $A_n \times \{m\}$.

*(ii)/(ii).* The trapezoid vertex $x$ of $\mathrm{fan}(e_1)$ (edges $xa, xy, xc, xd$ in $\mathrm{fan}(e_1)$, the apexes being $c, d$) must lie in $\mathrm{fan}(e_2)$ (otherwise the facet spanned by $\{xy, xc, xd\}$ contains $e_2$), so $x \in \{x', y'\}$ at distance 1 or $\sqrt{1 + \mathrm{ch}_m}$ from $c$. But $|x - c| = \sqrt{2}$, since $c$ is an apex of a prism over the $A_n$ cell containing $x$. Impossible. □

5.7. **Proof of Theorem 1.3.** Let $P$ have a large 2-face and let $s$ be the number of large 2-faces at a vertex. If $s = 1$, Theorem 5.10 gives $\{n\} \times Q$ or $A_n \times \{q\} = \{q\} \times A_n$. If $s \geq 2$, Proposition 5.16 gives $s = 2$ with both large 2-faces in one duoprism, and Theorem 5.17 gives $\{n\} \times \{m\} \times \{\}$, $\{n\} \times A_m$ or $\{m\} \times A_n$. □

5.8. **Proof of the Main Theorem.** A convex uniform 5-polytope either has all 2-faces with at most 16 sides (Theorem 1.2: one of the 628 polytopes of $\mathcal{K}_{16}$, all of type (K1)–(K3)) or has a 2-face with at least 17 sides (Theorem 1.3: of type (K3)). □

## 6. Verification and reproducibility

6.1. **Arithmetic and software.** The first implementations use ball arithmetic (Arb through python-flint [14]) at 200 bits unless stated, and exact algebra in $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$ (hand-written fields and SymPy minimal polynomials). In the proof routes (Section 6.2), every decision that cuts a branch, excludes a case or identifies two objects is either a certified inequality between enclosures, an exact algebraic identity, an exact equivalence certified by an explicit permutation, a combinatorial fact, or an equality of two vertices of one (possibly hypothetical) polytope decided by Corollary 2.3(a); an undecided case stops the computation or is reported `OPEN`, and the final runs report zero such cases. Floating-point arithmetic is used to propose (orderings, candidate correspondences, witnesses); the one remaining decision made from a double is covered by the proved coordinate lemma of Section 4.5.3. The `checks/primary/check_u3/` geometric and transcendental routines and `checks/check_caseb/` use `mpmath.iv`, exact arithmetic or outward-rounded binary64 intervals. Those routines avoid Arb, but the complete replay also calls exact model and root-audit code using Arb. Both `checks/independent-s1-b2/` and `checks/independent-s23/` use Arb and SymPy; their independence is in implementation, not arithmetic library. All public stages use the same Python environment, specified in `checks/requirements.txt`.

Exact angle decisions versus enclosures. Angles are compared exactly through algebraic cosines, positive sines and phase products, rather than by equating decimal arccosines. In stage 1, flat angle sums $\sum \alpha_i = 2\pi$ and degenerate closure roots use the exact operands of Computation 3.6 and Lemma 3.4: the non-snub squared cosines lie in $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{5})$, with positive square roots and, where needed, further algebraic number-field extensions fixed by the constructions. The phase identity $\prod_j(\cos\alpha_j + i\sin\alpha_j) = 1$ is exact; a certified interval for the sum in $(\pi, 3\pi)$ fixes its winding. In B2, polygon angle-sum inequalities $\sum(1-2/q_i) < 2$ are rational comparisons, and the three right-triangle degeneracies are identities in $\mathbb{Q}(\sqrt{5})$ (`b2check.py`, `angle_ok`, `exact_degenerate`; Lemma 5.5). General angles, strict sign/branch bounds, closure-root isolation, radius separation and continuous parameter ranges use outward-rounded Arb balls or certified intervals instead (Computations 3.3, 4.1, 4.9, 5.8, 5.11, 5.12). An interval containing zero never proves an algebraic equality; exact residuals and sign information supply those decisions.

6.2. **Computational evidence and shared dependencies.** The computations consist of exact model checks, the facet-library and stage-1 cycle checks, radius separation, the finite stage-2/3 searches, and Case B's fan and parameter-domain checks. The packaged commands and inputs are described in the bundle's `README.md`. Their paths are relative to the bundle root.

Shared inputs. Independently written programs do not give fully independent proofs of every premise. The exact-exclusion checks share the exact constructions used by the independent stage-1 library. The stage-2/3 implementations share stage-1 class intervals, facet and known-name lists, exceptional-model certificates, the separation lemma and other written geometry. The alias table and construction theorems are also shared. The implementations use common arithmetic libraries, including Arb and SymPy. P1's second implementation depends on the written angle, chord and monotonicity arguments for its finite lists and tails.

Written mathematical premises. Facts 1.4 and 1.5 are cited classification theorems, not re-proved here. Wythoff and product constructions, the geometric reductions, coverage and interpretation of the finite search, and the unbounded-parameter reductions use the written arguments in this paper. A successful executable check establishes its stated finite or interval claims; it does not replace these arguments or establish priority.

6.3. **The four-dimensional control.** A bounded control using the shared interval root solver and certified stage-2/3 engines with exact four-dimensional facet adapters completed 639 classes at family cutoff 14. It recovered all 152 known objects (185 construction names), including the grand antiprism and snub 24-cell, and refuted 33 other realizations with no unresolved, missing or extra object. All 1,476 support witnesses passed the local checker; deletion and support corruptions were rejected. This is a bounded non-proof control, separate from the public replay; it does not re-prove Möller's all-parameter classification.

6.4. **How to reproduce.** First follow the Python environment setup in the bundle's `checks/README.md`, creating the single `.venv` environment. The checks require Python 3.11 or later and Bash 4.4 or later. From the top of the bundle, run

```
PYTHON="$PWD/.venv/bin/python" bash checks/verify-public.sh full
```

Only `full` mode checks all computational claims used in the theorem; `quick` mode does not check the theorem. Full mode includes P=16 stages 2/3, all W fans, certificate authentication and corruption controls. Neither mode verifies the written reductions automatically.

## 7. What is checked in Lean

The directory `lean/` is a Lean project with dependencies specified in `lean/lean-toolchain` and `lean/lake-manifest.json`. It contains geometric lemmas and a conditional classification of genuine convex unit-edge uniform five-dimensional polytopes. The formalization is incomplete: the external numerical data and their geometric correspondence have not been jointly instantiated and discharged in Lean.

The conditional composition. With explicit hypotheses as parameters, `lean/Uniform5/CompositionMainClassification.lean` proves

$$\forall P : \mathrm{Set}(\mathrm{Space}\,5), \quad \mathrm{UnitUniform}(5, P) \implies K_1(P) \vee K_2(P) \vee K_3(P).$$

Here `UnitUniform` means a finite convex hull of full affine dimension five, ambient vertex transitivity, recursively uniform facets and unit exposed edges. The conclusions are the explicit Wythoff orbit hulls and two product families, up to intrinsic Euclidean congruence. The geometry-equivalence module connects the hull and vertex-set formulations. The composition supplies the finite fan realization and admissibility needed to apply W and combines the small-face and large-face arguments; it does not assume a Case A or Case B conclusion in place of that composition.

Assumed inputs. Both `mainClassification` and `classification_of_core_inputs` take twelve computational hypotheses and the two cited classifications, besides their raw data. The unused FILTER, RIDGES, QFOLLOW and LU hypotheses and their data parameters have been removed. Appendix A prints the twelve declarations verbatim.

Every computational input remains an assumed bound parameter in Lean. The external computations and written arguments establish the intended numerical and geometric assertions; no full external numerical search is discharged by the kernel. Checking report fields does not establish their geometric meaning or construct exact placements. Their correspondence to certificates, including coverage, canonical rigidity, refutation and forcing, remains an external semantic obligation.

The contracts distinguish coverage from existence. S1 asserts coverage and statuses without a merged-interval-width bound. S2 and U3 cover every actual figure or star, without asserting that every retained encoding is realizable. DEV is conditional on the hypothetical polytope's root, center and entire incident-facet star; it excludes that actual star, not every star having the same unlabeled figure. W is restricted to finite parameters, sufficient for actual polygonal faces, with real/integer ranges and exceptional slices explicit. The known models have separate existence arguments; the extra candidate needs no existence assertion.

Build and axiom audit. From `lean/`, run `lake exe cache get`, `lake build` and `lake env lean Audit.lean`. The audit rejects dependencies other than `propext`, `Classical.choice` and `Quot.sound`. These checks prove the stated implications, not their computational premises. The supplied modules express the sphere and separation lemmas, vertex-figure geometry, the centroid argument, the face lemma, rooted rigidity, local-to-global propagation, and conditional Case B arguments.

What remains outside Lean. The following written-to-computation links are not formalized: the numerical library/model correspondence (Section 3.2); stage-1 enumeration and root coverage (Section 4.1); the coverage of actual vertex figures by stage-2 signatures (Section 4.2); stage-3 encodings, identification and neighbor forcing (Section 4.4 and Proposition 4.14); the exhaustive stuck-development refutation of the extra candidate (Section 4.4.3); and the numerical B2, P1, partner-radius and W bounds used by the Case B structural proof (Section 5). The conditional geometric lemmas do not establish these hypotheses for the supplied certificates. In particular, rooted determinism (Lemma 4.13), congruence from discrete data (Lemma 4.10) and coset comparison (Lemma 4.11) are not formalized end to end for the concrete program encodings and candidate lists. The finite catalog coverage itself is derived conditionally from the cited classifications in `unitUniform4_smallFaces_catalogue`. No Lean theorem jointly instantiates the raw data and all twelve computational hypotheses with the two cited classifications; their joint truth rests on these written arguments and the external checkers. A kernel proof of their implication alone proves neither their truth nor their joint satisfiability. The classification is not a whole-theorem Lean proof.

7.1. **External hypothesis evidence map.** Table 2 maps the twelve declarations of Appendix A and four classification burdens to the written arguments and executable evidence. Paragraphs are identified by opening words, result names or numbered steps. The corresponding reader reference, `data/hypothesis-map.md`, includes the full command ledger and the count convention. "External" here means written proof together with certified computation, not a concrete Lean instance. The two cited lower-dimensional classifications are additional premises, not outputs of these scripts.

These are commands and expected outputs. The separate `data/replay-v2.json` records the earlier October 20 standalone full replay and its first v2 additions. It does not claim execution of the subsequently added primary-program and report-bridge checks C9 and C10.

Version 2 corrects an evidence omission in version 1: the primary programs named in the proof route and the source of the Lean report were absent from the bundle. The programs are now shipped under `checks/primary/`; C9 replays them. C10 regenerates the entire 629-row Lean file from the shipped source and compares every logical row with fresh independent stage-3 output. This closes the executable table correspondence. It does not construct a joint exact numerical instance of the twelve Lean hypotheses: their universal geometric interpretation still uses the written proofs. In particular, interval MATCH is never asserted to establish exact equality.

Table 2: External arguments and executable evidence. C1–C11 refer to the commands below.

| Input or burden | Written argument | Command and expected output | Scope and limitation |
|---|---|---|---|
| LIB | §3.2, LIB and LINK, paragraphs "What is computed" and "What is checked"; "A word determines its angle" proof; §3.3 construction paragraphs. | C1: `library_verify.py`, `library.py`, `parabolic.py`, `exceptional.py`; 181 types, 558 words, 1,277 polygon occurrences, containment and corruption checks pass. Exceptional models also run in C7. | External model/geometry correspondence. Construction proofs and cited 4-D classification remain premises; matching inventory counts alone is insufficient. |

| Input or burden | Written argument | Command and expected output | Scope and limitation |
|---|---|---|---|
| S1 | §4.1, S1 and "What is computed", then "Proof route"; exact exclusions in §3.2, "Exact inputs of the exclusions". | C1: `stage1.py`, `verify_stage1.py`, `compare.py`; 14,965 cycles, 7,080 flat, 57 degenerate, 537 rootless, 7,291 retained roots; full canonical-word comparison and corruptions pass. | Coverage on the extended domain, not a bound of $10^{-20}$ on merged intervals. The stopping tolerance applies to leaves only. |
| S2 | §4.2, S2, paragraph "Why such a search is exhaustive", and "Exact equivalence of placements" proof. | C2: `stage2.py`, `run.py`, `check_equivalence.py`; 563 complete classes, 629 signatures; 181 models and 5,108 ordered equivalences checked. Class 25 alone has no signature. | Coverage of all actual figures at every radius in each class; retained encodings need not be realizable. Uncertain predicates must remain OPEN. |
| ROOT | §4.4.1, ROOT and paragraph "Proof route"; §4.5.3, step 2. | C2: `s3_cycles.py`, `s3_radius.py`; Whole-class UNIQUE witnesses cover all 629 signatures, including the refuted signature. C6 tests wide intervals and corrupt closure constants. | Endpoint signs, derivative exclusion and continuity supply uniqueness. The final JSON audit checks coverage, not the derivative proof by itself. |
| DEV | §4.4.3, opening paragraph, DEV and "What is checked"; §4.5.3, step 5. | C2: `s3_develop.py`; extra radius-one signature REFUTED, zero OPEN; C6 recovers a known radius-one control and refutes the extra star, with uncertainty controls OPEN. | Refutes a full actual incident-facet star with the same root and center, conditional on the written geometry-to-encoding correspondence. C9 replays the primary encoding and C10 binds the same extra candidate to Lean row 12 by full cycle-word signature. Neither existence of the extra encoding nor exclusion based on an unlabeled figure alone is asserted. |
| B2 | §5.2, cycle-types proof; §5.3, three type-radius derivations; §5.4, "Reduction to a finite check", "Proof route", "Second implementation", "Input identity". | C1: `b2.py`, `verify_b2.sh`; 496,824 / 150,495 / 1,905 radii of types i/ii/iii, zero coincidences or undecided cases. `check_result.py`: 649,224 total, cutoff 398, disagreements 0. | Covers all integer $n \geq 17$ using the written tail bound $r_398 > \max \Omega$ and all polynomial roots in (-1,1). A finite loop without that reduction is insufficient. |

| Input or burden | Written argument | Command and expected output | Scope and limitation |
|---|---|---|---|
| P1 | §5.3, Type-(i) radius and closed-forms proofs; §5.6.1, P1 and paragraph "Hence the 67 configurations". | C3: `p1check.py`; 168 configurations, 137 closable, 28 nonclosable, 3 degenerate, 83 family, 67 excluded, 16 survivors. C4: `p1_literal_contract.py`; all 67 literal rational-range separations $> 1/2000$. | C4 closes an evidence omission: the earlier program could use tighter discrete/tail bounds instead of the literal wider intervals in P1Separated. Universal delta ranges use the written monotonicity proof, not samples of n. |
| W | §5.6.2, W; paragraphs "Scope of the two implementations", "Parameter domains", "Finite geometric realizations and the chart", "What is computed", "Completeness of computed results"; W-use proof. | C3: `fans.py`, `wcheck.py`, `w_bound.py` installed as `bnb.py`, `summary.py`; 127 full-domain fans, 120 certified, seven expected full-domain failures with seven certified n=m repairs. C5: `chartcheck.py`, four chart families and four rejected corruptions. | Finite geometric realizations only. New explicit chart paragraph supplies the coordinate-independent correspondence. Bounded partner parameters are 36 integer slices, not a real interval; seven exceptional fans are restricted to n=m. |
| Partner Radius Filter | §5.6.2, "The seven exceptional fans", Partner-radius filter lemma and proof beginning "From F". | C3: `summary.py` calls `partner.allowed`; all seven exceptional slices have exactly the matching singleton partner, and the full summary is ACCEPTED. | Both type-II exclusion and equality forcing use the radius equation, interval separation and written infinite-tail bounds. Running `partner.py` alone produces no audit. |
| U3 Realizations | §4.5.3, Rooted determinism proof and step 1 "Realization coverage"; §4.5.4, "Stages 2 and 3". | C2 + C9 + C10 + C11: `stage3.py`, primary `search.py` and `canon.py`, and `report_bridge.py`; 629 signatures and 657 independent realization groups, including 28 additional mirror placements excluded by separation; no OPEN. | C9 regenerates the primary signatures and raw encoding report; C10 checks its class/signature row correspondence with C2. Geometry-to-encoding completeness and fixed-radius rigidity remain written arguments, not concrete Lean instances. |
| U3 Identification | §3.3, construction paragraphs; §4.5.3, step 3 "Identification by elimination", and "stage 3, proof route" proposition proof. | C2 + C9 + C10: `s3_identify.py`, `models.py`, primary `identify.py` and the report bridge; 628 known signatures, 890 matched construction names, one wholly refuted signature, no OPEN. C6 rejects incomplete coverage. | Exact neighbor counts or exhaustive strict frame exclusions justify NO. POSSIBLE/MATCH interval overlap is not equality; existence of known constructions plus exhaustive elimination supplies identification. C10 checks all 890 MATCH, 230 NO-COUNT and 32 NO row/name pairs. Their exact-geometric soundness remains the written argument. |

| Input or burden | Written argument | Command and expected output | Scope and limitation |
|---|---|---|---|
| U3 Forcing | §4.5.3, step 4 "Local uniqueness", with Rooted determinism; independently §4.4.2, Coset comparison proof. | C2 + C9 + C10: `s3_stars.py` and primary `locuniq.py`; surviving known realizations have FORCED local uniqueness. C6 tests complete matching, symmetry and uncertainty controls. | The primary anchored-code route and independent coset route both run. C10 compares their FORCED flags for all 629 primary-corresponding rows, including the extra; 28 mirror alternatives are separate exclusions. Exact universal geometric meaning is supplied externally, not by the flags alone. |
| Exhaustiveness of local enumeration | §3.1 Library completeness; §4.1 complete closed walks and angle bound; §4.2 exhaustive opposite-hinge search; §4.5.3 step 1 and rooted rigidity. | C1 + C8 + C2 + C9 + C10 + C11, including equivalence/orientation coverage: 14,965 cycles → 563 classes → 629 signatures, no OPEN. | Enumerating known constructions alone would not prove completeness. Classification inputs, pruning soundness and geometry-to-encoding proofs remain external. |
| Identification and exclusion of retained signatures | §4.5.3 steps 2–4 and proposition proof; §4.6 proof of Case A. | C2: 628 known signatures, 890 names; 28 extra mirror realizations excluded by separation; all surviving known continuations forced. | Do not identify interval overlap with exact congruence or confuse 890 construction names with 628 objects. C9 and C10 now connect both executable routes to the Lean table; exact geometric soundness remains external. |
| Extra candidate | §4.4.3 DEV and "What is checked"; §4.5.3 step 5. | C2 + C6: the radius-one, eight-vertex figure's actual-star continuations all die; limits and uncertain identities remain OPEN. | C2 is required for full P16 coverage, beyond the small radius-one control. C10 checks the full signature and binds the extra candidate to Lean row 12; C9 replays its primary development. |
| At least 17 sides; unbounded parameters | §5.4 B2 corollary; §5.5 exactly-one-large-face product proof; §5.6.3 reduction to two paired faces; §5.6.4 paired theorem; §5.7 proof of Case B. | C1 + C3 + C4 + C5; optional Lean build checks their conditional geometric composition. | All-parameter coverage comes from the tail inequalities, finite charts, integer exceptions and structural product proof. Neither P=16 enumeration nor sampled large parameters supplies it. |

Literal contract clauses and data transport. The following field names refer to Appendix A. Each contract requires a written interpretation of numerical data as exact geometry; no Python command constructs a Lean proof term. P16 restricts polygon sizes in the finite library to at most 16. The audit field `historical_comparison=false` means that the S3 audit checks the new run's coverage and provenance rather than agreement with an old S3 result; it does not mean that inputs are unshared.

LIB.. `model_named`, `model_hull` and `vertices_exact` use the construction proofs and exact face certificates. `radius_enclosed`, `coordinates_enclosed` and `angles_enclosed` use rigorous coordinate and radius bounds and the word-angle proof. C1 rebuilds the local library and checks containment; C7 also checks exceptional whole models. Count agreement alone does not validate arbitrary real-valued Lean model data.

S1. `cycle_count`, `listed_admissible`, `distinct_classes` and `complete` use exhaustive canonical closed-walk enumeration and the angle-length bound. `reciprocal_enclosed` and `roots_cover` combine exact maximum-radius bounds, the extended reciprocal domain and certified root isolation. `with_roots`, `without_roots`, `flat_exact`, `flat_count`, `degenerate_exact` and `degenerate_count` use C1's exact status tests and inventories. Root intervals cover all zeros, including domain endpoints; small subdivision leaves may merge into larger intervals.

S2. `signature_count` and `empty_class` are finite accounting, including the explicit empty triangle-cycle class 25. `complete` covers every actual figure at every radius in each surviving class. Its proof requires exhaustive hinges and labels and exact placement equivalences, not merely the count 629. C8 checks the class and interval connection from C1 to C2.

ROOT.. `edge_root` covers every retained signature, including those not marked `KNOWN`. It requires strict opposite endpoint signs, a nonzero derivative throughout the reciprocal class hull, and a unique zero. C2 computes these witnesses; its final count script requires a whole-class `UNIQUE` witness for each of the 629 signatures, including the wholly refuted one. That script checks report coverage; `checks/independent-s23/s3_radius.py` supplies the interval argument.

DEV.. `refutation` concerns a hypothetical uniform polytope with the refuted encoding's actual root, center and full incident-facet star. U3Realizations' written coverage and encoding correspondence place it in the appropriate realization list; C2 exhausts its independent geometric realization/development route; C9 replays the primary route. C10 checks the full cyclic signature and binds the extra candidate to class 18, figure 0, Lean row 12. Figure shape alone and C6 alone do not establish full coverage.

B2. `type_i`, `type_ii` and `type_iii` quantify over integers $n \geq 17$ and their respective allowed configurations and parameters; the third includes every polynomial root in $(-1, 1)$. C1 checks the finite residual ranges; the written cutoff and exact $\Omega$ containment supply the infinite tail. The geometric definition of `stageOneOmega` is linked externally to the recorded enclosing intervals, rather than to an arbitrarily substituted $\Omega$.

P1. The data fields `configurations`, `closable`, `nonclosable`, `degenerate`, `family` and `excluded` are C3's six finite inventory sets. Their count fields `configurations_count`, `closable_count`, `nonclosable_count`, `degenerate_count`, `family_count` and `excluded_count` require, respectively, 168, 137, 28, 3, 83 and 67. The five equivalences `configurations_iff`, `closable_iff`, `nonclosable_iff`, `degenerate_iff`, `family_iff`, together with `excluded_subset`, link those sets to the stated predicates. `separation` uses C4's literal rational ranges and all 36 bounded tails. `delta44`, `delta333_range` and `delta36_range` require the written identities and monotonicity argument, not samples of $n$. `survivors` gives 16 literal family configurations. The other 54 closable configurations have no family-size polygon; together they form all 70 unpaired type-i W fans, so none is dropped.

W.. `no_long_pair` and `fan_distances` are distinct obligations on every finite `RealizesWFan`, with the exact `WListed` domains and allowed points. C3 checks them through the chart correspondence and supporting inequalities. C5 supplies only the free-box bound for four unbounded paired families, not all 127 fans. The chart correspondence remains a written proof obligation and does not follow from numerical success alone.

PartnerRadiusFilter. `exclude_ii` separates the antiprism-height alternative; `force_equal` compares each of seven exceptional slices with the three unbounded tail families and 36 bounded integer tails. C3 recomputes the seven singleton lists, and monotonic closed-form bounds cover the unbounded tails. The literal `PartnerTail` domain excludes the extra sampled nonclosable tails in diagnostic code.

U3Realizations. Row, class and signature accounting uses `class_valid`, `class_indices`, `signature_valid`, `signature_complete`, and `signature_indices`. `complete` covers all full geometric stars; `encoding_rigid` requires equal radius and equal full rooted encoding to imply star congruence. C2 checks its independent coverage and rigid template correspondences. C9 replays the primary raw encoding and C10 checks its full logical row projection against C2 and the Lean table. Rooted determinism supplies the external semantic argument.
U3Identification. `admitted` requires actual known uniform models, their roots, centers and radii, and full library-facet admission, using the construction proofs and C2's model checks. `soundNO` and `soundNOCOUNT` require strict exhaustive frame rejection and exact neighbor counts. C9 replays the primary raw encoding. C10 checks each row/name elimination against both the independent report and the Lean table.

There is no further MATCH or completeness field in U3Identification: coverage comes from U3Realizations, and finite other-row exclusions plus the soundness fields give the derived `identified_row` theorem.
U3Forcing. `checked` supplies `U3ForcingCheck` for every flagged row and admissible geometric star. Its four fields are `complete`, `retained_nonempty`, `encoding_unique` and `encoding_sound`. The primary argument uses complete centered candidate maps and anchored exact codes. C2 establishes geometric continuation forcing by another encoding/coset route. C9 replays the primary anchored-code implementation; C10 checks the row flags against the independent report and Lean table. Those checks do not construct the exact universal Lean contract.
Limits of the data correspondence. The primary programs, their source report and the table generator are now shipped. The former packaging limitation is corrected. Separate successful computations do not establish jointly satisfiable Lean data. Exporting floating-point or interval tables as Lean literals would not suffice: the fields also require exact real models, functions, isometries and universal geometry-to-data correspondences.

For the headline statement, see `lean/Uniform5/CompositionGeometry.lean` (`Uniform`, `UnitUniform`), `lean/Uniform5/CompositionKnownFiveNames.lean`,
and `lean/Uniform5/CompositionMainClassification.lean`, as explained in Section 7 and `lean/README.md`. The toolchain is Lean v4.19.0; `lean/lake-manifest.json` pins Mathlib to `c44e0c8ee63ca166450922a373c7409c5d26b00b`. `lean/Audit.lean` rejects every collected axiom other than the three listed foundations, including `sorryAx` and `Lean.ofReduceBool` if present. It audits logical dependencies, not every compiler or foreign-function execution path; no stronger audit of `extern` or `implemented_by` is asserted.

The lower-dimensional premise is Möller's 2004 thesis and corresponding *Results in Mathematics* article (doi:10.1007/BF03322887), following Conway–Guy's list, together with the cited convex uniform-polyhedron classification. These published classification premises are not re-proved or independently referee-reviewed by this bundle.
Commands C1–C11. Use the documented requirements and let `PY` denote their absolute Python interpreter, `OUT` a fresh absolute output directory. Commands start at the bundle root. C1 is

```
PYTHON="$PY" bash checks/independent-s1-b2/verify.sh
```

C2 is the full stage-2/3 replay and count extraction:

```
mkdir -p "$OUT"
(cd checks/independent-s23
 "$PY" unpack_inputs.py "$OUT/input" --pmax 16
 "$PY" run.py "$OUT/input/u2_classes_p16.json" "$OUT/s2.jsonl" --states
 "$PY" check_equivalence.py "$OUT/input/u2_classes_p16.json" \
   certificates/equivalences-p16.json
 "$PY" stage3.py --states "$OUT/s2.jsonl" --models models \
   --known "$OUT/input/u3_known_p16.json" \
```

```
   --output "$OUT/s3.jsonl" --jobs 2
 "$PY" audit_stage3.py --classes "$OUT/input/u2_classes_p16.json" \
   --states "$OUT/s2.jsonl" --results "$OUT/s3.jsonl" --models models \
   --known "$OUT/input/u3_known_p16.json" --pmax 16 \
   --receipt "$OUT/s3.jsonl.frozen.json")
"$PY" checks/count_convention.py "$OUT/s3.jsonl" \
  --known "$OUT/input/u3_known_p16.json" \
  --classes "$OUT/input/u2_classes_p16.json"
"$PY" checks/test_v2_outcomes.py "$OUT/s3.jsonl"
```

Every command must exit zero. The S3 audit expects PASS, 563 classes, 629 signatures and 657 realization groups: 628 known groups, 28 mirror alternatives excluded by strict full-star separation, and the additional candidate refuted by development. The raw schema uses REFUTED for both kinds of exclusion; their geometric reasons differ. C3 runs the full Case B check in a disposable copy:

```
mkdir -p "$OUT/caseb"
cp -R checks/check_caseb "$OUT/caseb/check_caseb"
cp checks/w_bound.py "$OUT/caseb/check_caseb/bnb.py"
PYTHON="$PY" bash "$OUT/caseb/check_caseb/run.sh" full
```

It must end with ACCEPTED and ALL OK, with precisely the seven expected full-domain failures and their restricted-point certifications. C4 checks the literal wider rational intervals of P1Separated:

```
"$PY" checks/p1_literal_contract.py
```

It requires all 67 clauses, with all selected separations greater than 1/2000. C5 checks four free-box chart bounds and rejects four corruptions:

```
"$PY" checks/check_caseb/chartcheck.py \
  checks/repair-caseb-w/chart-certificate.json
```

C6 runs small controls, not the P16 classification:

```
(cd checks/independent-s23
 "$PY" test_r1_stage3.py
 "$PY" test_stage3.py)
```

It requires the known/refuted radius-one controls, 74,963 brute-force matching comparisons, and the simplex/uncertainty corruption checks. C7 runs all public computational obligations:

```
PYTHON="$PY" bash checks/verify-public.sh full
```

The expected final line is `uniform-5-polytopes public full: PASS`. It includes the literal P1 check and the count extraction on fresh P16 output. It neither runs Lean nor automatically verifies these written mathematical arguments. C8 checks the connection from stage one to the packed stage-two inputs:

```
mkdir -p "$OUT/filter"
cp -R checks data "$OUT/filter/"
"$PY" "$OUT/filter/checks/check_filter.py"
```

Run C1 before C8, then C2. C1's `verify_stage1.py/equal_file` compares the fresh parsed cycles with `checks/independent-s1-b2/evidence/cycles.json.gz`; C8 reads that same authenticated file. Equality followed by recomputation connects these commands without an ephemeral filename. C2 alone is not full acceptance: all C7 parts must pass. C8 checks the complete canonical cycle sets, strict pairwise disjointness of the closed class hulls, and requires each consumer interval to contain the regenerated surviving stage-one roots, using exact rational endpoints. Expected: PASS, 4,224 initial P16 classes and 563 survivors; interval coverage counts P6/P10/P16 are 136/278/563; three

semantic filter corruptions and narrowed consumer intervals are rejected. C7 includes this check. Thus the data connection is C1–C8–C2, not an unexplained fixed list of 563 classes.

C9 replays the primary proof route in four independent parts:

```
"$PY" checks/replay_primary.py --part all --jobs 2 \
  --out "$OUT/primary"
```

Alternatively run `library`, `s23`, `caseb` and `wedge` with separate output directories. Expected: 181 models, 14,965 cycles, 563 surviving classes, 629 primary encodings (633 search states), 628 known, one refuted, 890 names and all root/forcing checks. The primary W inventory has 228 rows: 184 certified, 31 nonclosable, three degenerate and ten expected unrestricted failures. Seven admissible exceptional slices are repaired at $n = m$; the other three failed rows are supplementary fans $(3, 11, 13)$, $(4, 7, 9)$ and $(5, 6, 7)$, separately excluded by P1. No unrestricted failed fan is certified.

C10 regenerates and compares the Lean report:

```
"$PY" checks/report_bridge.py --write "$OUT/CompositionU3ReportP16.lean"
"$PY" checks/report_bridge.py --independent "$OUT/s3.jsonl" \
  --primary-report "$OUT/primary/s23/result_p16.json" \
  --signatures "$OUT/primary/s23/vfs.json"
"$PY" checks/test_report_bridge.py --independent "$OUT/s3.jsonl" \
  --primary-report "$OUT/primary/s23/result_p16.json" \
  --signatures "$OUT/primary/s23/vfs.json"
```

Expected: byte-identical regeneration of all 629 Lean rows; agreement of all class/full-cycle signatures, verdicts, ROOT and FORCED flags, all 890 MATCH, 230 NO-COUNT and 32 NO row/name pairs; nineteen corruptions rejected. Byte comparison is unconditional; `--write` also writes an inspectable copy. These paths follow C9's `--part all`; a separate `--part s23 --out DIR` writes directly under DIR.

Both source files are manifest-pinned and the full-signature source has its own decompressed SHA pin. The Lean header itself pins only the primary report. C9 regenerates full signatures, restores the pinned presentation order, then passes that actual ordered `vfs.json` to S3. C10 checks the report against that input, including interval identity. All indices are zero-based. The extra is class 18, figure 0, Lean row 12; A510001 belongs to the same class and its row-12 pair must be NO, not NO-COUNT. Coarse facet/count data alone cannot distinguish those figures. Their exact-star interpretation in Lean remains the external U3Realizations obligation; a signature hash is not that proof.

The candidate universe is every retained row of a class times every known name admitted by radius to that class: exactly 1,152 pairs. Both directions of the model/class inventory agree, and each of the 890 names MATCHes exactly one row. Other names are rejected by the radius/model-admission stage, not by an absent pair. C10 checks inventory consistency and all pair statuses; C2 and C9 perform the underlying geometric checks. Existence plus elimination gives the sole remaining row for each name, allowing several aliases in a row. MATCH is not an equality certificate.

The fresh-primary comparison includes every serialized Lean row field, including integer diagnostic counters. The independent comparison checks the semantic projection, since implementation-specific counters differ. No general Python-to-Lean contract translator is claimed: the other literal counts and P1 rational ranges are checked against their specified values by executable assertions and written inspection. `stageOneOmega` is a geometric data field, not a concrete interval table instantiated in Lean. See `data/report-provenance.md`.

C11 reconciles the search multiplicities and mirror exclusions:

```
"$PY" checks/encoding_correspondence.py \
  --primary-report "$OUT/primary/s23/result_p16.json" \
  --signatures "$OUT/primary/s23/vfs.json" \
  --independent "$OUT/s3.jsonl" --output "$OUT/accounting.json"
```

Primary 633 completed states reduce to 629 encodings: three duplicates are in the extra candidate and one in A511111. The independent 657 groups comprise 629 primary-corresponding groups and 28 additional snub-product mirror alternatives, excluded by strict full-star separation. Primary search cuts those alternatives earlier by cell-patch mismatch. C11 reruns all 28 normal searches, then a separately labeled diagnostic which bypasses only that cut: 56 distinct completed encodings, the original 28 preserved, and all 28 additional encodings with strict separation witnesses. Full signatures and class mappings agree with independent S3. This diagnostic does not replace normal exhaustive search or the written proof of pruning soundness. The two implementations use shared, separately audited inputs and written geometry; independence is in code, not every premise. C7 includes C9–C11 in full mode.

Known-name and model provenance. These have a C2 chain separate from C8's class/interval bridge. `unpack_inputs.py` checks the decompressed known-name list against both input manifests; `stage3.py` records its byte digest and each class's parsed-list digest; `audit_stage3.py` authenticates both and checks complete model-source and certificate inventories. Model admission checks construction binding, radius and facet types. The construction theorems still supply uniformity, transitivity and geometric interpretation: hashes establish none of those facts, nor a Lean instance. Structural/lazy model certification is not expansion and an all-pairs check of every full orbit. The two counts of 181 in LIB and S2 refer to the same canonical four-dimensional library, rebuilt and checked independently.

## Appendix A. The twelve Lean core hypotheses

The following declarations are copied verbatim from the shipped Lean sources. Names are in `Uniform5.Composition`, except B2, P1 and PartnerRadiusFilter in `Uniform5.CaseB`, and W in `Uniform5.Composition.Wedge`. Referenced types and predicates are in the cited source files; these declarations are assumptions, not verified instances of the certificates. P1 includes finite data as well as proof fields.

```
-- Core hypotheses: verbatim declarations from the accompanying Lean sources.

-- Uniform5.Composition.LIB
-- Source: lean/Uniform5/CompositionLibraryBalls.lean
structure LIB (D : LibraryNumericalData) : Prop where
  model_named : ∀ n, FacetLibraryModel n.val (D.model n)
  model_hull : ∀ n, D.model n = convexHull ℝ (D.vertexSet n : Set (Space 4))
  vertices_exact : ∀ n, (D.vertexSet n : Set (Space 4)) = extremePoints ℝ (D.model n)
  radius_enclosed : ∀ n, ∃ r, HasIntrinsicRadius (D.model n) r ∧
    (D.radiusBall n).Contains r
  coordinates_enclosed : ∀ n, ∀ v ∈ D.vertexSet n, ∀ j,
    (D.coordinateBall n v j).Contains (v j)
  angles_enclosed : ∀ n (hK : FinitePolytope (D.model n)) (k : ℕ)
    (G A B : Set (Space 4)) (hG : PolygonFace (D.model n) G k)
    (hA : Facet (D.model n) A) (hB : Facet (D.model n) B),
    G ⊆ A → G ⊆ B → A ≠ B → ∀ a b,
    Congruent A (cellTypeModel a) → Congruent B (cellTypeModel b) →
    (D.angleBall n k a b).Contains (geometricPolygonAngle hK hG hA hB)

-- Uniform5.Composition.S1
-- Source: lean/Uniform5/CompositionStageOneInputs.lean
structure S1 (N : LibraryNumericalData) (D : StageOneNumericalData) : Prop where
  cycle_count : D.listed.card = 14965
  listed_admissible : ∀ W ∈ D.listed, W.Admissible N
  distinct_classes : ∀ U ∈ D.listed, ∀ V ∈ D.listed,
    U.classCode = V.classCode → U = V
  complete : ∀ W : LibraryCycleWord, W.Admissible N → ∃ V ∈ D.listed, W.classCode = V.classCode
  reciprocal_enclosed : ∀ W ∈ D.listed,
    (D.reciprocalBall W).Contains (1 / W.maxRadius N)
  roots_cover : ∀ W ∈ D.listed, ∀ s ∈ Icc 0 (D.reciprocalBall W).upper,
    W.Domain N s → W.closure N s = 0 → ∃ I ∈ D.roots W, s ∈ I.carrier
```

```
  with_roots : D.count (fun W => D.roots W ≠ []) = 14428
  without_roots : D.count (fun W => D.roots W = []) = 537
  flat_exact : ∀ W ∈ D.listed, D.flat W = true ↔ W.angleSum N = 2 * Real.pi
  flat_count : D.count (fun W => D.flat W = true) = 7080
  degenerate_exact : ∀ W ∈ D.listed, ∀ i,
    D.degenerate W i = true ↔ D.EndpointInterval N W i
  degenerate_count : D.degenerateCount = 57

-- Uniform5.Composition.S2
-- Source: lean/Uniform5/CompositionStageTwoInputs.lean
structure S2 (N : LibraryNumericalData) (S : StageOneNumericalData)
    (T : StageTwoNumericalData S) : Prop where
  signature_count : stageTwoSignatureCount N S T = 629
  complete : ∀ c : ActiveRootClass S, c.val ∈ survivingRootClasses N S →
    ∀ R ∈ radiusClassRegion S c, ∀ D (hD : StageTwoFigureConditions N S c.val R D),
      D.signature hD ∈ T.signatures c
  empty_class : ∃ c : ActiveRootClass S, c.val ∈ survivingRootClasses N S ∧
    T.signatures c = ∅ ∧
    (∀ d : ActiveRootClass S, d.val ∈ survivingRootClasses N S → T.signatures d = ∅ → d = c) ∧
    ∃ W ∈ rootClassCycles S c.val, emptyStageTwoClassWord W

-- Uniform5.Composition.ROOT
-- Source: lean/Uniform5/CompositionRootInput.lean
structure ROOT (N : LibraryNumericalData) (S : StageOneNumericalData)
    (T : StageTwoNumericalData S) : Prop where
  edge_root : ∀ c, c.val ∈ survivingRootClasses N S →
    ∀ (σ : VertexFigureSignature), σ ∈ T.signatures c →
    ∃ W ∈ σ.edges,
      W.closure N (reciprocalClassHull S c).lo *
        W.closure N (reciprocalClassHull S c).hi < 0 ∧
      (∀ s ∈ (reciprocalClassHull S c).carrier, deriv (W.closure N) s ≠ 0) ∧
      ∃! s, s ∈ (reciprocalClassHull S c).carrier ∧ W.closure N s = 0

-- Uniform5.Composition.DEV
-- Source: lean/Uniform5/CompositionU3DevelopmentInput.lean
structure DEV (N : LibraryNumericalData) (S : StageOneNumericalData)
    (U : U3RealizationData S) (extra : U3StarData) : Prop where
  refutation : ∀ X, U3GeometricStar N S (U.rowClass u3RefutedRow).val X →
    X.radius ∈ radiusClassRegion S (U.rowClass u3RefutedRow) →
    (U.rowSignature u3RefutedRow).RadiusCompatible N X.radius →
    U.canonicalEncoding X = U.rowEncoding u3RefutedRow →
    ∀ P : UnitUniform5, ∀ v ∈ P.val.extremePoints ℝ,
      X.root = v → X.center = polytopeCenter P →
      X.star N = facetStar P.val v → False

-- Uniform5.CaseB.B2
-- Source: lean/Uniform5/CaseBInputs.lean
structure B2 (Ω : Set ℝ) : Prop where
  type_i : ∀ (n : ℕ), 17 ≤ n → ∀ c : List ℕ,
    AngleCondition (realConfig c) → Closable (realConfig c) →
    Real.sqrt (polygonRadius n ^ 2 + configRadius (realConfig c) ^ 2) ∉ Ω
  type_ii : ∀ (n q : ℕ), 17 ≤ n → 3 ≤ q →
    Real.sqrt (antiprismRadiusSq n + polygonRadius q ^ 2) ∉ Ω
  type_iii : ∀ (n q₁ q₂ : ℕ), 17 ≤ n → TypeIIIPair q₁ q₂ →
    ∀ γ : ℝ, -1 < γ → γ < 1 → TypeIIIPolynomial n q₁ q₂ γ = 0 →
    Real.sqrt (antiprismRadiusSq n + 1 / (2 * (1 + γ))) ∉ Ω

-- Uniform5.CaseB.P1
-- Source: lean/Uniform5/CaseBInputs.lean
structure P1 where
  configurations : Finset (List ℕ)
  closable : Finset (List ℕ)
  nonclosable : Finset (List ℕ)
  degenerate : Finset (List ℕ)
  family : Finset (List ℕ)
```

```
  excluded : Finset (List ℕ)
  configurations_iff : ∀ c, c ∈ configurations ↔ SmallConfiguration c
  closable_iff : ∀ c, c ∈ closable ↔ c ∈ configurations ∧ Closable (realConfig c)
  nonclosable_iff : ∀ c, c ∈ nonclosable ↔ c ∈ configurations ∧
    configArcSum (realConfig c) (configMax (realConfig c)) < 2 * Real.pi
  degenerate_iff : ∀ c, c ∈ degenerate ↔ c ∈ configurations ∧
    configArcSum (realConfig c) (configMax (realConfig c)) = 2 * Real.pi
  family_iff : ∀ c, c ∈ family ↔ c ∈ closable ∧ ∃ q ∈ c, FamilyPolygon q
  excluded_subset : excluded ⊆ family
  configurations_count : configurations.card = 168
  closable_count : closable.card = 137
  nonclosable_count : nonclosable.card = 28
  degenerate_count : degenerate.card = 3
  family_count : family.card = 83
  excluded_count : excluded.card = 67
  separation : ∀ c ∈ excluded, ∃ q ∈ c, FamilyPolygon q ∧ P1Separated c q
  delta44 : ∀ n : ℕ, 17 ≤ n → delta [4, 4] n = 1 / 4
  delta333_range : ∀ n : ℕ, 17 ≤ n →
    delta [3, 3, 3] n ∈ Set.Icc (186963 / 1000000 : ℝ) (3 / 16)
  delta36_range : ∀ n : ℕ, 17 ≤ n →
    delta [3, 6] n ∈ Set.Icc (1 / 3 : ℝ) (337262 / 1000000)
  survivors : ∀ c, c ∈ family ∧ c ∉ excluded ↔
    ∃ q, FamilyPolygon q ∧ (c = [4, 4, q] ∨ c = [3, 3, 3, q])

-- Uniform5.Composition.Wedge.W
-- Source: lean/Uniform5/CompositionWedgeInput.lean
structure W (p1 : CaseB.P1) : Prop where
  no_long_pair : ∀ n f, CaseB.LargeParameter n → CaseB.WListed p1 n f →
    ∀ F, RealizesWFan p1 n f F → ∀ p q,
    (F.Admissible p ∨ F.Allowed p) → (F.Admissible q ∨ F.Allowed q) →
    CaseB.chord 17 ≤ dist p q → p ∈ F.vertices ∧ q ∈ F.vertices
  fan_distances : ∀ n f, CaseB.LargeParameter n → CaseB.WListed p1 n f →
    ∀ F, RealizesWFan p1 n f F → ∀ p ∈ F.vertices, ∀ q ∈ F.vertices,
    ¬ F.specialPair p q → dist p q < CaseB.chord 17

-- Uniform5.CaseB.PartnerRadiusFilter
-- Source: lean/Uniform5/CaseBInputs.lean
structure PartnerRadiusFilter : Prop where
  exclude_ii : ∀ m c₁, ExceptionalPaired m c₁ →
    delta c₁ m ≠ antiprismHeightSq m / 4
  force_equal : ∀ (n m : ℕ) (c₁ c₂ : List ℕ), 17 ≤ n → ExceptionalPaired m c₁ →
    PartnerTail n c₂ →
    AngleCondition ((n : ℝ) :: realConfig c₂) → Closable ((n : ℝ) :: realConfig c₂) →
    delta c₁ m = delta c₂ n → c₂ = c₁ ∧ n = m

-- Uniform5.Composition.U3Realizations
-- Source: lean/Uniform5/CompositionU3RealizationInputs.lean
structure U3Realizations (N : LibraryNumericalData) (S : StageOneNumericalData)
    (T : StageTwoNumericalData S) (U : U3RealizationData S) : Prop where
  class_valid : ∀ r ∈ u3ReportP16,
    (U.rowClass r).val ∈ survivingRootClasses N S
  class_indices : ∀ r ∈ u3ReportP16, ∀ s ∈ u3ReportP16,
    U.rowClass r = U.rowClass s → r.classIndex = s.classIndex
  signature_valid : ∀ r ∈ u3ReportP16,
    U.rowSignature r ∈ T.signatures (U.rowClass r)
  signature_complete : ∀ c : ActiveRootClass S,
    c.val ∈ survivingRootClasses N S → ∀ σ ∈ T.signatures c,
      ∃ r ∈ u3ReportP16, U.rowClass r = c ∧ U.rowSignature r = σ
  signature_indices : ∀ r ∈ u3ReportP16, ∀ s ∈ u3ReportP16,
    U.rowClass r = U.rowClass s → U.rowSignature r = U.rowSignature s →
      r.signatureIndex = s.signatureIndex
  complete : ∀ (c : ActiveRootClass S), c.val ∈ survivingRootClasses N S →
    ∀ X (hX : U3GeometricStar N S c.val X), X.radius ∈ radiusClassRegion S c →
      X.figure.signature hX.figure ∈ T.signatures c →
      ∃ r ∈ u3ReportP16, U.rowClass r = c ∧
```

```
        U.rowSignature r = X.figure.signature hX.figure ∧
        U.canonicalEncoding X = U.rowEncoding r
  encoding_rigid : ∀ (c : ActiveRootClass S), c.val ∈ survivingRootClasses N S →
    ∀ X Y, U3GeometricStar N S c.val X → U3GeometricStar N S c.val Y →
      X.radius ∈ radiusClassRegion S c → Y.radius ∈ radiusClassRegion S c →
      X.radius = Y.radius → U.canonicalEncoding X = U.canonicalEncoding Y →
        U3StarCongruent N X Y

-- Uniform5.Composition.U3Identification
-- Source: lean/Uniform5/CompositionU3IdentificationInputs.lean
structure U3Identification (N : LibraryNumericalData) (S : StageOneNumericalData)
    (U : U3RealizationData S) (M : U3KnownModelData) : Prop where
  admitted : ∀ r ∈ u3ReportP16, ∀ n : KnownFiveName,
    n.sourceName ∈ r.known_names →
      KnownFiveModel n (M.model n) ∧
      ∃ hQ : UnitUniform 5 (M.model n),
        M.root n ∈ (M.model n).extremePoints ℝ ∧
        M.center n = vertexCentroid (vertices (M.model n) (uniform_finitePolytope hQ.1)) ∧
        intrinsicRadius (M.model n) ∈ radiusClassRegion S (U.rowClass r) ∧
        ∀ F, Facet (M.model n) F →
          ∃ s ∈ u3ReportP16, ∃ k ∈ (U.rowSignature s).facets, Congruent F (N.model k)
  soundNO : ∀ r ∈ u3ReportP16, ∀ n : KnownFiveName,
    (n.sourceName, U3EmbeddingVerdict.no) ∈ r.embedding →
      ∀ X, U3GeometricStar N S (U.rowClass r).val X →
        X.radius ∈ radiusClassRegion S (U.rowClass r) →
        U.canonicalEncoding X = U.rowEncoding r → ¬ U3MatchesKnown N M n X
  soundNOCOUNT : ∀ r ∈ u3ReportP16, ∀ n : KnownFiveName,
    (n.sourceName, U3EmbeddingVerdict.noCount) ∈ r.embedding →
      ∀ X, U3GeometricStar N S (U.rowClass r).val X →
        X.radius ∈ radiusClassRegion S (U.rowClass r) →
        U.canonicalEncoding X = U.rowEncoding r → ¬ U3MatchesKnown N M n X

-- Uniform5.Composition.U3Forcing
-- Source: lean/Uniform5/CompositionU3ForcingInputs.lean
structure U3Forcing (N : LibraryNumericalData) (S : StageOneNumericalData)
    (U : U3RealizationData S) (D : U3ForcingData) : Prop where
  checked : ∀ r ∈ u3ReportP16, r.local_uniqueness.forced = true →
    ∀ X, U3GeometricStar N S (U.rowClass r).val X →
      X.radius ∈ radiusClassRegion S (U.rowClass r) →
      U.canonicalEncoding X = U.rowEncoding r →
      ∀ w, starNeighbor X.root w (X.star N) → U3ForcingCheck N D X w
```

## Appendix B. Names

Table 3. Coxeter diagrams and node orders used in the names.

| Group | Diagram (nodes $0, 1, \dots$) |
|---|---|
| $A_4$, $A_5$ | linear, all branches 3 |
| $B_4$ | linear $3, 3, 4$; the 4 between nodes 2 and 3 |
| $B_5$ | linear $3, 3, 3, 4$; the 4 between nodes 3 and 4 |
| $D_4$ | node 1 joined to nodes 0, 2, 3 (all 3) |
| $D_5$ | node 2 joined to nodes 1, 3, 4; node 0 joined to node 1 |
| $F_4$ | linear $3, 4, 3$; the 4 between nodes 1 and 2 |
| $H_4$ | linear $5, 3, 3$; the 5 between nodes 0 and 1 |

A name such as `B40001` lists the rings on nodes $0, 1, 2, 3$ (here the tesseract; `B41000` is the 16-cell). Cell labels: `tet`, `oct`, `cube`, `ico`, `doe` (Platonic); `tut`, `co`, `tic`, `toe`, `sirco`, `girco`, `id`, `tid`, `ti`, `srid`, `grid`, `snic`, `snid` (truncated tetrahedron, cuboctahedron, truncated cube, truncated octahedron, rhombicuboctahedron, truncated cuboctahedron, icosidodecahedron, truncated icosahedron,

truncated dodecahedron, rhombicosidodecahedron, truncated icosidodecahedron, snub cube, snub dodecahedron, in this order; note that `tid` is the truncated icosahedron and `ti` the truncated dodecahedron, the reverse of Bowers' usual names); `P`$n$ the $n$-gonal prism ($n \neq 4$) and `A`$n$ the $n$-gonal antiprism ($n \neq 3$).

TABLE 4. The 33 aliases of the facet library and their canonical names (Computation 3.2).

| Canonical | Aliases | Canonical | Aliases |
|---|---|---|---|
| `A40001` | `A41000` | `B41000` | `D40001`, `D40010`, `D41000` |
| `A40010` | `A40100` | `B41010` | `D41011`, `F40010`, `F40100` |
| `A40011` | `A41100` | `B41100` | `D40101`, `D40110`, `D41100` |
| `A40101` | `A41010` | `B41110` | `D41111`, `F40011`, `F41100` |
| `A40111` | `A41110` | `F40101` | `F41010` |
| `A41011` | `A41101` | `F40111` | `F41110` |
| `B40001` | `pr-cube`, `duoprism-4-4` | `F41011` | `F41101` |
| `B40010` | `D40011`, `D41001`, `D41010` | `pr-oct` | `antiprism-prism-3` |
| `B40100` | `D40100`, `F40001`, `F41000` | | |
| `B40110` | `D40111`, `D41101`, `D41110` | | |

## REFERENCES


[1] W. A. Wythoff, *A relation between the polytopes of the $C_{600}$-family*, Proceedings, Koninklijke Akademie van Wetenschappen Amsterdam **20** (1918), 966–970.

[2] H. S. M. Coxeter, *Wythoff's construction for uniform polytopes*, Proceedings of the London Mathematical Society **s2-38** (1935), 327–339. https://doi.org/10.1112/plms/s2-38.1.327.

[3] J. Bowers, *Uniform polytera*, online catalog, https://www.polytope.net/hedrondude/polytera.htm.

[4] R. Klitzing, *Polytopes and their relations*, online catalog, https://bendwavy.org/klitzing/home.htm.

[5] M. L. Balinski, On the graph structure of convex polyhedra in $n$-space, *Pacific J. Math.* 11 (1961), 431–434. doi:10.2140/pjm.1961.11.431.

[6] G. Blind and R. Blind, The semiregular polytopes, *Comment. Math. Helv.* 66 (1991), 150–154. doi:10.1007/BF02566640.

[7] M. Càllisto, *The 28 convex uniform honeycombs: a completeness theorem*, Zenodo, September 21, 2026. doi:10.5281/zenodo.22881686. Lean companion: doi:10.5281/zenodo.22868902.

[8] J. H. Conway and M. J. T. Guy, Four-dimensional Archimedean polytopes, in: *Proc. Colloquium on Convexity, Copenhagen 1965*, Københavns Univ. Mat. Inst., 1967, pp. 38–39. Zbl 0149.17806.

[9] H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover, New York, 1973.

[10] H. S. M. Coxeter, M. S. Longuet-Higgins and J. C. P. Miller, Uniform polyhedra, *Philos. Trans. Roy. Soc. London Ser. A* 246 (1954), 401–450. doi:10.1098/rsta.1954.0003.

[11] E. L. Elte, *The Semiregular Polytopes of the Hyperspaces*, Gebroeders Hoitsema, Groningen, 1912. https://books.google.com/books?id=_UzAHuloksIC.

[12] T. Gosset, On the regular and semi-regular figures in space of $n$ dimensions, *Messenger of Mathematics* 29 (1900), 43–48.

[13] B. Grünbaum, An enduring error, *Elem. Math.* 64 (2009), 89–101. doi:10.4171/EM/120.

[14] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* 66 (2017), 1281–1292. doi:10.1109/TC.2017.2690633.

[15] N. W. Johnson, *The Theory of Uniform Polytopes and Honeycombs*, Ph.D. thesis, University of Toronto, 1966.

[16] N. W. Johnson, *Uniform Polytopes*, unpublished manuscript, 1991.

[17] M. Möller, *Vierdimensionale Archimedische Polytope*, Dissertation, Universität Hamburg, 2004. https://ediss.sub.uni-hamburg.de/handle/ediss/689.

[18] M. Möller, 4-dimensionale Archimedische Polytope, *Results Math.* 46 (2004), 271–360. doi:10.1007/BF03322887.

[19] Open Problem Garden, Convex uniform 5-polytopes (posted May 24, 2012), http://www.openproblemgarden.org/op/convex_uniform_5_polytopes.

[20] Polytope Wiki, Uniform polytope, section 5D, https://polytope.miraheze.org/wiki/Uniform_polytope.

[21] E. Schulte, Symmetry of polytopes and polyhedra, Chapter 18 in *Handbook of Discrete and Computational Geometry*, 3rd ed. (J. E. Goodman, J. O'Rourke, C. D. Tóth, eds.), CRC Press, 2017.

[22] J. Skilling, The complete set of uniform polyhedra, *Philos. Trans. Roy. Soc. London Ser. A* 278 (1975), 111–135. doi:10.1098/rsta.1975.0022.
[23] S. P. Sopov, A proof of the completeness of the list of elementary homogeneous polyhedra, *Ukrain. Geom. Sb.* 8 (1970), 139–156 (in Russian).
[24] M. Winter, *Spectral Realizations of Symmetric Graphs, Spectral Polytopes and Edge-Transitivity*, Dissertation, TU Chemnitz, 2021, Appendix E.3, urn:nbn:de:bsz:ch1-qucosa2-752155.